

Face selection for slowly switching Markov chains and the planar persistent walk

Arjun Pemmasani

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Abstract

Kagey’s Problem 131 asks when the middle bin of a Galton board with persistent bounces becomes as likely as an endpoint. We study the same question for a Markov chain on d states whose switching probabilities are T/N times fixed rates, with $T = \tau \log N$. The counts of the N visits play the role of the bin. To first order the mode lies on a face F of the simplex that minimizes $\tau\lambda_F + |F| - 1$ among the faces that the chain can fill. Here λ_F is the rate at which the chain leaves F , and this holds for every starting law. On faces whose switching graph is strongly connected and that carry start mass we prove a local law with explicit constants K_F . Then every crossing between 2 such faces F and G satisfies $\delta T + \frac{\Delta}{2} \log T = \Delta \log N - \log(K_F/K_G) + O(1/\log N)$, where $\Delta = |F| - |G|$ and $\delta = \lambda_G - \lambda_F$. The 2 companion papers are special cases. A directed 3-cycle shows that the coefficient $\frac{1}{2}$ can fail for other faces. We also treat sticky priors for hidden Markov models, the planar persistent walk, and how much the endpoint alone tells us about the switching rate.

1 Introduction

Problem 131 of Kagey’s Open Problem Collection [1] asks about a Galton board where the ball tends to keep going the same way. Each bounce repeats the one before it with probability p , and the question is when the middle bin becomes as likely as an endpoint. The companion papers treat this question for 2 directions [2] and for d directions with uniform switching [3]. In both, the crossing comes when N times the probability of switching to a given other direction is $\log N - \frac{1}{2} \log \log N + O(1)$, with an explicit constant in the $O(1)$ term. In this paper we ask the same question when the switching rule is a general finite Markov chain.

The model. Let Q be a generator on $[d] = \{1, \dots, d\}$, with rates $q_{ij} \geq 0$ for $i \neq j$ and exit rates $q_i = \sum_{j \neq i} q_{ij}$, and let μ be a starting law on $[d]$. We run the chain $X_1, \dots, X_N$ with $X_1 \sim \mu$ and transition matrix $I + hQ$, where

$$h = \frac{T}{N}, \quad T = \tau \log N, \quad \tau > 0.$$

So at each step the chain leaves state i with probability hq_i , and N steps look like the continuous-time chain with generator Q run for time T . The composition $n = (n_1, \dots, n_d)$ counts the visits to each state, and it plays the role of the bin. For $d = 2$, $q_{12} = q_{21} = 1$ and a uniform start, this is the walk of Paper A with $h = 1 - p$. With $q_{ij} = 1$ for all $i \neq j$ and a uniform start it is the model of Paper B (Example 2.5). The support of n is a face F of the simplex, and Kagey’s question becomes which face holds the most likely composition, and at which T the mode moves from one face to another. The middle bin against an endpoint is the full face against a vertex.

The switching graph. Let G_Q be the switching graph, with an edge $i \rightarrow j$ when $q_{ij} > 0$. For a face F let Q_F be the principal submatrix of Q on F , and put $\lambda_F = -\varrho(Q_F)$, where ϱ is the largest real eigenvalue. This is the exit rate of F . The continuous-time chain stays in F up to time t with probability about $e^{-t\lambda_F}$. We show that if some path of positive probability has support exactly F , then the most likely composition with support F has probability $N^{-\Phi_\tau(F)+o(1)}$, where

$$\Phi_\tau(F) = \tau\lambda_F + |F| - 1.$$

The reason is that a path with support F must enter each of the other $|F| - 1$ states of F , and each first entry is a switch of probability about $h = T/N$. So entering the new states is a factor of about $N^{-(|F|-1)}$, up to powers of T . A later run in a state needs another switch, but it also multiplies the number of ways to split the visits to that state into runs by a factor of order N , so it is not penalized. Staying inside F for the whole time T is a factor about $e^{-T\lambda_F}$. With $T = \tau \log N$ both factors are powers of N . So a sticky walk prefers to sit in a well-separated community, a set of states with a small exit rate for its size. The winning faces are read off from the lower-left boundary of the convex hull of the points $(\lambda_F, |F| - 1)$, and the winning face changes at values of τ equal to minus the slopes of its edges (Theorem 3.6 and Figure 1).

Results. Theorem 3.3 proves this for every start μ . Call a face admissible if some path of positive probability has support exactly F . With $L = \log N$ we prove

$$\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L),$$

where the minimum is over admissible faces, and that for large N every global mode has its support on a minimizing face. Note that Φ_τ does not depend on μ , so to first order the start enters only through the set of admissible faces. If the minimizing faces are irreducible, meaning that their switching graphs are strongly connected, the mode lies within $O(1/L)$ of the quasi-ergodic law of the chain killed on leaving the face. This holds for example when Q has symmetric support or μ has full support (Theorem 3.6).

Section 4 finds the $\log \log N$ term. Say that F satisfies (H) if it is irreducible and $\mu(F) > 0$, write $k = |F|$, and let M_F be the largest probability of a composition with support F . We prove a local law for the compositions in the interior of F (Theorem 4.3 and Lemma 4.4) and deduce that

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} (1 + O(1/T + T^2/N))$$

uniformly for $1 \leq T \leq N^{1/3}$, with an explicit constant K_F (Theorem 4.5). The factor $T^{(k-1)/2}$ appears because the occupation fractions fluctuate on the scale $T^{-1/2}$ in $k - 1$ directions inside the face. Now take 2 faces F and G with (H), and put $\Delta = |F| - |G| \geq 1$ and $\delta = \lambda_G - \lambda_F > 0$. Then every $T_N \in [1, N^{1/3}]$ with $M_F = M_G$ satisfies

$$\delta T_N + \frac{\Delta}{2} \log T_N = \Delta L - \log \frac{K_F}{K_G} + O\left(\frac{1}{L}\right)$$

(Theorem 4.6). Hence $T_N = \frac{\Delta}{\delta} [L - \frac{1}{2} \log L + O(1)]$, and the coefficient $\frac{1}{2}$ of $\log \log N$ in Papers A and B holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility. When all the faces that tie to first order satisfy (H), the global mode follows the upper envelope of explicit lines in a window of this form (Theorem 4.7). Papers A and B are special cases (Corollaries 4.9 and 4.10). For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 4.2), and for Paper B a weighted form of Cayley's

formula evaluates it. Without (H) the coefficient can change. On a directed 3-cycle the mode passes through a reducible face, and the 2 crossings have coefficients 1 and 0 (Proposition 4.12).

Section 5 works out examples. On the complete graph every face has the same first-order exponent at $\tau = 1$, which is the degeneracy of Paper B, and the second order decides. There the start matters. On K_3 with $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$ an edge is the mode on a window of width about 0.026 in the offset of Theorem 4.7, and with a uniform start this window is empty (Proposition 5.2). On the cycle C_d with $d \geq 4$ the winners are arcs of $1, 2, \dots, \lfloor 2d/3 \rfloor$ states and then the whole cycle (Theorem 5.4). Once an arc covers two thirds of the cycle, closing the cycle beats growing the arc. Two triangles joined by a link of small rate ε form 2 communities, and a triangle wins from $\tau \approx 1$ to $\tau \approx 9/\varepsilon$ (Proposition 5.8).

Section 6 applies face selection to sticky priors for hidden Markov models. We fix the exit probability of every state at $h = c \log N/N$ and let the switching kernel Π be a stochastic matrix with zero diagonal and positive off-diagonal entries. Then for large N the most likely composition under the prior, which we call the MAP composition, has its support on a minimizing face, with no reversibility needed (Corollary 6.1). For $d \geq 3$ the minimizing faces go from single states straight to all d states exactly when $\varrho(\Pi_F) \leq (|F| - 1)/(d - 1)$ for every face F (Proposition 6.2). For a symmetric kernel this happens only for the uniform kernel of Paper B. Under the weak-limit sticky HDP-HMM prior with 3 states and uniform base weights, it has probability $\frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$ as the concentration $\omega \rightarrow \infty$ (Proposition 6.3). So the degeneracy of Paper B is not typical. For a symmetric pair of states with equal start weights, a single state against the pair is exactly Kagey's crossing (Proposition 6.4). The MAP composition can also mislead. Below the first threshold it uses a single state, while the event that the chain uses a single state has probability about N^{-c} (Proposition 6.5).

Section 7 treats the planar persistent walk, which is Kagey's Galton board in 2 dimensions. The walk moves in the directions E, N, W, S . At each step it keeps its direction with probability $1 - (2r + s)h$, turns left or right with probability rh each, and reverses with probability sh . For even N we compare the corner $(N, 0)$, which is reached only by going east at every step, with the origin. The crossing is unique (Proposition 7.2). For $r > 0$ it happens at $h = T_N^*/N$ with $T_N^*/\log N \rightarrow \min(2/(2r + s), 1/s)$ (Theorem 7.3). The first value comes from paths that use all 4 directions and the second from paths along a single axis, so the regime switches at $s = 2r$. When reversals dominate ($s > 2r$), sT_N^* follows the expansion of Paper A with $\log N$ replaced by $\log N - \log 2$, since the 2 axes double the mass at the origin (Theorem 7.4). When turns dominate ($s < 2r$), a two-dimensional local limit (Theorem 7.5) gives

$$(2r + s)T_N^* + \log T_N^* = 2 \log N + \log \frac{\pi}{8(r + s)} + o(1)$$

(Corollary 7.6). With turns only ($s = 0$), rT_N^* agrees with the crossing $N(1 - p_N)$ of Paper A up to $O((\log N)^4/N)$ (Proposition 7.7).

Section 8 asks how much the endpoint alone tells us about the switching rate. For the walk of Paper A with flip probability q , let $e_N(q)$ be the Fisher information about q in the endpoint divided by the Fisher information in the whole path. With $x = Nq$ we prove $e_N(q) = \frac{1}{2x} + \frac{1}{4x^2} + O(x^{-3} + x^3/N)$ for $1 \leq x \leq N^{1/4}$ (Theorem 8.3). At Kagey's crossing this gives $e_N \approx 1/(2 \log N)$ (Corollary 8.4), but the correction is large. At $N = 3200$ the product $2e_N \log N$ is still about 1.35.

Relation to established results. The rate I_F of Section 3 is the Donsker–Varadhan rate of the occupation measure of the chain killed on leaving F . For reversible jump processes Dupuis and

Liu give the Donsker–Varadhan rate explicitly [4, Eq. (3.2)]. On a finite reversible chain it reads $\sum_i \alpha_i q_i - \sum_{i \neq j} \sqrt{\alpha_i \alpha_j q_{ij} q_{ji}}$ for a law α on the states, and for α carried by F this is $I_F(\alpha)$. Guillin, Nectoux and Wu [5] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set (their Thm 2), and its rate vanishes only at the quasi-ergodic law (their (2.13)). When the process is reversible they prove that the exit rate of the set is the least value of the Donsker–Varadhan rate over laws on the set (their Cor. 2). In general they prove an inequality between their rate, the Donsker–Varadhan rate and the exit rate (their (4.6)), and they conjecture equality in (4.6) (their (2.15)). For a finite chain their condition (C5) requires Q_F to be irreducible. For such a chain the exit rate is the least value of I_F also without reversibility. We give a short self-contained proof in Proposition 3.1(a), because we also need the other properties of I_F proved there. It does not address their conjecture (2.15), which concerns their conditional rate. The quasi-ergodic law goes back to Darroch and Seneta [6, 7] for irreducible absorbing chains in discrete and continuous time, and Colonius and Rasmussen [8] treat the reducible case.

Dietz and Sethuraman [9] study chains that switch from i to j at time n with probability $G(i, j)/n$, where every $G(i, j) > 0$. Their occupation fractions converge in law to a distribution with full support on the simplex, and it is a Dirichlet law when $G(i, j)$ depends only on j . Their chains also make about $\log n$ switches by time n , but the switches are spread log-uniformly in time and the switching graph is complete. In an earlier paper [10] they prove large deviations at speed n for chains whose transition matrices converge to a reducible matrix, and they remark that scales up to $\log n$ are of a different kind.

Exact distributions of transition counts go back to Whittle [11], Dawson and Good [12] and Goodman [13]. Poletti [14] and Carugno, Vivo and Coghi [15] derive such counts from the BEST theorem and the matrix-tree theorem and study the statistics of transition counts when every count grows linearly with the length of the path. We group the words by runs instead (Lemma 2.3), which suits a regime with only $O(\log N)$ switches. Bertini, Gabrielli and Landim [16], and Landim [17] for non-reversible chains, expand the level-two large deviation rate functions of finite chains whose rates depend on a parameter. They describe the time scales of such chains but do not ask where the mode of the composition sits. The sticky HDP-HMM of Fox, Sudderth, Jordan and Willsky [18] adds a self-transition bias to the hierarchical Dirichlet process hidden Markov model, and Section 6 studies an idealized form of its prior.

On the lattice the correlated random walk goes back to Gillis [19]. Chen and Renshaw study the Gillis–Domb–Fisher walk and its n -step characteristic function [20], and they show that the characteristic function of any correlated random walk satisfies a recurrence [21], so in principle the law of our planar walk is known. Marris and Giuggioli [22] give lattice propagators for walks with forward, backward and lateral steps and solve their first-passage problem. In the continuum, Santra, Basu and Sabhapandit [23] find the exact marginals of a run-and-tumble particle with 4 directions that turns by $\pm\pi/2$ and never reverses, and they describe from simulations how its mass moves from the boundary to the center. Angelani [24] treats general switching among 4 orthogonal directions, and his walk with turns only factorizes in coordinates rotated by 45° [24, Eq. (36)]. Proposition 7.7 uses a lattice analogue of this factorization. In the continuum the corners carry atoms and the origin only a density, so the comparison of a corner with the origin is a lattice question. Smiga, Radaelli, Binder and Landi [25] compute the Fisher information of the empirical distribution of a stationary process under a Gaussian approximation. Their leading term depends on how the stationary law moves with the parameter. For the symmetric chain of Paper A the stationary law does not depend on q , so their leading term vanishes here. We did not find the face-selection question or the planar crossing of the corner and the origin in the literature.

Organization. Section 2 sets up the model and the exact law. Section 3 proves first-order face selection and describes the winners, and Section 4 gives the local law, the face maxima and the crossing equation. Sections 5, 6, 7 and 8 treat the examples, the sticky priors, the planar walk and the Fisher information. Section 9 describes the numerical checks and lists open questions, and the longer proofs are in the appendices. We do not treat compositions near the boundary of a face, where a local law that holds uniformly is open (Example 3.5), and we do not prove in general that a crossing is unique.

2 The model and the exact law

Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$, with rates $q_{ij} \geq 0$ for $i \neq j$, exit rates $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$ and $q_{\max} = \max_i q_i$. The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. For $N \geq 2$ and $T > 0$ put $h = T/N$, $L = \log N$ and $\tau = T/L$. The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So it switches from i to j with probability hq_{ij} and stays with probability $1 - hq_i$. We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise, and the exact identities below only need $hq_{\max} \leq 1$. The composition is $n = (n_1, \dots, n_d)$, where n_i is the number of $t \leq N$ with $X_t = i$. We write $P_N(n)$ for its probability and $\alpha = n/N$. We are interested in T of order $\log N$, so that the switching probabilities are of order $(\log N)/N$.

A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. We use F and G for faces. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F , and Δ_F° the set of those that are positive on F . So $n/N \in \Delta_F^\circ$ exactly when $\text{supp } n = F$. For coordinates we number the states of F as $1, \dots, k$ and use $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : \sum_i t_i = T\}$. Any other $k-1$ coordinates give the same Lebesgue measure. For $x, y \in \mathbb{R}^F$ we write $\langle x, y \rangle = \sum_{i \in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i \in F}$. Let $A_F = (q_{ij})_{i, j \in F}$ with zero diagonal, and let $Q_F = A_F - \text{diag}(q_i)_{i \in F}$ be the principal submatrix of Q on F . Note that q_i includes the rates of leaving F , so Q_F generates the chain killed when it leaves F . The switching graph G_F is the directed graph on F with an edge $i \rightarrow j$ whenever $i \neq j$ and $q_{ij} > 0$. A matrix is Metzler if its off-diagonal entries are nonnegative. The Perron root $\varrho(B)$ of a Metzler matrix B is its largest real eigenvalue. It exists, and if B is irreducible it is a simple eigenvalue with positive left and right eigenvectors, by the Perron–Frobenius theorem for $B + cI \geq 0$ with c large [27].

Definition 2.1 (faces). Let F be a face. We call F *irreducible* if G_F is strongly connected, so every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F . We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F \rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [6, 7]. The *face maximum* is $M_F = \max\{P_N(n) : \text{supp } n = F\}$.

The pair (r_F, l_F) is unique up to $(cr_F, l_F/c)$, and $\alpha_F^* \in \Delta_F^\circ$ does not depend on c . When $hq_{\max} < 1$, a path has positive probability exactly when $\mu_{x_1} > 0$ and $q_{x_t x_{t+1}} > 0$ at every switch, and repeating its last state keeps it positive. So $M_F > 0$ for all large N if F is admissible, and $M_F = 0$ otherwise. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between different components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j} (its Frobenius normal form [27]), so $\lambda_F = \min_j \lambda_{C_j}$. If F is irreducible, the row sums of $e^{TQ_F} \geq 0$ are the probabilities that the continuous-time chain started in F stays in F up to time T , and $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$. Comparing **1** with r_F shows that these probabilities are $e^{-T\lambda_F}$ up to a factor between $\min r_F / \max r_F$ and its inverse. So λ_F is the rate at which the chain leaves F .

Lemma 2.2 (admissible faces). *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F . (a) F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$. (b) An irreducible face is admissible if and only if $\mu(F) > 0$. So (H) holds exactly for the irreducible admissible faces. (c) If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof. The states of the maximal runs of a path of positive probability with support F form a walk in G_F that starts in $\text{supp } \mu$ and visits every state of F . For N at least its length, every such walk arises in this way. So F is admissible if and only if G_F has such a walk. (a) Along such a walk, a component C cannot be visited again after a state $y \notin C$, since then y and C would be reachable from each other. So the walk passes through each component in one block. The first block contains the start, and consecutive blocks are joined by an edge. Conversely, given the numbering and edges $x_j \rightarrow y_{j+1}$ from C_j to C_{j+1} , take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$. Since C_j is strongly connected, some walk from y_j to x_j visits every state of C_j . Joining these walks by the edges $x_j \rightarrow y_{j+1}$ gives the required walk. (b) is (a) with $r = 1$. (c) With symmetric support an edge from C_1 to C_2 gives an edge back, which would merge C_1 and C_2 . So (a) forces $r = 1$. $\square$

For a face F and $m \in \mathbb{Z}_{\geq 1}^F$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}}$, where the sum runs over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ for $j < M$ in which each $i \in F$ occurs m_i times. We do not show the dependence on F in W , or in H and f_T below.

Lemma 2.3 (run expansion). *Let $hq_{\max} \leq 1$ and let n have support F . Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}_{\geq 1}^F} W(m) h^{M-1} \prod_{i \in F} \binom{n_i-1}{m_i-1} (1 - hq_i)^{n_i-m_i}.$$

Proof. Split a path with support F into maximal runs. Their states $c_1 \cdots c_M$ form a word as in $W(m)$, where m_i is the number of runs of state i . The path has $M - 1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i , of probability $1 - hq_i$ each. So its probability is $\mu_{c_1} h^{M-1} \prod_{j < M} q_{c_j c_{j+1}} \prod_{i \in F} (1 - hq_i)^{n_i-m_i}$. For a fixed c the path is determined by the run lengths, and the lengths of the runs of state i form one of the $\binom{n_i-1}{m_i-1}$ compositions of n_i into m_i positive parts, independently over i . Summing over c gives the claim. $\square$

The continuous-time analogue replaces the binomial coefficients by powers. For $t \in (0, \infty)^F$ put $H(t) = \sum_{m \in \mathbb{Z}_{\geq 1}^F} W(m) \prod_{i \in F} t_i^{m_i-1} / (m_i - 1)!$. There are at most k^M words of length M , so $W(m) \leq k^M q_{\max}^{M-1}$ and $H(t) \leq k^k q_{\max}^{k-1} e^{kq_{\max} \sum_i t_i} < \infty$.

Lemma 2.4 (the continuum density). *Let $(Y_s)_{s \geq 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\} ds$, and let E_F be the event that Y stays in F during $[0, T]$ and visits every state of F . Then $\mathbb{P}(E_F, \mathcal{O} \in \cdot)$ has the density $e^{-\langle q, t \rangle} H(t)$ in $(t_2, \dots, t_k)$ on $\{t \in (0, \infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F, \mathcal{O}/T \in \cdot)$ has the density*

$$f_T(\alpha) = T^{k-1} e^{-T \langle q, \alpha \rangle} H(T\alpha)$$

in $(\alpha_2, \dots, \alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t \rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

Proof. For a word $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ that contains every state of F , let E_c be the event that Y visits exactly $c_1, \dots, c_M$ in this order during $[0, T]$. Since Y jumps finitely often in $[0, T]$ almost surely, E_F is the disjoint union of the events E_c up to a null set. Fix c , let $s_1, \dots, s_{M-1}$ be the first $M-1$ holding times and $s_M = T - \sum_{j < M} s_j$. The holding time in c_j is exponential with rate q_{c_j} , the jump to c_{j+1} has probability $q_{c_j c_{j+1}}/q_{c_j}$, and the last holding time is censored. It only has to exceed s_M , which has probability $e^{-q_{c_M} s_M}$. So on $\{s_j > 0 \text{ for } j \leq M\}$ the event E_c has density $\mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}} e^{-\langle q, \mathcal{O} \rangle}$ in $(s_1, \dots, s_{M-1})$, where $\mathcal{O}_i = \sum_{j: c_j = i} s_j$. (If $q_{c_j} = 0$ for some $j < M$, then E_c is null and the formula gives 0.) Let j_i be the position of the last letter i in c , so $j_{c_M} = M$. Replacing s_{j_i} by $\mathcal{O}_i$ for $i \neq c_M$ is a triangular linear change of variables with unit diagonal. For fixed $\mathcal{O} = t$ the other $m_i - 1$ holding times in state i range over $\{x \in (0, \infty)^{m_i-1} : \sum x < t_i\}$, for every $i \in F$. For $i = c_M$ this is the condition $s_M > 0$, with $t_{c_M} = T - \sum_{i \neq c_M} t_i$. The set has volume $t_i^{m_i-1}/(m_i-1)!$ and is empty unless $t_i > 0$. Integrating out these variables gives the density of $\mathcal{O}$ on E_c in $(t_i)_{i \neq c_M}$, and hence in $(t_2, \dots, t_k)$. Summing over c by monotone convergence gives $e^{-\langle q, t \rangle} H(t)$. The rest follows from the change of variables $t = T\alpha$, with Jacobian T^{k-1} , and from $T/N = h$. For $k = 1$ the only word is $c = i$, so $H = \mu_i$. $\square$

Example 2.5 (the 4 models). (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain is the persistent walk of [2] with $p = 1 - h$, so $T = N(1 - p)$. Bin k is the composition $(k, N - k)$, and Lemma 2.3 is the run count of [2, Thm. 2.1]. (ii) For $Q = J - dI$, with J the all-ones matrix, and uniform μ the chain is the model of [3] with its parameters $r = h$ and $p = 1 - (d - 1)h$. So $T = Nr$, and its crossing variable $Nu = Nr/p$ is $T(1 + O(T/N))$. Here $\lambda_F = d - |F|$ and α_F^* is uniform on F . (iii) For $Q = \Pi - I$, with $\Pi = (\pi_{ij})$ stochastic with zero diagonal, $I + hQ = (1 - h)I + h\Pi$ and $\lambda_F = 1 - \varrho(\Pi_F)$, where Π_F is the principal submatrix of Π on F . This is the sticky chain of Section 6. (iv) Let $d = 4$, number the directions east, north, west and south as 0, 1, 2, 3 modulo 4, and put $q_{j,j\pm 1} = r$, $q_{j,j+2} = s$ and μ uniform. Then X_t is the direction of step t of a walk on $\mathbb{Z}^2$ that turns left or right with probability rh each and reverses with probability sh . Its position $(n_0 - n_2, n_1 - n_3)$ is a linear function of the composition (Section 7). In (i), (ii) and (iv), Q has symmetric support and μ has full support. So by Lemma 2.2 the admissible faces are the irreducible ones, and they all satisfy (H).

Remark 2.6 (the power of N). Since $\sum_{i \in F} (m_i - 1) = M - k$, we can write $h^{M-1} = h^{k-1} \prod_i h^{m_i-1}$ in Lemma 2.3. With $t = hn$, $h^{m_i-1} \binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!} \prod_{l < m_i} (1 - \frac{l}{n_i})$, and $(1 - hq_i)^{n_i-m_i}$ is close to $e^{-q_i t_i}$. So each term with every m_i^2 small compared with n_i is close to the matching term of $h^{k-1} e^{-\langle q, t \rangle} H(t) = N^{-(k-1)} f_T(n/N)$. The terms that matter have M of order T , and under (H) Lemma 4.4 shows that the relative error is $O(T^2/N)$ on compact subsets of Δ_F° . So a run after the first in state i brings a factor h for its switch and a factor of order n_i for its placement, and together these give a factor of order T . Only the $k - 1$ switches that enter a new state leave a factor $h = T/N$ each.

3 First-order face selection

In this section $T = \tau L$ with $\tau > 0$ fixed, and we find the face that carries the mode. For a face F put

$$\Phi_\tau(F) = \tau \lambda_F + |F| - 1.$$

We call $\Phi_\tau(F)$ the exponent of F . Theorem 3.3 shows that for every start the mode lies on an admissible face that minimizes Φ_τ , and Theorem 3.6 describes these faces by a convex hull. First

we need the rate inside a face. For a face F and $\alpha \in \Delta_F$ let

$$I_F(\alpha) = \sup_{u \in (0, \infty)^F} \sum_{i \in F} \alpha_i \left(-\frac{(Q_F u)_i}{u_i} \right) = \sup_{u \in (0, \infty)^F} \sum_{i \in F} \alpha_i \left(q_i - \sum_{j \in F \setminus \{i\}} q_{ij} \frac{u_j}{u_i} \right),$$

and let $q_i^{\text{out}}(F) = \sum_{j \notin F} q_{ij}$. Taking $u = \mathbf{1}$ gives $I_F(\alpha) \geq \sum_i \alpha_i q_i^{\text{out}}(F)$, and dropping the terms with q_{ij} gives $I_F(\alpha) \leq \sum_i \alpha_i q_i$. As we said in the introduction, I_F is the Donsker–Varadhan rate of the occupation times of the chain killed on leaving F . Dupuis and Liu [4] give it explicitly for reversible jump processes, and Guillin, Nectoux and Wu [5] relate it to the rate of their conditional large deviation principle when the process is reversible. We need its properties without reversibility, and for a finite chain the proofs are short.

Proposition 3.1 (the rate on a face). *Let F be irreducible with $k = |F| \geq 2$.*

- (a) $I_F \geq \lambda_F$ on Δ_F , with equality exactly at $\alpha_F^* = l_F \circ r_F$.
- (b) I_F is convex on Δ_F and strictly convex on Δ_F° . For $\alpha \in \Delta_F^\circ$ the supremum defining $I_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij} r_{\alpha, j} / r_{\alpha, i}$.
- (c) If there is $\nu \in (0, \infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$I_F(\alpha) = \sum_{\{i, j\} \subseteq F, i \neq j} (\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2 + \sum_{i \in F} \alpha_i q_i^{\text{out}}(F).$$

- (d) For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $I_F(\alpha) = I_{F'}(\alpha) \geq \lambda_{F'} > \lambda_F$.

Proof. Write $u = e^\psi$, so that $I_F(\alpha) = \sup_{\psi \in \mathbb{R}^F} \Psi_\alpha(\psi)$ with $\Psi_\alpha(\psi) = \sum_i \alpha_i q_i - \sum_{i \neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i}$, where i, j run over F . Note that Ψ_α is concave in ψ , affine in α , and unchanged when a constant is added to ψ . Let G^ψ be the generator on F with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. Then $\nabla_\psi \Psi_\alpha(\psi) = -\alpha^\top G^\psi$, so ψ maximizes Ψ_α exactly when α is stationary for G^ψ . The switching graph of G^ψ is G_F , so G^ψ is irreducible and has one stationary law.

The lower bound in (a), for every face. Let F' be a nonempty face and write $Q_{F'} = B - cI$ with $B \geq 0$. For $s > \varrho(Q_{F'})$ we have $s + c > \varrho(B)$, the spectral radius of B . So $(sI - Q_{F'})^{-1} = \sum_{n \geq 0} B^n (s + c)^{-n-1} \geq (s + c)^{-1} I$ entrywise, and $u = (sI - Q_{F'})^{-1} \mathbf{1}$ is positive with $Q_{F'} u = su - \mathbf{1} < su$. This u shows that $I_{F'} > -s$ on $\Delta_{F'}$. Letting $s \downarrow \varrho(Q_{F'})$ gives $I_{F'} \geq \lambda_{F'}$.

The value at α_F^ .* Let $\psi = \log r_F$ and $R = \text{diag}(r_F)$. Since $Q_F r_F = -\lambda_F r_F$, we get $G^\psi = R^{-1}(Q_F + \lambda_F I)R$, and then $(\alpha_F^*)^\top G^\psi = l_F^\top (Q_F + \lambda_F I)R = 0$. So $\log r_F$ maximizes $\Psi_{\alpha_F^*}$, and $I_F(\alpha_F^*) = \Psi_{\alpha_F^*}(\log r_F) = \lambda_F$.

(b) Convexity holds because I_F is a supremum of affine functions. Let $\alpha \in \Delta_F^\circ$ and let a be the least positive rate q_{ij} with $i, j \in F$. Normalize $\min_i \psi_i = 0$ and put $M = \max_i \psi_i$. Some edge $i \rightarrow j$ of a shortest path in G_F from a minimum of ψ to a maximum has $\psi_j - \psi_i \geq M/(k-1)$, so $\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a(\min_i \alpha_i) e^{M/(k-1)}$, which tends to $-\infty$ as $M \rightarrow \infty$. Hence the supremum is attained at some ψ , and α is stationary for G^ψ . If $\psi + \delta$ also attains it, then $\Psi_\alpha(\psi + t\delta)$ is constant for $0 \leq t \leq 1$ by concavity. It is a constant minus a sum of convex functions of t , so each term $\alpha_i q_{ij} e^{\psi_j - \psi_i + t(\delta_j - \delta_i)}$ is affine in t . So $\delta_j = \delta_i$ for every edge $i \rightarrow j$ of G_F , and δ is constant. For strict convexity let $\alpha^0 \neq \alpha^1$ lie in Δ_F° , let $\alpha^t = (1-t)\alpha^0 + t\alpha^1$ with $0 < t < 1$, and let ψ attain $I_F(\alpha^t)$. Then

$$I_F(\alpha^t) = (1-t)\Psi_{\alpha^0}(\psi) + t\Psi_{\alpha^1}(\psi) \leq (1-t)I_F(\alpha^0) + tI_F(\alpha^1),$$

and equality would make both α^0 and α^1 stationary for G^ψ , which is impossible.

(d) For $\alpha \in \Delta_{F'}$ only rows $i \in F'$ enter, and there the terms with $j \in F \setminus F'$ are nonnegative and subtracted. Dropping them gives $I_F(\alpha) \leq I_{F'}(\alpha)$, and $u_j = \varepsilon \rightarrow 0$ for $j \notin F'$ gives equality. The lower bound gives $I_{F'}(\alpha) \geq \lambda_{F'}$. Finally $\lambda_{F'} > \lambda_F$, because the Perron root of an irreducible nonnegative matrix, here $Q_F + cI$ for a large c , is larger than that of each proper principal submatrix [27, Ch. 2].

(a) Let $\alpha \neq \alpha_F^*$. If α lies on the boundary of Δ_F , then (d) with $F' = \text{supp } \alpha$ gives $I_F(\alpha) > \lambda_F$. If $\alpha \in \Delta_F^\circ$ and $I_F(\alpha) = \lambda_F$, strict convexity gives $I_F(\frac{1}{2}(\alpha + \alpha_F^*)) < \lambda_F$, which contradicts the lower bound.

(c) Since $q_i = \sum_{j \in F \setminus \{i\}} q_{ij} + q_i^{\text{out}}(F)$, the objective equals $\sum_i \alpha_i q_i^{\text{out}}(F)$ plus a sum over pairs $\{i, j\}$ of $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM–GM inequality each such term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$. For the reverse inequality put $u_i = \sqrt{\alpha_i/\nu_i}$ if $\alpha_i > 0$ and $u_i = \varepsilon$ if $\alpha_i = 0$. If $\alpha_i, \alpha_j > 0$, reversibility gives $\alpha_i q_{ij} u_j/u_i = \alpha_j q_{ji} u_i/u_j = \sqrt{\alpha_i \alpha_j q_{ij} q_{ji}}$, and the term equals its bound. If $\alpha_i > 0 = \alpha_j$ the term tends to $\alpha_i q_{ij}$ as $\varepsilon \rightarrow 0$, and if $\alpha_i = \alpha_j = 0$ both sides are 0. $\square$

When Q_F is symmetric, $l_F = r_F$ and the minimizer α_F^* is the entrywise square of the Perron vector normalized in ℓ^2 .

Away from α_F^* we need a bound that holds for all compositions on a face at once. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \text{diag } \theta) = 0\}$. Note that $q - \lambda_F \mathbf{1} \in V_F$, since $A_F - \text{diag}(q - \lambda_F \mathbf{1}) = Q_F + \lambda_F I$.

Lemma 3.2 (a uniform upper bound). *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ , depending on θ , Q and μ , such that for every $n \in \mathbb{Z}_{\geq 0}^F$ with $\text{supp } n = F$ and every h with $h q_{\max} \leq \frac{1}{2}$, the probability that the first $|n|$ visits of the chain have composition n is at most $C_\theta h^{k-1} \exp(-h \langle n, q - \theta \rangle + C_\theta h^2 |n|)$.*

The proof is in Appendix A. With $\theta = q - \lambda_F \mathbf{1}$ it gives $P_N(n) \leq C h^{k-1} e^{-T \lambda_F + C T^2/N}$ for every n with support F . The bound depends only on n and h , and not on N , which we use for the blocks of a path below.

Theorem 3.3 (first-order face selection). *Fix Q and μ .*

(i) *Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\text{supp } n = F$ with $n/N \in \mathcal{K}$,*

$$\log P_N(n) = -(k-1)L - T I_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

(ii) *Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ , T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$*

$$\begin{aligned} \log M_F &\geq -(k_F - 1)L + (k_F - k^*) \log T - T \lambda_F - \kappa, \\ \log M_F &\leq -(k_F - 1)L + (k_F - 1) \log T - T \lambda_F + \kappa. \end{aligned}$$

If F is not admissible, $M_F = 0$.

(iii) *Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ . Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\text{supp } \hat{n} \in \mathcal{F}_\tau$. This holds for every start μ .*

(iv) If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \text{supp } \hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 3.6(c)).

Theorem 4.3 and Lemma 4.4 sharpen the error term in (i) to $\frac{k-1}{2} \log T + \log C_F(n/N) + O(1/T + T^2/N)$, with C_F from Definition 4.1.

Remark 3.4 (the exponents and reducible faces). The factor $N^{-(k-1)}$ in (i) and (ii) comes from the $k-1$ switches that enter new states (Remark 2.6), and $e^{-T\lambda_F}$ is the chance of staying in F for a time T . For a reducible face, the path in our lower bound spends about N/T visits, a time of order one, in every component except one with the least exit rate. For such faces we only give bounds, and Proposition 4.12 shows that the power of T can differ from $(k_F - 1)/2$.

We prove (ii) and (iii) here. Parts (i) and (iv) follow from the second-order results, Theorems 4.3 and 4.5 and Lemma 4.4, and we derive them at the end of the proof.

Proof. (ii) If F is not admissible, $M_F = 0$ by definition. Let F be admissible, let a be the least positive rate q_{ij} , and take N_0 so large that $h \leq N^{-2/3}$ is as small as we need. Note that $T^2/N \leq 1$. A path of positive probability with support F cannot leave a component and come back to it, since the states visited in between would then belong to that component. So it visits each component in one block of consecutive times and passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern with blocks on $C_{\pi(1)}, \dots, C_{\pi(r)}$, first states y_j and switches $x_j \rightarrow y_{j+1}$, and let $\text{supp } n = F$. Block j has length $N_j = \sum_{i \in C_{\pi(j)}} n_i$. The paths with this pattern and composition n have total probability $\mu_{y_1} \prod_{j < r} h q_{x_j y_{j+1}}$ times the product over j of the total weight of the block paths from y_j to x_j inside $C = C_{\pi(j)}$ with the right composition. Without the condition on the last state, this weight is the probability that the chain started at y_j has that composition in its first N_j visits. By Lemma 3.2 with start y_j and $\theta = q - \lambda_C \mathbf{1}$ it is at most $\kappa_1 h^{|C|-1} e^{-h\lambda_C N_j + \kappa_1 h^2 N_j}$ when $|C| \geq 2$. For $C = \{i\}$ it is $(1 - h q_i)^{N_j-1} \leq e^{1/2 - h\lambda_C N_j}$. Since $\sum_j (|C_{\pi(j)}| - 1) + r - 1 = k_F - 1$, $\sum_j N_j = N$ and $\lambda_C \geq \lambda_F$ for every component, summing over the patterns gives $P_N(n) \leq \kappa_2 (T/N)^{k_F-1} e^{-T\lambda_F + \kappa_1}$.

Lower bound. By Lemma 2.2(a) we can order the components as $C_1, \dots, C_r$ and choose $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_j y_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, let $\lambda_{C_{j^*}} = \lambda_F$ and $k^* = |C_{j^*}|$, and put $m = \lceil N/T \rceil$, so that $1 \leq hm \leq 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A . Indeed, take a walk $y = v_0, \dots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \leq |C|^2$. The paths that make exactly these switches, at any J of the $m-1$ transition times, have total probability at least $\binom{m-1}{J} (ha)^J (1 - h q_{\max})^m \geq (1 - Jh)^J a^J e^{-4q_{\max}} / J!$, since $1 \leq hm \leq 2$. Also $(1 - Jh)^J \geq 2^{-J}$ for small h .

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \geq 2m$, the same event with C_{j^*} and ℓ visits has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Indeed, let $w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + h Q_{C_{j^*}} \geq 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*} , and $\sum_{z'} (B^s)_{zz'} \geq (B^s w)_z / \max w \geq (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y , then $s = \ell - 2m + 1$ steps inside C_{j^*} , then Claim A for the last m visits, ending at x . This gives at least $c_A^2 (\min w / \max w) (1 - h\lambda_F)^\ell$. Also $\lambda_F \leq q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \leq \frac{1}{2}$ and $\log(1 - h\lambda_F) \geq -h\lambda_F - h^2 \lambda_F^2$.

Give block $j \neq j^*$ the length m and block j^* the length $N - (r - 1)m$, which is at least $2m$ for $T \geq T_0 = r + 2$ and $N \geq N_0$. The paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j + 1$ by the switch $x_j \rightarrow y_{j+1}$ all have support F . By the Markov property their total probability is at least $\mu_{y_1}(ha)^{r-1}c_A^{r-1}c_B e^{-T\lambda_F - q_{\max}^2}$. Their compositions are fixed by the restrictions to the blocks, so there are at most $(N + 1)^{k^*-1} \prod_{j \neq j^*} (m + 1)^{|C_j|-1}$ of them, and $m + 1 \leq 3N/T$. Since $\sum_{j \neq j^*} (|C_j| - 1) = k_F - k^* - (r - 1)$, dividing the probability by this number gives $M_F \geq c N^{-(k_F-1)} T^{k_F-k^*} e^{-T\lambda_F}$.

(iii) Let $T = \tau L$. For large N we have $T_0 \leq T \leq N^{1/3}$, and by (ii) $\log M_F = -L \Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$ over finitely many faces, the first claim follows. Let $m_\tau = \min_F \Phi_\tau(F)$ and let $\varepsilon_0 > 0$ be the least value of $\Phi_\tau(F) - m_\tau$ over admissible $F \notin \mathcal{F}_\tau$. Each such face has $M_F \leq N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n) = N^{-m_\tau + o(1)}$ for large N . A global mode has positive probability, so its support is admissible, and hence it is in $\mathcal{F}_\tau$.

Proof of (i) and (iv). For (i) with $k = 1$, $P_N(Ne_i) = \mu_i(1 - hq_i)^{N-1}$ with $\mu_i > 0$ and $I_{\{i\}} = q_i$, so (i) holds because $Nh^2 \leq 1$. Let $k \geq 2$, and let f_T be the density of Lemma 2.4. It is continuous and positive in (T, α) , as we note after Theorem 4.3. Also C_F is positive and continuous on Δ_F° . So the ratio of $f_T(\alpha)$ to $T^{(k-1)/2} C_F(\alpha) e^{-TI_F(\alpha)}$ is positive and continuous, and by Theorem 4.3 it tends to 1 uniformly on $\mathcal{K}$. Hence it is bounded above and below by positive constants for $T \geq 1$ and $\alpha \in \mathcal{K}$, and $\log f_T(\alpha) = -TI_F(\alpha) + O_{\mathcal{K}}(1 + \log T)$. By Lemma 4.4, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O(T^2/N))$, and the last factor lies in $[\frac{1}{2}, 2]$ for $N \geq N_0$ since $T^2/N \leq N^{-1/3}$. This proves (i).

For (iv), let $F \in \mathcal{F}_\tau$. It is admissible and irreducible, so $\mu(F) > 0$ by Lemma 2.2(b), and F satisfies (H). By (iii), for large N a global mode $\hat{n}$ has its support F in $\mathcal{F}_\tau$, and it maximizes P_N over the compositions with support F . Since $T = \tau L \leq N^{1/3}$ for large N , Theorem 4.5 gives $\hat{n}/N = \alpha_F^* + O(1/T)$, and there are finitely many faces. $\square$

The law in (i) holds only on compact subsets of the open face, and the next example shows that the boundary really is different.

Example 3.5 (the boundary of a face). Let $d = \ell + 1$ with $\ell \geq 3$, and let the only positive rates be q_{jd} and q_{dj} for $j \leq \ell$. So the switching graph is a star with center d and leaves $1, \dots, \ell$. Let μ have full support and $F = [d]$, so that F satisfies (H). A leaf can only switch to the center, so the runs of a path alternate between the center and the leaves. A path with support F has at least ℓ runs at leaves, and hence at least $\ell - 1$ runs at the center. So $P_N(n) = 0$ whenever $\text{supp } n = F$ and $n_d < \ell - 1$, although n/N lies in Δ_F° and $-(k - 1)L - TI_F(n/N)$ is finite. Hence the law in Theorem 3.3(i) cannot hold uniformly up to the boundary of Δ_F .

By Theorem 3.3(iii), first-order face selection is a finite minimization. For an admissible face F put $P_F = (\lambda_F, |F| - 1) \in \mathbb{R}^2$, and let S be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$, so $\mathcal{F}_\tau$ is the set of faces whose points minimize $\tau x + y$ on S . For each $\tau > 0$ the points of $\text{conv } S$ where $\tau x + y$ is least form a vertex or an edge of $\text{conv } S$. We call the union of these vertices and edges over all $\tau > 0$ the lower-left boundary of $\text{conv } S$.

Theorem 3.6 (geometry of the winners). *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ .*

- (a) *The lower-left boundary of $\text{conv } S$ is a path whose vertices lie in S . Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \leq i \leq m$, with $\lambda^{(0)} > \dots > \lambda^{(m)}$ and $y^{(0)} < \dots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \leq i \leq m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \dots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.*

(b) If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \geq \lambda_G$ and $|F| \leq |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.

(c) If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.

Proof. (a) Let $g(\tau) = \min_{(x,y) \in S} (\tau x + y)$ for $\tau > 0$. It is a minimum of finitely many affine functions, so it is concave and piecewise affine. Let $0 < \tau_1 < \dots < \tau_m$ be the points where its slope changes, with $\tau_0 = 0$ and $\tau_{m+1} = \infty$. On (τ_i, τ_{i+1}) we have $g(\tau) = \tau \lambda^{(i)} + y^{(i)}$ for some $v_i = (\lambda^{(i)}, y^{(i)}) \in S$. If $p = (x, y) \in S$ has $\tau x + y = g(\tau)$ for some τ in this interval, then $\tau' \mapsto \tau' x + y - g(\tau')$ is affine and nonnegative on the interval and vanishes inside it. So it vanishes on the interval and $p = v_i$. The slope of g decreases, so $\lambda^{(0)} > \dots > \lambda^{(m)}$, and continuity at τ_i gives $\tau_i \lambda^{(i-1)} + y^{(i-1)} = \tau_i \lambda^{(i)} + y^{(i)}$. This is the formula for τ_i , and it gives $y^{(i)} > y^{(i-1)}$. Now let $p = (x, y) \in S$ with $\tau_i x + y = g(\tau_i)$. It lies on the line $\tau_i x + y = g(\tau_i)$, which contains v_{i-1} and v_i . Comparing $\tau x + y$ with $g(\tau)$ just below and just above τ_i gives $\lambda^{(i)} \leq x \leq \lambda^{(i-1)}$, so p lies on the closed edge $[v_{i-1}, v_i]$. Conversely every point of S on this edge attains $g(\tau_i)$. For each τ the minimizers of $\tau x + y$ on $\text{conv } S$ form the convex hull of its minimizers on S , which is v_i or $[v_{i-1}, v_i]$. So the lower-left boundary is the path $v_0 v_1 \dots v_m$. Its edges are exposed by different values of τ , so they have different directions and the v_i are its vertices.

(b) By optimality $\Phi_\tau(F) \leq \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \leq \Phi_{\tau'}(F)$. Adding them gives $(\tau' - \tau)(\lambda_F - \lambda_G) \geq 0$, so $\lambda_F \geq \lambda_G$, and then the first inequality gives $|F| - |G| \leq \tau(\lambda_G - \lambda_F) \leq 0$. If the sets are disjoint and $\lambda_F = \lambda_G$, the inequalities give $|F| = |G|$, so $\Phi_{\tau'}(F) = \Phi_{\tau'}(G)$ and $F \in \mathcal{F}_{\tau'}$, which is a contradiction. So $\lambda_F > \lambda_G$, and then $|F| < |G|$.

(c) With symmetric support every admissible face is irreducible by Lemma 2.2(c). Let μ have full support, and let F be admissible and reducible. Take a component C of F with $\lambda_C = \lambda_F$. It is irreducible with $\mu(C) > 0$, so it is admissible by Lemma 2.2(b). Since $|C| \leq |F| - 1$, we get $\Phi_\tau(C) \leq \Phi_\tau(F) - 1$, so F is not a minimizer. $\square$

A face with small λ_F for its size is a set of states that the chain rarely leaves, a well-separated community. As τ grows the exit rate counts for more against the size, and the mode moves to larger communities that hold the chain better. It need not grow around its first vertex, and it can move to a different community (Remark 5.9). Figure 1(a) shows S and its lower-left boundary for the 6-cycle.

For symmetric generators the exit rates have a variational form. For $1 \leq s \leq d$ let $\underline{\lambda}_s = \min\{\lambda_F : F \text{ admissible, } |F| = s\}$ when such a face exists. A face of size s with $\lambda_F > \underline{\lambda}_s$ has a larger value of Φ_τ than one with $\lambda_F = \underline{\lambda}_s$, for every $\tau > 0$, so only the points $(\underline{\lambda}_s, s - 1)$ matter. In part (c) of the next lemma we plot them with the axes swapped, as $(s - 1, \underline{\lambda}_s)$, so that the winners are read off a lower convex hull. We call a face connected if its switching graph is connected.

Lemma 3.7 (symmetric generators). *Let Q be symmetric and irreducible, and put $\mathcal{E}(f) = \sum_{i < j} q_{ij}(f_i - f_j)^2$ for $f \in \mathbb{R}^d$.*

(a) *For every face F , $\lambda_F = \min\{\mathcal{E}(f) : \text{supp } f \subseteq F, \|f\|_2 = 1\}$, and the minimum is attained at some $f \geq 0$.*

(b) *If $F \subsetneq G$ and G is connected, then $\lambda_G < \lambda_F$.*

(c) *If μ has full support, then $\underline{\lambda}_1 > \underline{\lambda}_2 > \dots > \underline{\lambda}_d = 0$. Let $1 = s_0 < s_1 < \dots < s_m = d$ be the sizes s at which $(s - 1, \underline{\lambda}_s)$ is a vertex of the lower convex hull of these d points, and put $\tau_i = (s_i - s_{i-1}) / (\underline{\lambda}_{s_{i-1}} - \underline{\lambda}_{s_i})$ for $1 \leq i \leq m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \dots < \tau_m$, and for $\tau_i < \tau < \tau_{i+1}$ the set $\mathcal{F}_\tau$ consists of the faces F with $|F| = s_i$ and $\lambda_F = \underline{\lambda}_{s_i}$.*

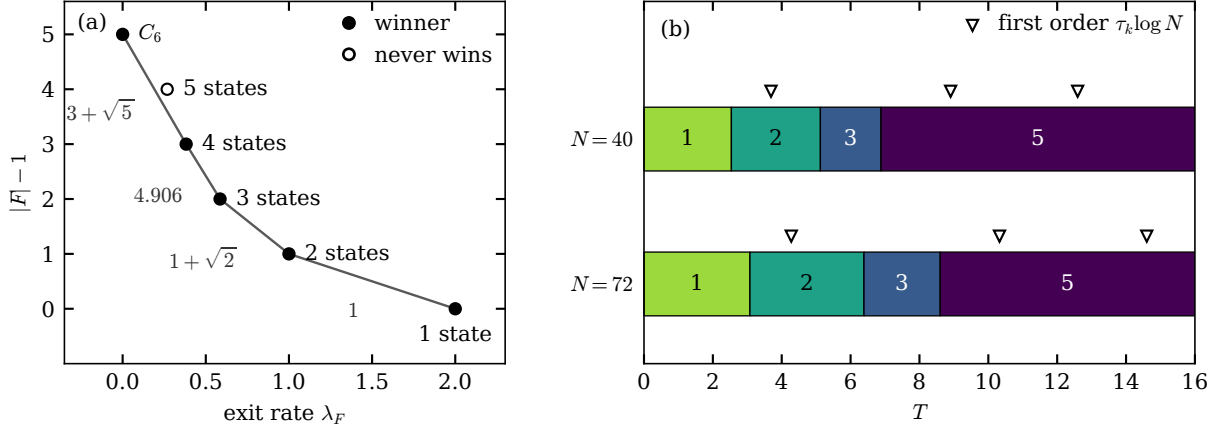


Figure 1: (a) The points $(\lambda_F, |F| - 1)$ of the connected faces of the 6-cycle C_6 . These are the arcs of 1 to 5 states and C_6 itself, and all arcs of one size give the same point. The line is the lower-left boundary of the convex hull. The winners are the arcs of 1 to 4 states and then C_6 (filled), while the arc of 5 states never wins (open). The switch values $1, 1 + \sqrt{2}, 4.906$ and $3 + \sqrt{5}$ are minus the slopes of the hull edges. (b) The support of the exact mode on C_5 with the uniform start, against T , at $N = 40$ and $N = 72$. The digits give the number of states in the support. The mode changes at $T = 2.538, 5.123$ and 6.884 for $N = 40$ and at $T = 3.078, 6.389$ and 8.602 for $N = 72$. The triangles mark the first-order switch points $\tau \log N$ with $\tau = 1, 1 + \sqrt{2}$ and $2 + \sqrt{2}$ (Theorem 5.4).

Proof. (a) Since Q is symmetric with zero row sums, $\langle f, -Qf \rangle = \mathcal{E}(f)$. If $\text{supp } f \subseteq F$ this equals $\langle f_F, -Q_F f_F \rangle$, and the least eigenvalue of the symmetric matrix $-Q_F$ is λ_F . So (a) is the Rayleigh–Ritz principle, together with $\mathcal{E}(|f|) \leq \mathcal{E}(f)$.

(b) Monotonicity follows from (a). Let $\varphi \geq 0$ attain the minimum for F , and suppose that $\lambda_G = \lambda_F$. Then φ also minimizes the Rayleigh quotient of $-Q_G$, so $-Q_G \varphi = \lambda_G \varphi$ on G . Since $\text{supp } \varphi \subseteq F \neq G$ and G is connected, some $j \in G \setminus \text{supp } \varphi$ has $q_{ji} > 0$ for an $i \in \text{supp } \varphi$. Then $(-Q_G \varphi)_j = -\sum_{i'} q_{ji'} \varphi_{i'} < 0 = \lambda_G \varphi_j$, which is a contradiction.

(c) Here the admissible faces are the connected faces, by Lemma 2.2(b),(c). Also $\lambda_{[d]} = 0$ because $Q\mathbf{1} = 0$ and $\mathbf{1} > 0$, so $\lambda_d = 0$. Let $s < d$ and let F be connected with $|F| = s$ and $\lambda_F = \lambda_s$. The switching graph of Q is connected, so some $j \notin F$ has a neighbor in F . Then $F \cup \{j\}$ is connected, and (b) gives $\lambda_{s+1} \leq \lambda_{F \cup \{j\}} < \lambda_s$. The lower convex hull of the points $(s - 1, \lambda_s)$ is the graph of a convex piecewise linear function with corners at the points $(s_i - 1, \lambda_{s_i})$. Its slopes $\sigma_i = (\lambda_{s_i} - \lambda_{s_{i-1}})/(s_i - s_{i-1}) = -1/\tau_i$ increase with i , and they are negative because λ_s decreases. Minimizing $\tau \lambda_s + s - 1$ over s means finding where a line of slope $-1/\tau < 0$ supports this graph. With $\sigma_0 = -\infty$ and $\sigma_{m+1} = 0$, the line touches only the corner at s_i exactly when $\sigma_i < -1/\tau < \sigma_{i+1}$, that is when $\tau_i < \tau < \tau_{i+1}$. $\square$

4 The second-order theory

Theorem 3.3 decides which face holds the mode up to factors $N^{o(1)}$. To locate a crossing between 2 faces up to $o(1)$ in T we need the constant in front of $N^{-(k-1)} e^{-T\lambda_F}$, and in this section we find it for every face that satisfies (H). Inside a face F with k states we number the states of F as $1, \dots, k$ and use $\alpha' = (\alpha_2, \dots, \alpha_k)$ as coordinates on Δ_F . Gradients and Hessians in α are taken in these coordinates, $|\cdot|$ is the Euclidean norm, $q = (q_i)_{i \in F}$, and $\mu \cdot v = \sum_{i \in F} \mu_i v_i$.

4.1 The tilted chain and the constant

Definition 4.1 (tilt and constants). Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $I_F(\alpha)$ (Proposition 3.1(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and $l_\alpha = \alpha / r_\alpha$ (entrywise). The *tilted generator* is $\hat{Q}_\alpha = R^{-1}(A_F - \text{diag } \theta(\alpha))R$ with $R = \text{diag } r_\alpha$. With $D = \text{diag } \alpha$ let

$$Z_\alpha = (\mathbf{1}\alpha^\top - \hat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \quad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let $K_{\{i\}} = \mu_i$ for a single state, and

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2} \sqrt{\det \Sigma_\alpha}}, \quad K_F = C_F(\alpha_F^*).$$

The vector r_α is unique up to a positive factor, and nothing in Definition 4.1 changes when it is scaled. Note that $l_\alpha \cdot r_\alpha = 1$ and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$. Since $(A_F - \text{diag } \theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, and a positive eigenvector of an irreducible Metzler matrix belongs to its Perron root [27], $\theta(\alpha)$ lies in the set V_F defined before Lemma 3.2. The vector r_α minimizes $u \mapsto \sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$ over $u > 0$, and at $u = r_\alpha$ the derivative in u_j is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j} \theta_j(\alpha)$. Hence

$$l_\alpha^\top (A_F - \text{diag } \theta(\alpha)) = 0. \quad (1)$$

The off-diagonal entries $q_{ij} r_{\alpha,j} / r_{\alpha,i}$ of $\hat{Q}_\alpha$ have the pattern of A_F , and $\hat{Q}_\alpha \mathbf{1} = 0$. By (1), $\alpha^\top \hat{Q}_\alpha = 0$. So $\hat{Q}_\alpha$ is an irreducible generator with stationary law α . If $(\mathbf{1}\alpha^\top - \hat{Q}_\alpha)v = 0$, then $\alpha^\top v = 0$ and $\hat{Q}_\alpha v = 0$, so $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \quad \alpha^\top Z_\alpha = 0, \quad -\hat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \quad (2)$$

So $\Gamma_\alpha \mathbf{1} = 0$. (The matrix Γ_α is the asymptotic covariance, per unit time, of the occupation times of the chain with generator $\hat{Q}_\alpha$, but we do not use this.) As Γ_α is symmetric with $\Gamma_\alpha \mathbf{1} = 0$, its adjugate is a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* we have $I_F = \lambda_F$ (Proposition 3.1(a)), and $u = r_F$ gives the value λ_F in the supremum. So $r_{\alpha_F^*}$ is a multiple of r_F , $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F . In the local law below, $\mu \cdot r_\alpha$ accounts for the start of the path, $l_\alpha \cdot \mathbf{1}$ for its free end, and Σ_α for the Gaussian spread of the occupation fractions.

For conductances $c_{ij} = c_{ji}$ on F let $\text{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F .

Proposition 4.2 (the constant). *Let F be irreducible with $k \geq 2$.*

(a) *The maps $\alpha \mapsto \theta(\alpha)$, $I_F(\alpha)$, Σ_α , r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F° , Σ_α is positive definite, and $\nabla^2 I_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 I_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) \sqrt{\det \nabla^2 I_F(\alpha) / (2\pi)^{k-1}}$.*

(b) *Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0, \infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\hat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j} q_{ij} \sqrt{\nu_i / \nu_j}$. Also $I_F(\alpha) = \sum_i q_i \alpha_i - \sum_{i \neq j} c_{ij}(\alpha)$, and*

$$\det \Sigma_\alpha = \frac{2^{k-1} (\prod_i \alpha_i)^2}{\text{tree}(c(\alpha))}, \quad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\text{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

Proof. (a) For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \text{diag } \vartheta)$, so $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$. This is a simple eigenvalue with positive right and left eigenvectors $\hat{r}, \hat{l}, \hat{l} \cdot \hat{r} = 1$ [27]. By the implicit function theorem applied to the characteristic polynomial, Λ and these vectors are real-analytic in ϑ . Put $\pi = \hat{l} \circ \hat{r}$ and $G = \hat{R}^{-1}(Q_F + \text{diag } \vartheta - \Lambda(\vartheta)I)\hat{R}$ with $\hat{R} = \text{diag } \hat{r}$. As above, G is an irreducible generator with stationary law π , and we define Z and Γ from (G, π) as Z_α and Γ_α are defined from $(\hat{Q}_\alpha, \alpha)$. Then (2) holds for them.

Step 1. $\nabla\Lambda(\vartheta) = \pi$ and $\nabla^2\Lambda(\vartheta) = \Gamma$. Since $\hat{R}$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \kappa(\eta)$ with $\kappa(\eta) = \varrho(G + \text{diag } \eta)$. Let $v(\eta)$ be its eigenvector with $\pi^\top v = 1$, so $v(0) = \mathbf{1}$ and $\kappa(0) = 0$, and write subscripts for derivatives at $\eta = 0$. Differentiating $(G + \text{diag } \eta)v = \kappa v$ in η_j gives $Gv_j + e_j = \kappa_j \mathbf{1}$. Multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$, and then $v_j = Ze_j$ by (2), since $\pi^\top v_j = 0$. Differentiating again, in η_i , gives $Gv_{ij} + \text{diag}(e_i)v_j + \text{diag}(e_j)v_i = \kappa_{ij} \mathbf{1} + \kappa_i v_j + \kappa_j v_i$, and multiplying by $\pi^\top$ gives $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

Step 2. $\eta^\top \Gamma \eta > 0$ unless $\eta \in \mathbb{R}\mathbf{1}$. Let $u = Z\eta$ and $f = \eta - (\pi \cdot \eta)\mathbf{1}$. By (2), $-Gu = f$ and $\pi \cdot u = 0$, so $\eta^\top \Gamma \eta = 2 \sum_i \pi_i f_i u_i = -2 \sum_i \pi_i u_i (Gu)_i = \sum_{i \neq j} \pi_i G_{ij} (u_i - u_j)^2$, using that the rows of G sum to zero and $\pi^\top G = 0$. The graph of G is strongly connected, so this vanishes only if u is constant, hence $u = 0$ and $f = 0$. In particular Γ without the row and column of state 1 is positive definite.

Step 3. Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. By Steps 1 and 2 its Hessian is positive definite everywhere. So $\nabla\Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \text{diag } \vartheta(\alpha) = A_F - \text{diag } \theta(\alpha)$ has Perron root 0 and, by (1), Perron vectors r_α and l_α . So at $\vartheta(\alpha)$ we have $\pi = \alpha$, $G = \hat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. By the shift property, $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2\Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence $\beta(\alpha) = (\nabla\Lambda_0)^{-1}(\alpha')$ is real-analytic, and so are $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$, $\theta(\alpha)$, its Perron vectors and Σ_α . Moreover $I_F(\alpha) = \langle \vartheta(\alpha), \alpha \rangle - \Lambda(\vartheta(\alpha)) = \langle \beta(\alpha), \alpha' \rangle - \Lambda_0(\beta(\alpha))$. Differentiating and using $\nabla\Lambda_0(\beta) = \alpha'$ gives $\nabla I_F = \beta$, and then $\nabla^2 I_F = D\beta = \Sigma_\alpha^{-1}$.

(b) Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$. Then $\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i \nu_i} q_{ij}$ and $l_j (A_F u)_j / u_j = \nu_j \sum_i q_{ji} \sqrt{\alpha_i / \nu_i}$ agree because $\nu_j q_{ji} = \nu_i q_{ij}$. So u is a critical point of $\sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$, which is convex in $w = \log u$, and r_α is a multiple of u . Then $\alpha_i (\hat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$, which is symmetric by reversibility, and $\alpha_i \theta_i(\alpha) = \sum_j c_{ij}(\alpha)$ gives I_F , in agreement with Proposition 3.1(c).

For the determinant let $\mathcal{L} = -D\hat{Q}_\alpha$, the Laplacian of the conductances. Fix η with $\eta_1 = 0$ and let $u = Z_\alpha \eta$ and $\zeta = D\eta - (\alpha \cdot \eta)\alpha$. By (2), $\mathcal{L}u = \zeta$ and $\alpha \cdot u = 0$, so $\eta^\top \Gamma_\alpha \eta = 2(D\eta)^\top u = 2\zeta^\top u$. Let $v = u - u_1 \mathbf{1}$. As $\zeta \cdot \mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, $\zeta^\top u = \zeta^\top v$ and $\mathcal{L}v = \zeta$, and since $v_1 = 0$ the rows $2, \dots, k$ give $\mathcal{L}_{(1)} v' = \zeta'$. Here $\mathcal{L}_{(1)}$ is $\mathcal{L}$ without row and column 1, primes denote the coordinates $2, \dots, k$, and $\zeta' = (D' - \alpha' \alpha'^\top) \eta'$ with $D' = \text{diag } \alpha'$. Hence $\Sigma_\alpha = 2(D' - \alpha' \alpha'^\top) \mathcal{L}_{(1)}^{-1} (D' - \alpha' \alpha'^\top)$. By the matrix determinant lemma, $\det(D' - \alpha' \alpha'^\top) = \prod_{i \geq 2} \alpha_i (1 - \sum_{i \geq 2} \alpha_i) = \prod_i \alpha_i$, and by the matrix-tree theorem [28] $\det \mathcal{L}_{(1)} = \text{tree}(c(\alpha))$. (The weighted form follows from the form for multigraphs, since both sides are polynomials in the c_{ij} .) $\square$

4.2 The local law and the face maxima

Inside the face the occupation fractions fluctuate on the scale $T^{-1/2}$ in $k-1$ directions. So a single composition near α gets the Gaussian factor $T^{(k-1)/2}$ times the volume $N^{-(k-1)}$ of one lattice cell. This factor gives the $\frac{1}{2}$ in the crossings below.

Theorem 4.3 (local law). *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly*

for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} \left(1 + \frac{c(\alpha)}{T} + O_{\mathcal{K}}(T^{-3/2}) \right),$$

where $c(\alpha)$ is real-analytic on Δ_F° and is given by Lemma B.1 applied to the torus integral of Lemma A.1. For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$ exactly.

The proof is in Appendix A. We state only the first 2 terms, which is all we use. The hypothesis $\mu(F) > 0$ is needed for the relative form, since $f_T = C_F = 0$ when $\mu(F) = 0$. Note that $f_T(\alpha)$ is continuous and positive in (T, α) . Indeed H in Lemma 2.4 is a power series with coefficients $0 \leq W(m) \leq k^M q_{\max}^{M-1}$, so it converges everywhere, and some $W(m)$ with every $m_i \geq 1$ is positive since F is irreducible and $\mu(F) > 0$.

Lemma 4.4 (discrete and continuum). *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\text{supp } n = F$ with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N) (1 + O_{\mathcal{K}}(T^2/N))$.*

The proof is in Appendix A.

Theorem 4.5 (face maxima). *Let F satisfy (H). If $k \geq 2$, then uniformly for $1 \leq T \leq N^{1/3}$,*

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T \lambda_F} (1 + O(1/T + T^2/N)),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/T)$. For $k = 1$, $M_{\{i\}} = \mu_i (1 - h q_i)^{N-1} = \mu_i e^{-T q_i} (1 + O(T^2/N))$. In both cases, uniformly for $1 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \frac{k-1}{2} \log T + \log K_F - T \lambda_F + O(1/T + T^2/N). \quad (3)$$

Proof. For $k = 1$ the only path is i^N , and $(N-1) \log(1 - h q_i) = -T q_i + O(T^2/N)$. Let $k \geq 2$, put $X = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T \lambda_F}$ and $x = \alpha' - \alpha_F^*$. By Proposition 4.2(a), I_F and $\log C_F$ are real-analytic on Δ_F° (note that $\mu \cdot r_\alpha > 0$), $\nabla I_F(\alpha_F^*) = 0$ at the interior minimum, and $\nabla^2 I_F > 0$. So there are a closed ball $U = \{|x| \leq \rho_0\} \subset \Delta_F^\circ$ and constants c_U, C_U, L', L'' such that on U we have $c_U |x|^2 \leq I_F - \lambda_F \leq C_U |x|^2$, $|\log(C_F/K_F) - \gamma \cdot x| \leq L' |x|^2$ with $\gamma = \nabla \log C_F(\alpha_F^*)$, and $\log C_F$ is L'' -Lipschitz.

Outside U . As a supremum of continuous functions, I_F is lower semicontinuous on Δ_F . By Proposition 3.1(a) it is at least $\lambda_F + 3\eta$ on $\Delta_F \setminus U^\circ$ for some $\eta > 0$. So each $\beta \in \Delta_F \setminus U^\circ$ has some $\theta \in V_F$ with $\langle q - \theta, \cdot \rangle > \lambda_F + 2\eta$ near β , and by compactness finitely many $\theta_1, \dots, \theta_J$ suffice. With the right θ_j and $T^2/N \leq 1$, Lemma 3.2 gives $P_N(n) \leq C(T/N)^{k-1} e^{-T(\lambda_F + 2\eta)} = X \cdot O(T^{(k-1)/2} e^{-2\eta T})$ for $\text{supp } n = F$ and $n/N \notin U^\circ$.

Upper bound inside U . By Theorem 4.3 and Lemma 4.4 with $\mathcal{K} = U$, for $\alpha = n/N \in U$,

$$P_N(n) = X \frac{C_F(\alpha)}{K_F} e^{-T(I_F(\alpha) - \lambda_F)} (1 + E), \quad |E| \leq C_2(1/T + T^2/N). \quad (4)$$

For $T \geq 2L'/c_U$ the middle factors are at most $\exp(\sup_x [\gamma \cdot x - \frac{c_U T}{2} |x|^2]) = e^{|\gamma|^2/(2c_U T)} = 1 + O(1/T)$, and for smaller T they are bounded. So $M_F \leq X(1 + O(1/T + T^2/N))$.

Lower bound. Let n^0 have support F and $|n_i^0 - N \alpha_{F,i}^*| < 1$ for all i , so $|x^0| \leq k/N$. Then $C_F(n^0/N) \geq K_F e^{-kL''/N}$ and $T(I_F(n^0/N) - \lambda_F) \leq C_U k^2 T/N^2$, and (4) gives $P_N(n^0) \geq X(1 - O(1/T + T^2/N))$, since $1/N$ and T/N^2 are at most T^2/N . This is useful for $T \geq T_2$ with T_2 large. For $T \leq T_2$, $T^2/N \rightarrow 0$, so Lemma 4.4 gives $P_N(n^0) \geq \frac{1}{2} N^{-(k-1)} f_T(n^0/N) \geq cX$, because f_T

is continuous and positive on $[1, T_2] \times U$ and $X \leq CN^{-(k-1)}$ there. So M_F/X lies in $[c, C]$ for $T \leq T_2$ and in $[\frac{1}{2}, \frac{3}{2}]$ for $T \geq T_2$, which gives the first display and (3).

The maximizer. For T bounded the claim is trivial. For $T \geq T_3$ with T_3 large, the bound outside U is below $X/2 \leq P_N(n^0)$, so $\hat{\alpha} = \hat{n}/N \in U$, and $|E| \leq \frac{1}{2}$. Taking logarithms in $P_N(\hat{n}) \geq P_N(n^0)$ gives, with $\hat{x} = \hat{\alpha}' - \alpha_F^*$,

$$c_U T |\hat{x}|^2 \leq L'' |\hat{x}| + C_4(1/T + T^2/N) \leq L'' |\hat{x}| + 2C_4/T,$$

because $T \leq N^{1/3}$ gives $T^2/N \leq 1/T$. This step uses the exponent $\frac{1}{3}$. If $L'' |\hat{x}| \geq 2C_4/T$ then $|\hat{x}| \leq 2L''/(c_U T)$, and otherwise $|\hat{x}| < 2C_4/(L'' T)$. $\square$

4.3 Crossings and tie windows

Theorem 4.6 (the crossing equation). *Let F and G satisfy (H), with $\Delta = k_F - k_G \geq 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) *For large N , D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.*

(ii) *For large N , every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies*

$$\delta T_N + \frac{\Delta}{2} \log T_N = \Delta L - \log \frac{K_F}{K_G} + O\left(\frac{1}{L}\right). \quad (5)$$

(iii) *All such zeros lie in one interval of length $O(1/L)$, and*

$$T_N = \frac{\Delta}{\delta} \left[L - \frac{1}{2} \log L + b_{FG} \right] + O\left(\frac{\log L}{L}\right), \quad b_{FG} = -\frac{1}{2} \log \frac{\Delta}{\delta} - \frac{1}{\Delta} \log \frac{K_F}{K_G}.$$

Proof. By (3), uniformly on $[1, N^{1/3}]$, $D_N(T) = -\Delta L + \frac{\Delta}{2} \log T + \delta T + \log \frac{K_F}{K_G} + O(1/T + T^2/N)$.

(i) At $T = \frac{\Delta}{2\delta}L$ this is $-\frac{\Delta}{2}L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta}L$ it is $\Delta L + O(\log L) > 0$, for large N . For fixed N , M_F is a maximum of finitely many polynomials in $h = T/N$, and it is positive because F is admissible. So D_N is continuous and has a zero in between.

(ii) Let $\psi(T) = \delta T + \frac{\Delta}{2} \log T$, which is increasing. At a zero $T_N \in [1, N^{1/3}]$, $\psi(T_N) = \Delta L - \log(K_F/K_G) + O(1)$. This is larger than $\psi(\frac{\Delta}{2\delta}L) = \frac{\Delta}{2}L + O(\log L)$ for large N , and $\delta T_N \leq \psi(T_N)$. So $\frac{\Delta}{2\delta}L < T_N \leq \frac{2\Delta}{\delta}L$, and then $O(1/T_N + T_N^2/N) = O(1/L)$.

(iii) Let $\psi(T^\circ) = \Delta L - \log(K_F/K_G)$. As $\psi' \geq \delta$, every zero is within $O(1/L)$ of T° . First $T^\circ = \frac{\Delta}{\delta}L + O(\log L)$, so $\log T^\circ = \log L + \log \frac{\Delta}{\delta} + O(\log L/L)$, and substituting this into $\psi(T^\circ)$ gives the explicit form. $\square$

The error of the explicit form comes only from the last expansion, and it is large at every N we can compute. So computations should solve the implicit equation (5). Take the 4-cycle with unit rates and uniform start, F an arc of 3 states and G a vertex, so that $\Delta = 2$, $\delta = \sqrt{2}$ and $\log(K_F/K_G) = 0.964591$ (Remark 5.6). Numerically, at $N = 10^{14}$ the exact crossing is $T_N = 42.2633$, with $\Delta L - \frac{\Delta}{2} \log T_N - \delta T_N = 0.959070$. The root of the implicit equation is within 0.004 of T_N , while the explicit form gives 42.2059. We do not prove that the zero of D_N is unique in general. For their faces Papers A and B prove uniqueness, and Corollaries 4.9 and 4.10 use that.

Theorem 4.7 (tie windows). *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 3.6(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2} \log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F-1}{2} \log \tau^* - \tau^* \lambda_{Ft}$ and $E(t) = \max_{G \in \mathcal{F}^*} \ell_G(t)$.*

- (i) Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^*L + \frac{m^*}{2} \log L + \ell_F(t) + O(\log L/L)$.
- (ii) There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \leq \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \leq N^{-\varepsilon_0} \max_{G \in \mathcal{F}^*} M_G$.
- (iii) There are $\varepsilon, K > 0$ such that for large N and $K \leq |t| \leq \varepsilon L$, every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[K, \infty)$ if $t > 0$, or on $(-\infty, -K]$ if $t < 0$. Such a face has the least λ_F in $\mathcal{F}^*$ if $t > 0$ and the largest if $t < 0$.

Consequently, for every $\eta > 0$ and large N , if $|t| \leq \varepsilon L$ is at distance at least η from every kink of E , then every global mode lies on a face whose line equals E near t . So the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2} \log L + t_j) + o(1)$, where t_j runs over the kinks of E , and it changes near each kink. Faces with identical lines (the same k , λ and K) tie at this order and give no crossing.

Proof. Every $F \in \mathcal{F}^*$ is admissible and irreducible, so $\mu(F) > 0$ by Lemma 2.2(b), and F satisfies (H). Write $T = \tau^*L(1 + y)$ with $y = (t - \frac{1}{2} \log L)/L$, so that $|y| \leq \frac{1}{2}$ for $|t| \leq L/4$ and large N . By (3) and $\tau^*\lambda_F + k_F - 1 = m^*$, uniformly for $|t| \leq L/4$,

$$\log M_F = -m^*L + \frac{m^*}{2} \log L + \ell_F(t) + \frac{k_F - 1}{2} \log(1 + y) + O(1/L) \quad (F \in \mathcal{F}^*). \quad (6)$$

For t in a compact set, $\log(1 + y) = O(\log L/L)$, which gives (i).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. As $0 \leq \lambda_F \leq q_{\max}$, for $|\tau - \tau^*| \leq \varepsilon_1 = g/(4q_{\max} + 4)$ we have $\Phi_{\tau}(F) - \Phi_{\tau}(G) \geq g/2$ for such F and all $G \in \mathcal{F}^*$. With $\tau = T/L$, Theorem 3.3(ii) gives $\log M_F \leq -L\Phi_{\tau}(F) + (k_F - 1) \log T + C$ and $\log M_G \geq -L\Phi_{\tau}(G) - C$. So $\log M_F - \log M_G \leq -\frac{g}{2}L + O(\log L) \leq -\frac{g}{4}L$.

(iii) Take $K \geq 1$ larger than every $|t_j|$, and $\varepsilon \leq \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$, so that (ii) applies. Let $t \geq K$, let $\ell_F = E$ on $[K, \infty)$, and let $G \in \mathcal{F}^*$ with $\ell_G \neq \ell_F$. The line with the least λ has the largest slope, so $\lambda_F \leq \lambda_G$. By (6), $\log M_F - \log M_G = \ell_F(t) - \ell_G(t) + \frac{k_F - k_G}{2} \log(1 + y) + O(1/L)$. If $\lambda_G = \lambda_F$, then $k_G = k_F$ because both faces are tied, and the difference is $\log(K_F/K_G) + O(1/L)$, where $K_F > K_G$ because the 2 lines are parallel and $\ell_G \leq \ell_F$. Otherwise $\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) > 0$, again because both faces are tied, and $\ell_F(t) - \ell_G(t) = \Delta t + \ell_F(0) - \ell_G(0)$. As $|\log(1 + y)| \leq 2|y| \leq t/2$ for large N , the difference is at least $\frac{\Delta}{2}t - |\ell_F(0) - \ell_G(0)| - O(1/L) > 0$ if K is large. The case $t \leq -K$ is the same with the order of the λ 's reversed.

For the consequence, fix $\eta > 0$ and let $S_{\eta} = [-K, K] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$. For $G \in \mathcal{F}^*$, $E - \ell_G$ is convex, piecewise linear and nonnegative, so its zero set is an interval with kinks of E as endpoints, or a single kink. So on each component of S_{η} either $\ell_G = E$ or $E - \ell_G \geq g_{\eta}$ for some $g_{\eta} > 0$. By (i) and (ii), for large N and $t \in S_{\eta}$ a global mode lies on a face with $\ell_F(t) = E(t)$, and for $K \leq |t| \leq \varepsilon L$ we use (iii). The line equal to E is constant on each component of $[-\varepsilon L, \varepsilon L] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$ and changes at each kink. Since η is arbitrary, the claim follows. $\square$

Remark 4.8 (scope of the $\frac{1}{2}$). The coefficient $\frac{1}{2}$ of $\log \log N$ in Theorems 4.6 and 4.7 is due to the power $T^{(k-1)/2}$ in Theorem 4.5. It is the same for every pair of faces that satisfy (H), whatever the graph, the rates, the sizes and the start, and it needs no reversibility or aperiodicity. At a kink of E where the line of G gives way to the line of F as t increases, we have $\lambda_F < \lambda_G$ and $\Delta = \tau^*\delta$, and solving $\ell_F = \ell_G$ gives $t_j = b_{FG}$. So when every tied face is irreducible, every change of the line of the face that carries the global mode in a tie window is at $T = \frac{\Delta}{\delta}(L - \frac{1}{2} \log L + b_{FG}) + o(1)$, a crossing of Theorem 4.6. Without this assumption the $\frac{1}{2}$ can fail, and Proposition 4.12 gives the coefficients 1 and 0.

4.4 Papers A and B

We recall the notation of Paper A [2]. For positive integers a, b let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a letters R and b letters L , where $s(w)$ is the number of switches of w . Paper A shows that $S_{a,b}(u) = P_{a+b}(a)/P_{a+b}(0)$ for its walk with $u = (1-p)/p$, and writes it as an explicit polynomial [2, Eq. (9)]. Let $S_N = S_{\lfloor N/2 \rfloor, \lfloor N/2 \rfloor}$, let u_N be the positive root of $S_N(u_N) = 1$, and let $p_N = 1/(1+u_N)$ be the unique crossing of the middle and the endpoints [2, Theorem 5.1]. Put $c_0 = \frac{1}{2} \log(\pi/8)$.

Corollary 4.9 (Paper A). *Let $d = 2$, $q_{12} = q_{21} = 1$ and $\mu = (\frac{1}{2}, \frac{1}{2})$, the walk of Paper A with $h = 1 - p$ and $T = N(1 - p)$. Then $K_{\{1\}} = K_{\{2\}} = \frac{1}{2}$ with $\lambda = 1$, and $K_{\{1,2\}} = \sqrt{2/\pi}$ with $\lambda = 0$ and $\det \Sigma = \frac{1}{4}$. For every $N \geq 2$, $M_{\{1,2\}} = M_{\{1\}}$ holds exactly at $T = N(1 - p_N)$, and for large N*

$$T + \frac{1}{2} \log T = L + c_0 + O(1/L) \quad \text{at } T = N(1 - p_N).$$

So $b_{FG} = c_0$, the constant of [2, Theorem 5.2].

Proof. The full face is reversible with $\nu = \mathbf{1}$, $\alpha^* = (\frac{1}{2}, \frac{1}{2})$, $r_F = \mathbf{1}$ and $l_F = \frac{1}{2}\mathbf{1}$, so $(\mu \cdot r)(l \cdot \mathbf{1}) = 1$. By Proposition 4.2(b), $\text{tree}(c) = c_{12} = \frac{1}{2}$ and $\det \Sigma = 2 \cdot \frac{1}{16} \cdot 2 = \frac{1}{4}$, so $K_{\{1,2\}} = 2/\sqrt{2\pi}$. By [2, Prop. 6.1] and the symmetry $S_{a,b} = S_{b,a}$, every $S_{a,N-a}$ is dominated coefficientwise by S_N , so the middle bin is the largest bin with support $\{1, 2\}$ for every p . Hence $M_{\{1,2\}} = M_{\{1\}} S_N(u)$ with $u = h/(1-h)$, and the crossing is at p_N . By [2, Theorem 5.1], $N(1 - p_N) \sim L$. Theorem 4.6 with $\Delta = \delta = 1$ and $\log(K_{\{1,2\}}/K_{\{1\}}) = \log(2\sqrt{2/\pi}) = -c_0$ gives the equation. $\square$

Theorem 4.6(iii) then gives $N(1 - p_N) = L - \frac{1}{2} \log L + c_0 + O(\log L/L)$, which is [2, Eq. (7)]. With $c_{\{1,2\}} - c_{\{1\}} = -\frac{1}{8}$, Proposition 4.13(ii) turns the equation into $T + \frac{1}{2} \log T = L + c_0 + \frac{1}{8T} + O(T^{-3/2})$. This is the equation behind [2, Eq. (8)], and solving it recovers that expansion, with the weaker error $O(L^{-3/2})$ in place of the $O((\log L)^2/L^2)$ of Paper A.

Corollary 4.10 (Paper B). *Let $Q = J - dI$, where J is the all-ones matrix, and let μ be uniform. This is the model of Paper B [3] with $r = h$ and $T = Nh$. For $|F| = k$, $\lambda_F = d - k$, α_F^* is uniform on F , $\det \Sigma_{\alpha_F^*} = 2^{k-1} k^{1-2k}$ and $K_F = \frac{1}{d} k^{k+1/2} (4\pi)^{-(k-1)/2}$. The crossing of a face of k states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k = \frac{1}{2} \log(4\pi) - \frac{k+1/2}{k-1} \log k$, the constant of [3, Theorem 3.1]. At $\alpha \in \Delta_F^\circ$, with $A = \sum_{i \in F} \sqrt{\alpha_i}$,*

$$I_F(\alpha) = d - A^2, \quad dC_F(\alpha) = A^{k/2+1} \left(\prod_{i \in F} \alpha_i \right)^{-3/4} (4\pi)^{-(k-1)/2},$$

which is the constant $D(\alpha)$ of [3, Theorem 3.5]. At $\tau^* = 1$ every face ties, and Theorem 4.7 gives the window of [3, Theorem 3.2].

Proof. All rates are 1 and $q_i = d - 1$, so every face is reversible with $\nu = \mathbf{1}$. By Proposition 4.2(b), r_α and l_α are multiples of $\sqrt{\alpha}$, so $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = A^2/d$, and $c_{ij}(\alpha) = x_i x_j$ with $x_i = \sqrt{\alpha_i}$. So $I_F(\alpha) = d - 1 - \sum_{i \neq j} x_i x_j = d - A^2$. For $c_{ij} = x_i x_j$ and $S = \sum_i x_i$, the Laplacian grounded at state 1 is $\text{diag}(x_i S)_{i \geq 2} - x' x'^\top$, and the matrix determinant lemma gives the weighted Cayley formula $\text{tree}(c) = S^{k-1} \prod_{i \geq 2} x_i (1 - \sum_{i \geq 2} x_i/S) = x_1 \cdots x_k S^{k-2}$. So $\sqrt{\text{tree}(c(\alpha))} = (\prod_i \alpha_i)^{1/4} A^{(k-2)/2}$, which gives $dC_F(\alpha)$. By Cauchy-Schwarz, $A^2 \leq k$ with equality only at the uniform α . So α_F^* is uniform and $\lambda_F = d - k$, and $A = \sqrt{k}$ and $\text{tree}(c) = 1/k$ there give K_F and $\det \Sigma$. A vertex has $K = 1/d$ and $\lambda = d - 1$, so $b_{FG} = -\frac{1}{k-1} \log(dK_F) = a_k$.

Since $p = 1 - (d-1)h \geq \frac{1}{2} \geq \frac{1}{d}$, the balanced bins are the largest bins on a face [3, Theorem 3.6], so M_F is Paper B's balanced face bin and $M_{\{i\}}$ its vertex bin. Their crossing is unique by [3, Theorem 3.1], and there Theorem 4.6 gives $T + \frac{1}{2} \log T = L + a_k + O(1/L)$, in agreement with Paper B, whose variable is $Nu = T/p = T + O(L^2/N)$. By Theorem 4.3 and Lemma 4.4, and since the vertex bin is $\frac{1}{d}e^{-(d-1)T}(1 + O(T^2/N))$, the ratio of an interior bin to a vertex bin is $dC_F(\alpha)T^{(k-1)/2}e^{(A^2-1)T}N^{-(k-1)}(1 + O(1/T + T^2/N))$, as in [3, Theorem 3.5]. At $\tau^* = 1$, $\Phi_1(F) = d-1$ for every face, and Q has symmetric support. Theorem 4.7(i) gives $\log(M_F/M_{\{i\}}) = \ell_F(t) - \ell_{\{i\}}(t) + O(\log L/L) = (k-1)(t - a_k) + O(\log L/L)$, which is [3, Theorem 3.2], since the 2 ways of writing T in the window differ by $o(1)$ in t . $\square$

So Paper B's factor $A^{k/2+1}(\prod_i \alpha_i)^{-3/4}$ equals $A^2 \sqrt{\text{tree}(c(\alpha))} / \prod_i \alpha_i$. The part that depends on the graph is the square root of a weighted count of spanning trees.

4.5 Exact examples

Proposition 4.11 (an exact pair identity). *Let $i \neq j$ with $q_{ij} = q_{ji} = a$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$. For every $N \geq 2$, $1 \leq m \leq N-1$ and $h > 0$ such that $I + hQ$ is stochastic and $qh < 1$,*

$$\frac{P_N(me_i + (N-m)e_j)}{P_N(Ne_i)} = S_{m,N-m}(u), \quad u = \frac{ah}{1-qh}.$$

Proof. A path with this composition is a word in i, j with m letters i . Each stay has probability $1 - qh$ and each switch ah , so a word with s switches has probability $\mu_i(1 - qh)^{N-1-s}(ah)^s = \mu_i(1 - qh)^{N-1}u^s$, while $P_N(Ne_i) = \mu_i(1 - qh)^{N-1}$. Sum over the words. $\square$

The reason is that each switch replaces a factor $1 - qh$ by ah , and nothing else about the graph enters. So one state against a symmetric pair is Paper A's crossing at every N . We use this for K_3 and the cycles in Section 5 and for sticky kernels in Proposition 6.4.

The next example shows that the irreducibility assumption cannot simply be dropped from Theorem 4.7. The best composition of a reducible face can visit a state only once. Then there is no Gaussian window in that direction, and the power of T is 1 rather than $\frac{1}{2}$. The assumption fails at both tie points below.

Proposition 4.12 (a directed 3-cycle). *Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu = \delta_1$.*

- (i) *The admissible faces are $\{1\}$, $\{1, 2\}$ (reducible, $\lambda = 1$) and $\{1, 2, 3\}$ (irreducible, $\lambda = 0$). Exactly $M_{\{1\}} = (1 - bh)^{N-1}$ and $M_{\{1,2\}} = bh(1 - h)^{N-2}$, and the latter is attained only at $n = (1, N-1, 0)$. For the full face, $\alpha^* = (1, b, b)/(1 + 2b)$, $\det \nabla^2 I = (1 + 2b)^4/(3b^2)$ and $K_{\{1,2,3\}} = (1 + 2b)^2/(2\sqrt{3}\pi b)$.*
- (ii) *$M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$. It satisfies $(b-1)T + \log T = L - \log b + O(T^2/N)$, so $T_N^{(1)} = \frac{1}{b-1}[L - \log L + \log \frac{b-1}{b}] + O(\log L/L)$, and the coefficient of $\log \log N$ in the bracket is 1.*
- (iii) *Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log \frac{2\sqrt{3}\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log \log N$ is 0. These roots lie in an interval of length $O(1/L)$.*

(iv) Both are changes of the global mode. For large N and $T \leq N^{1/3}$, the mode is $(N, 0, 0)$ for $T < T_N^{(1)}$, it is $(1, N-1, 0)$ between $T_N^{(1)}$ and the smallest root in (iii), and it lies on the full face above the largest root in (iii).

Proof. (i) The chain moves only $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ and starts at 1, so the support of a path is $\{1\}$, $\{1, 2\}$ or $\{1, 2, 3\}$. The components of $\{1, 2\}$ are $\{1\}$ and $\{2\}$ with exit rates b and 1. A path with support $\{1, 2\}$ is $1^{n_1}2^{n_2}$, with probability $(1-bh)^{n_1-1}bh(1-h)^{n_2-1}$, and raising n_1 by one multiplies this by $(1-bh)/(1-h) < 1$. For the full face, $\lambda = 0$, $r = \mathbf{1}$, and the stationary law solves $\pi_2 = b\pi_1 = \pi_3$. So $\alpha^* = l = (1, b, b)/s$ with $s = 1 + 2b$, and $(\mu \cdot r)(l \cdot \mathbf{1}) = 1$. The ratios u_2/u_1 , u_3/u_2 , u_1/u_3 range over the positive triples with product 1, so by the inequality of arithmetic and geometric means $I(\alpha) = \sum_i q_i \alpha_i - 3(b\alpha_1\alpha_2\alpha_3)^{1/3}$. Let $g = (\alpha_1\alpha_2\alpha_3)^{1/3}$ and $w = 1/\alpha$. The Hessian of g in all 3 coordinates is $-gM$ with $M = \frac{1}{3} \text{diag}(w)^2 - \frac{1}{9}ww^\top$, so in the coordinates (α_2, α_3) , $\nabla^2 I = 3b^{1/3}g J^\top M J$, where J is the Jacobian of $(\alpha_2, \alpha_3) \mapsto (1 - \alpha_2 - \alpha_3, \alpha_2, \alpha_3)$. At α^* , $3b^{1/3}g = 3b/s$ and $w = (s, s/b, s/b)$. With $y = 1/b$, $J^\top M J$ has diagonal entries $\frac{2}{9}s^2(1+y+y^2)$ and off-diagonal entries $\frac{1}{9}s^2(2+2y-y^2)$, so

$$\det J^\top M J = \frac{s^4}{81} [4(1+y+y^2)^2 - (2+2y-y^2)^2] = \frac{s^4}{81} 3y^2(2+y)^2 = \frac{s^6}{27b^4}.$$

Hence $\det \nabla^2 I = (3b/s)^2 s^6 / (27b^4) = s^4 / (3b^2)$, and Proposition 4.2(a) gives $K_{\{1,2,3\}}$.

(ii) Let $D_1 = \log M_{\{1,2\}} - \log M_{\{1\}} = \log(bh) + (N-2)\log(1-h) - (N-1)\log(1-bh)$. Then $D_1'(h) = \frac{1}{h} - \frac{N-2}{1-h} + \frac{b(N-1)}{1-bh} > 0$ for $bh < 1$, while $D_1 \rightarrow -\infty$ as $h \rightarrow 0$ and $D_1 \rightarrow \infty$ as $bh \rightarrow 1$. So there is exactly one root. Expanding, $D_1 = \log b + \log T - L + (b-1)T + O(T^2/N)$, which is positive at $T = L/(b-1)$ and negative at $T = L/(2(b-1))$. So the root lies between them, and substituting $\log T = \log L - \log(b-1) + O(\log L/L)$ gives the expansion.

(iii) Here $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$, and Theorem 4.5 for the full face ($k = 3$, $\lambda = 0$, $\mu(F) = 1$) gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/T + T^2/N)$. So $\log M_{\{1,2\}} - \log M_{\{1,2,3\}} = L - T + \log b - \log K_{\{1,2,3\}} + O(1/T + T^2/N)$, and every zero in $[L/2, 2L]$ is within $O(1/L)$ of $L + \log b - \log K_{\{1,2,3\}}$.

(iv) Only the 3 faces in (i) are admissible. Lemma 3.2 with $\theta = q$ gives $M_{\{1,2,3\}} \leq C(T/N)^2$ for $T \leq N^{1/3}$. For $T < T_N^{(1)}$, $M_{\{1\}} > M_{\{1,2\}}$ by the monotonicity of D_1 , and $M_{\{1\}} \geq e^{-bT-b^2T^2/N} \geq N^{-b/(b-1)-o(1)}$ since $T_N^{(1)} \leq L/(b-1)$. This beats $C(T/N)^2$ because $b/(b-1) < 2$. For $T > T_N^{(1)}$, $M_{\{1,2\}} > M_{\{1\}}$, and we compare $\{1, 2\}$ with the full face. For $T \leq L/2$, $\log M_{\{1,2\}} - \log M_{\{1,2,3\}} \geq L - T - \log T - C > 0$. On $[L/2, 2L]$ the sign is given by the formula in (iii), which is positive at $L/2$ and negative at $2L$. For $2L \leq T \leq N^{1/3}$ the same formula is at most $-L + C < 0$. $\square$

4.6 The next term

The face maxima also have a $1/T$ term. It collects the $1/T$ term of the local law at α_F^* and the gain from moving the maximizer by $\Sigma\gamma/T$ toward larger C_F .

Proposition 4.13 (the next term). *Let F satisfy (H).*

(i) *The limit $c_F = \lim_{T \rightarrow \infty} T[\sup_{\alpha \in \Delta_F^\circ} f_T(\alpha)/(T^{(k-1)/2} K_F e^{-T\lambda_F}) - 1]$ exists. For $k \geq 2$, $c_F = c(\alpha_F^*) + \frac{1}{2}\gamma^\top \Sigma\gamma$, where c is the coefficient in Theorem 4.3, $\gamma = \nabla \log C_F(\alpha_F^*)$ and $\Sigma = \Sigma_{\alpha_F^*}$, in the coordinates α' .*

(ii) *Let G also satisfy (H), with Δ and δ as in Theorem 4.6. At every zero $T_N \in [1, N^{1/3}]$ of $\log M_F - \log M_G$,*

$$T_N \left(\Delta L - \frac{\Delta}{2} \log T_N - \delta T_N - \log \frac{K_F}{K_G} \right) = c_F - c_G + O(T_N^{-1/2} + T_N^3/N).$$

- (iii) $c_F = 0$ if F is a single state, $c_F = -1/(8q_{ij})$ if $F = \{i, j\}$ with $q_{ij} = q_{ji}$, $q_i = q_j$ and $\mu_i = \mu_j$ (so $c_F = -\frac{1}{8}$ for unit rates), and $c_F = -(k-1)/8$ if $Q = J - dI$ and μ is constant on F .

The proof is in Appendix B. For the arc of 3 states and the whole 4-cycle, the formula in (i) is evaluated in Remark 5.6.

5 Examples

We now work out the winners for a few switching graphs, and the second-order constants where they are explicit. In every example Q is symmetric. So the admissible faces are the connected faces with positive start mass (Lemma 2.2), Lemma 3.7 applies, and Theorem 4.7 applies at every tie. We call the faces in $\mathcal{F}_\tau$ the *winners* at τ , and their list as τ runs from 0 to ∞ the *first-order winner sequence*. By Theorem 3.3(iii), for fixed τ and large N every global mode lies on a winner.

5.1 The complete graph and the start

Proposition 5.1 (complete graph and single jumps). *(a) Let $Q = J - dI$, the complete graph K_d with unit rates. Then $\lambda_F = d - |F|$ for every face F , so $\Phi_\tau(F) = d\tau - 1 + |F|(1 - \tau)$. At $\tau = 1$ every face has $\Phi_1(F) = d - 1$. For $\tau < 1$ the winners are the admissible vertices, and for $\tau > 1$ the only winner is $[d]$.*

(b) Let Q be symmetric and irreducible, and let $x \in [d]$. Then $\lambda_{[d] \setminus \{x\}} \leq q_x/(d-1)$, with equality if and only if $q_{yx} = q_x/(d-1)$ for every $y \neq x$.

(c) Let Q be symmetric and irreducible and let the start have full support. Suppose that the first-order winner sequence is a single jump, that is, for some $\tau_1 > 0$ the winners are vertices for $0 < \tau < \tau_1$ and $[d]$ alone for $\tau > \tau_1$. Then every vertex x that wins for small τ has $q_{yx} = q_x/(d-1)$ for every $y \neq x$.

(d) Let Q have unit rates on a connected graph G_Q and let the start have full support. Then the first-order winner sequence is a single jump if and only if G_Q is complete.

Proof. (a) On a face with k states, $-Q_F = dI - J$ has the eigenvalue $d - k$ on $\mathbf{1}_F$ and d on the vectors with zero sum. So $\Phi_\tau(F) = \tau(d - k) + k - 1$. It equals $d - 1$ at $\tau = 1$, increases with k for $\tau < 1$ and decreases with k for $\tau > 1$.

(b) Put $R = [d] \setminus \{x\}$. In Lemma 3.7(a) the test vector $\mathbf{1}_R/\sqrt{d-1}$ has $\mathcal{E} = \sum_{y \in R} q_{yx}/(d-1) = q_x/(d-1)$, by symmetry. This gives the bound. Note that $(-Q_R \mathbf{1}_R)_y = q_{yx}$ for $y \in R$. If equality holds, then $\mathbf{1}_R$ minimizes the Rayleigh quotient, so it is an eigenvector of $-Q_R$ for λ_R , and every q_{yx} equals λ_R . Conversely, let $q_{yx} = c$ for all $y \in R$. Since $\mathcal{E}(|\varphi|) \leq \mathcal{E}(\varphi)$, there is an eigenvector $\varphi \geq 0$ of $-Q_R$ for λ_R . Then $c\langle \varphi, \mathbf{1}_R \rangle = \langle \varphi, -Q_R \mathbf{1}_R \rangle = \lambda_R \langle \varphi, \mathbf{1}_R \rangle$, so $c = \lambda_R$.

(c) A vertex x that wins for small τ has the least exit rate, so it wins on all of $(0, \tau_1)$. The least exponent $\min_F \Phi_\tau(F)$ is continuous in τ , so $\tau_1 q_x = d - 1$. Let C be a component of $[d] \setminus \{x\}$ with $\lambda_C = \lambda_{[d] \setminus \{x\}}$. It is admissible, so $\tau_1 \lambda_C + |C| - 1 \geq d - 1$, and $|C| \leq d - 1$ gives $\lambda_C \geq 1/\tau_1 = q_x/(d-1)$. So equality holds in (b).

(d) If G_Q is complete, then (a) gives a single jump. Conversely, with unit rates q_x is the degree of x , and by (c) a vertex of least degree is joined to every other state. So G_Q is complete. $\square$

Part (b) is the reason for (d). For a vertex x of least exit rate, the point of $[d] \setminus \{x\}$ lies on or to the left of the chord from $\{x\}$ to $[d]$, and on it only when every other state feeds x at the same

rate. Otherwise a face with between 2 and $d - 1$ states wins on an interval. Proposition 6.3(a) is the analogue for sticky kernels. On K_d itself the first order decides nothing at $\tau = 1$. With a uniform start the second order makes the mode jump from a vertex straight to $[d]$ [3, Cor. 3.7], as Corollary 4.10 explains, although at finite N an intermediate face can hold the mode [3, Ex. 3.8]. The start enters K_F only through the factor $(\mu \cdot r_F)(l_F \cdot \mathbf{1})$, and on K_3 this can open a window for an edge. The reason is that a start with no mass on state 3 raises the constants of the vertices 1, 2 and of the edge $\{1, 2\}$ by the same factor, compared with the uniform start, and leaves the full face alone. The edge line has the middle slope, so it can rise above the point where the other 2 lines cross.

Proposition 5.2 (the start matters on K_3). *Let $Q = J - 3I$, let μ be any start, and put $T = L - \frac{1}{2} \log L + t$. At $\tau^* = 1$ all admissible faces tie, and the lines of Theorem 4.7 for the admissible faces are*

$$\ell_{\{i\}}(t) = \log \mu_i - 2t, \quad \ell_{\{i,j\}}(t) = \log \left((\mu_i + \mu_j) \sqrt{2/\pi} \right) - t, \quad \ell_{[3]}(t) = \log \frac{3^{5/2}}{4\pi}.$$

(a) *There is a nonempty open interval of t on which $\max_F \ell_F(t)$ is attained only by edges if and only if $\max_{i < j} (\mu_i + \mu_j)^2 > \frac{3^{5/2}}{8} \max_l \mu_l$. The edges that attain it are then those with the largest $\mu_i + \mu_j$.*

(b) *For $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$ this interval is*

$$(t_{ve}, t_{ef}) = \left(\frac{1}{2} \log \frac{\pi}{8}, \frac{1}{2} \log \frac{2}{\pi} - \log \frac{3^{5/2}}{4\pi} \right) = (-0.4673558, -0.4412978),$$

of width 0.0260580. For the uniform start there is no such interval.

(c) *Let $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$. For every compact $J \subset (t_{ve}, t_{ef})$ and all large N , every global mode with $t \in J$ has support $\{1, 2\}$. For every compact J' disjoint from $[t_{ve}, t_{ef}]$ and all large N , no global mode with $t \in J'$ has support $\{1, 2\}$.*

Proof. By Proposition 5.1(a) all admissible faces tie at $\tau^* = 1$, and Theorem 4.7 applies since K_3 has symmetric support. Each Q_F has constant row sums, so $r_F = \mathbf{1}_F$, $l_F = \mathbf{1}_F/|F|$ and $(\mu \cdot r_F)(l_F \cdot \mathbf{1}) = \mu(F)$. This is the only factor of K_F that depends on the start, so Corollary 4.10 gives $K_F = \mu(F)|F|^{(|F|-1)/2}(4\pi)^{-(|F|-1)/2}$, and $\lambda_F = 3 - |F|$ gives the lines.

(a) Let $S = \max_{i < j} (\mu_i + \mu_j)$ and $\bar{\mu} = \max_l \mu_l$. Only the top line of each slope matters, namely $v(t) = \log \bar{\mu} - 2t$, $e(t) = \log(S\sqrt{2/\pi}) - t$ and $f = \log(3^{5/2}/(4\pi))$. Let $v(t_{vf}) = f$. If $e(t_{vf}) > f$, then e is the strict maximum near t_{vf} . Otherwise $e \leq v$ for $t \leq t_{vf}$ and $e \leq f$ for $t \geq t_{vf}$, because $e - v$ increases and $e - f$ decreases. As $e^{t_{vf}} = (\bar{\mu}e^{-f})^{1/2}$, the condition $e(t_{vf}) > f$ reads $S^2 \cdot \frac{2}{\pi} > \bar{\mu}e^f$, which is the criterion.

(b) Here $S = 1$, $\bar{\mu} = \frac{1}{2}$ and $1 > 3^{5/2}/16 = 0.9743$. The ends solve $e = v$ and $e = f$. For the uniform start $S^2 = \frac{4}{9} < \frac{3^{5/2}}{8} \cdot \frac{1}{3} = 0.6495$.

(c) By Theorem 4.7(i), $\log M_F = -2L + \log L + \ell_F(t) + O(\log L/L)$ for every admissible face, uniformly for t in compacts. On J the line of $\{1, 2\}$ beats every other line by a fixed margin, and on J' some other line beats it by a fixed margin. $\square$

The window is thin, since the criterion holds with ratio 0.97428. At finite N its ends also drift by $O(\log L/L)$ (Figure 2). At the left end the balanced composition of $\{1, 2\}$ overtakes a vertex. By Proposition 4.11 this is exactly the binary crossing of [2, Thm. 5.1] with $u = h/(1 - 2h)$. The equation of Theorem 4.6 with the term of Proposition 4.13, here $T + \frac{1}{2} \log T = L + \frac{1}{2} \log \frac{\pi}{8} + \frac{1}{8T}$, gives this left end to within 2×10^{-5} at $N = 10^{20}$.

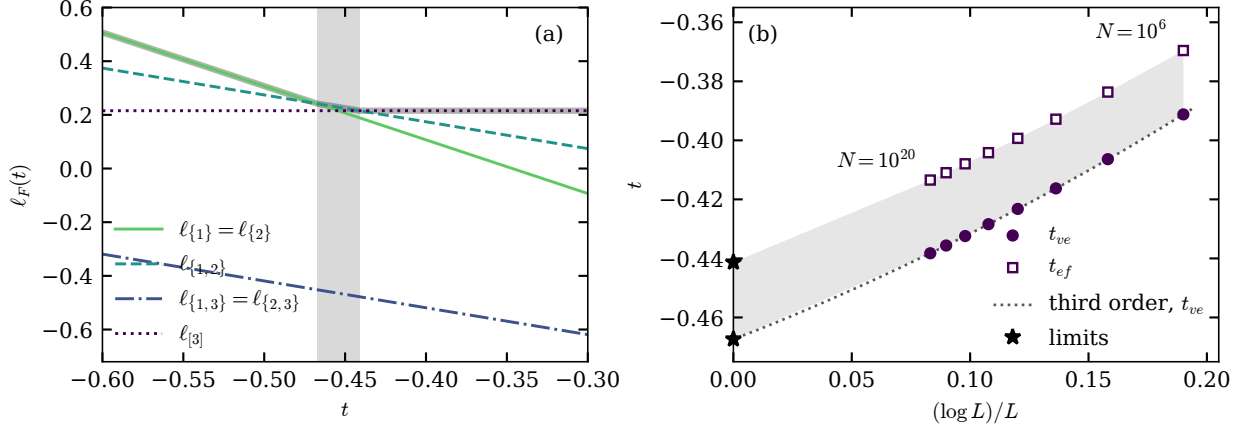


Figure 2: (a) The lines $\ell_F(t)$ of Proposition 5.2 on K_3 with the start $(\frac{1}{2}, \frac{1}{2}, 0)$. The vertex $\{3\}$ has no start mass and is not admissible. The upper envelope is bold, and the edge window $(-0.4674, -0.4413)$ is shaded. (b) The ends t_{ve} and t_{ef} of the window at $N = 10^6, 10^8, \dots, 10^{20}$, computed from the exact face maxima, against $(\log L)/L$. The stars are the limits from (a). The dotted curve solves $T + \frac{1}{2} \log T = L + \frac{1}{2} \log \frac{\pi}{8} + \frac{1}{8T}$ for the left end. At $N = 10^{20}$ the window is $(-0.43825, -0.41345)$.

5.2 Cycles and paths

Let C_d ($d \geq 3$) be the cycle with $q_{i,i\pm 1} = 1$, indices mod d . An *arc* is a set of $k \leq d-1$ consecutive states. For $k \geq 1$ put

$$\lambda_k = 2 - 2 \cos \vartheta_k, \quad \vartheta_k = \frac{\pi}{k+1}, \quad \tau_k = \frac{1}{\lambda_{k-1} - \lambda_k} \quad (k \geq 2), \quad D_k = k + \lambda_k \tau_k.$$

Lemma 5.3 (arcs). (a) On C_d an arc of k states has exit rate λ_k . A proper face that is not an arc is not admissible, and its exponent is larger than that of its longest arc.

(b) The map $k \mapsto \lambda_k$ is strictly decreasing and strictly convex on $k \geq 1$. Hence $\tau_2 < \tau_3 < \dots$.

(c) $D_2 = 3$, and $\frac{3}{2}k - \frac{1}{2} < D_k < \frac{3}{2}k$ for every $k \geq 3$. In particular D_k is not an integer for $k \geq 3$.

Proof. (a) For $k \leq d-1$ the graph on an arc is the path P_k , and $-Q_F = 2I - \text{Adj}(P_k)$ has the eigenvalues $2 - 2 \cos(j\pi/(k+1))$, $1 \leq j \leq k$. A proper face that is not an arc has at least 2 components, each an arc. So it is not admissible (Lemma 2.2), and by (b) its exit rate is that of its longest component, which has fewer states.

(b) With $g(x) = 2 - 2 \cos(\pi/x)$ we have $\lambda_k = g(k+1)$, and for $x \geq 2$

$$g'(x) = -\frac{2\pi \sin(\pi/x)}{x^2} < 0, \quad g''(x) = 2\pi \left(\frac{\pi \cos(\pi/x)}{x^4} + \frac{2 \sin(\pi/x)}{x^3} \right) > 0.$$

(c) Put $a = \pi/(2k)$, $b = \pi/(2(k+1))$ and $\rho = \lambda_k/\lambda_{k-1} = (\sin b/\sin a)^2$. Since $\lambda_k \tau_k = \rho/(1-\rho)$, the claim is $\frac{k-1}{k+1} < \rho < \frac{k}{k+2}$. For $k = 2$, $\rho = \sin^2(\pi/6)/\sin^2(\pi/4) = \frac{1}{2}$ gives $D_2 = 3$. As $\sin x/x$ decreases on $(0, \pi]$, $\sin b/\sin a \geq b/a = k/(k+1)$, so $\rho \geq k^2/(k+1)^2 > (k-1)/(k+1)$. For the upper bound let $k \geq 3$, $m = k+1$ and $\text{sinc } x = \sin x/x$. The claim reads $\text{sinc } b/\text{sinc } a < m/\sqrt{m^2-1}$,

and the right side exceeds $1 + 1/(2m^2)$. By $1 - x^2/6 \leq \operatorname{sinc} x \leq 1 - x^2/6 + x^4/120$ it suffices that $(1 - \frac{b^2}{6} + \frac{b^4}{120}) - (1 - \frac{a^2}{6}) \leq (1 - \frac{a^2}{6})/(2m^2)$, that is

$$\frac{\pi^2}{12} \cdot \frac{2m-1}{(m-1)^2} + \frac{\pi^4}{960m^2} \leq 1 - \frac{\pi^2}{24(m-1)^2}.$$

The left side decreases in m and the right side increases. At $m = 4$ they are 0.6460 and 0.9543. Finally, $(\frac{3}{2}k - \frac{1}{2}, \frac{3}{2}k)$ has length $\frac{1}{2}$ and an integer end, so it contains no integer. $\square$

Theorem 5.4 (cycle cascade). *Let the start have full support.*

- (a) *On $C_3 = K_3$ the winners are the vertices for $\tau < 1$ and $[3]$ for $\tau > 1$, and all 3 sizes tie at $\tau = 1$.*
- (b) *Let $d \geq 4$ and $K = \lfloor 2d/3 \rfloor$. The winners are the d vertices for $0 < \tau < 1$, the d arcs of k states for $\tau_k < \tau < \tau_{k+1}$ ($2 \leq k \leq K-1$), the d arcs of K states for $\tau_K < \tau < \tau_{\text{full}}$, and C_d alone for $\tau > \tau_{\text{full}} = \lceil d/3 \rceil / \lambda_K$. All these intervals are nonempty. Arcs of more than K states never win, and at each switch value exactly 2 sizes tie.*
- (c) $\tau_2 = 1$, $\tau_3 = 1 + \sqrt{2}$, $\tau_4 = 2/(1 - 2\sqrt{2} + \sqrt{5}) = 4.906280$ and $\tau_5 = 1/(\sqrt{3} - \frac{1+\sqrt{5}}{2}) = 8.770636$. *The arcs of 2 states win on an interval of τ if and only if $d \geq 4$. For $k \geq 3$ the arcs of k states win if and only if $d \geq \lceil 3k/2 \rceil$.*
- (d) *As $k \rightarrow \infty$, $\tau_k = \frac{k^3}{2\pi^2} + \frac{3k^2}{4\pi^2} + O(k)$, and as $d \rightarrow \infty$, $\tau_{\text{full}} \sim \frac{4d^3}{27\pi^2}$.*

The cascade stops at two thirds of the cycle. The reason is a comparison per added state. Growing an arc of k states by one state lowers the exit rate by $\lambda_k - \lambda_{k+1} \approx 2\lambda_k/k$. Closing the cycle removes all of λ_k but adds $d - k$ new states, and this is better once $\lambda_k/(d - k) > 2\lambda_k/k$, that is once $k > 2d/3$.

Proof. By Lemma 5.3(a) the admissible faces are the arcs and C_d , with exponents $c_k(\tau) = \tau\lambda_k + k - 1$ and $d - 1$. Part (a) is Proposition 5.1(a) with $d = 3$.

(b) Let $d \geq 4$. Since $c_k(\tau) - c_{k-1}(\tau) = 1 - \tau/\tau_k$ and the τ_k increase, $k \mapsto c_k(\tau)$ is strictly smallest at the k with $\tau_k < \tau < \tau_{k+1}$, where $\tau_1 = 0$ and $\tau_d = \infty$. At τ_k exactly the sizes $k - 1$ and k tie. The least arc exponent $g(\tau) = \min_k c_k(\tau)$ increases continuously and strictly from 0 to ∞ , so C_d alone wins exactly for τ above the root τ_{full} of $g = d - 1$. The arcs of $k \geq 2$ states win on an interval if and only if $\tau_k < \tau_{\text{full}}$, that is $c_k(\tau_k) < d - 1$, that is $D_k < d$. As the τ_k increase, the winning sizes are $1, \dots, K'$ for some K' . Let $k_0 = \lfloor 2d/3 \rfloor$, so $2 \leq k_0 \leq d - 2$. By Lemma 5.3(c), $D_{k_0} < \frac{3}{2}k_0 \leq d$ if $k_0 \geq 3$, and $D_2 = 3 < 4 = d$ if $k_0 = 2$. Also $D_{k_0+1} > \frac{3}{2}(k_0 + 1) - \frac{1}{2} \geq d$, because $3k_0 \geq 2d - 2$. So $K' = k_0 = K$. Then $\tau_K < \tau_{\text{full}} < \tau_{K+1}$, so $c_K(\tau_{\text{full}}) = d - 1$ and $\tau_{\text{full}} = (d - K)/\lambda_K = \lceil d/3 \rceil / \lambda_K$. An arc of $k > K$ states has a larger exponent than the arc of K states for $\tau < \tau_{K+1}$, and one larger than $d - 1$ for $\tau \geq \tau_{\text{full}}$, so it never wins. Arcs of 2 sizes tie only at the distinct values τ_k , and C_d ties with 2 sizes only if $\tau_{\text{full}} = \tau_{K+1}$, that is $D_{K+1} = d$, which Lemma 5.3(c) excludes.

(c) The values follow from $\lambda_1 = 2$, $\lambda_2 = 1$, $\lambda_3 = 2 - \sqrt{2}$, $\lambda_4 = (3 - \sqrt{5})/2$ and $\lambda_5 = 2 - \sqrt{3}$. By (a) and (b), the arcs of k states win on an interval if and only if $d > D_k$, and for $k \geq 3$ they do not win at all otherwise. For $k \geq 3$ no integer lies in $[D_k, \frac{3}{2}k)$, so an integer d exceeds D_k if and only if $d \geq \frac{3}{2}k$.

(d) Since $2 - 2\cos x = x^2 + O(x^4)$, with $m = k + 1$ we get

$$\lambda_{k-1} - \lambda_k = \pi^2 \left(\frac{1}{(m-1)^2} - \frac{1}{m^2} \right) + O(m^{-5}) = \frac{2\pi^2}{m^3} \left(1 + \frac{3}{2m} + O(m^{-2}) \right).$$

k	λ_k	τ_k	least d	b_k
2	1	1	4	-0.4673558
3	0.585786	2.414214	5	-0.9379218
4	0.381966	4.906280	6	-1.3514657
5	0.267949	8.770636	8	-1.7011216
6	0.198062	14.308827	9	-2.0012059
7	0.152241	21.823898	11	-2.2632478
8	0.120615	31.619377	12	-2.4955119

Table 1: Arc thresholds on the cycle C_d . The columns λ_k , τ_k and b_k do not depend on d . The arcs of k states win on an interval of τ if and only if d is at least the value in the fourth column (Theorem 5.4(c)), and b_k is the offset of Corollary 5.5 for the uniform start.

So $\tau_k = \frac{m^3}{2\pi^2} - \frac{3m^2}{4\pi^2} + O(m)$, which is the expansion. Also $\lambda_K \sim \pi^2/K^2$ and $K \sim 2d/3$, so $\tau_{\text{full}} \sim \frac{d}{3} \cdot \frac{4d^2}{9\pi^2}$. $\square$

Table 1 lists the first thresholds. For $d = 4, 5, 6, 8$ we get $\tau_{\text{full}} = 2, 2 + \sqrt{2}, 3 + \sqrt{5}$ and $6 + 3\sqrt{3}$.

Corollary 5.5 (second order on cycles). *Let the start be uniform on C_d . Every arc satisfies (H). The arc of $k \leq d - 1$ states has*

$$K_{\text{arc}}(k) = \frac{2}{d(k+1)} \frac{(1 + \cos \vartheta_k)^2}{\sin^3 \vartheta_k} \left(\frac{k+1}{2\pi} \right)^{(k-1)/2},$$

and the whole cycle has $K_{C_d} = d^{(d+2)/2} (4\pi)^{-(d-1)/2}$. Let $d \geq 4$ and $2 \leq k \leq \lfloor 2d/3 \rfloor$, and put

$$b_k = \frac{1}{2} \log(\lambda_{k-1} - \lambda_k) - \log \frac{K_{\text{arc}}(k)}{K_{\text{arc}}(k-1)},$$

which does not depend on d . Then for large N the global mode passes from an arc of $k-1$ states to an arc of k states, and there is $\varepsilon > 0$ such that every change in the size of the arc that carries the global mode with $|T/L - \tau_k| \leq \varepsilon$ happens at $T = \tau_k(L - \frac{1}{2} \log L + b_k) + o(1)$. Also $b_2 = \frac{1}{2} \log(\pi/8)$.

Proof. Arcs and C_d are connected and have positive start mass, so they satisfy (H), and Proposition 4.2(b) applies with $\nu = \mathbf{1}$. On an arc of k states write $\vartheta = \vartheta_k$. The Perron vectors are proportional to $(\sin j\vartheta)_{j=1}^k$. As $\sum_j \sin^2 j\vartheta = (k+1)/2$, we get $\alpha_j^* = \frac{2}{k+1} \sin^2 j\vartheta$, and we may take $r = l = \sqrt{\alpha^*}$, so $(\mu \cdot r)(l \cdot \mathbf{1}) = \frac{1}{d} (\sum_j \sqrt{\alpha_j^*})^2$. The only spanning tree of a path is the path, so $\text{tree}(c) = \prod_j \alpha_j^* / (\alpha_1^* \alpha_k^*)^{1/2}$. Now $\sum_j \sin j\vartheta = \cot(\vartheta/2) = (1 + \cos \vartheta) / \sin \vartheta$ and $\alpha_1^* = \alpha_k^* = \frac{2}{k+1} \sin^2 \vartheta$. Also $\prod_j \sin j\vartheta = (k+1)/2^k$, which follows from $\prod_{j=1}^k (1 - \omega^j) = k+1$ with $\omega = e^{2\pi i/(k+1)}$ and $|1 - \omega^j| = 2 \sin j\vartheta$. Inserting these values gives $K_{\text{arc}}(k)$. On C_d the law α^* is uniform, $r = \mathbf{1}$ and $l = \mathbf{1}/d$, so $(\mu \cdot r)(l \cdot \mathbf{1}) = 1$. Every edge has conductance $1/d$, and the d spanning trees give $\text{tree}(c) = d^{2-d}$. So $K_{C_d} = (4\pi)^{-(d-1)/2} d^{(2-d)/2} d^d$.

By Theorem 5.4(b) the faces tied at $\tau^* = \tau_k$ are the arcs of $k-1$ and of k states, and all arcs of one size have the same line. As $\tau_k(\lambda_{k-1} - \lambda_k) = 1$, the lines of Theorem 4.7 satisfy $\ell_k(t) - \ell_{k-1}(t) = \log(K_{\text{arc}}(k)/K_{\text{arc}}(k-1)) + \frac{1}{2} \log \tau_k + t$, which vanishes exactly at $t = b_k$. So Theorem 4.7 gives the claim. The factor $1/d$ cancels in b_k , and $K_{\text{arc}}(1) = 1/d$ and $K_{\text{arc}}(2) = 4/(d\sqrt{2\pi})$ give $b_2 = \log(\sqrt{2\pi}/4)$. $\square$

The first step is the binary problem exactly. By Proposition 4.11, for every N and d a balanced composition of 2 adjacent states beats a vertex if and only if $S_N(u) > 1$ with $u = h/(1 - 2h)$, as in [2, Thm. 5.1], and b_2 is the constant c_0 of [2]. Figure 1(b) shows the support of the exact mode on C_5 at $N = 40$ and 72. The phases come in the order that first order predicts, but much earlier than $\tau_k \log N$. Most of the gap is the shift $-\frac{1}{2} \log L + b_k$. At $N = 40$, for example, $\tau_3(L - \frac{1}{2} \log L + b_3) = 5.07$, while the exact switch from 2 to 3 states is at $T = 5.12$ and $\tau_3 \log N = 8.91$.

Remark 5.6 (the 4-cycle). On C_4 with the uniform start the winners are a vertex, then an adjacent pair from $\tau = 1$, then C_4 from $\tau = 2$. Corollary 5.5 gives $K_{\text{pair}} = 1/\sqrt{2\pi}$, $K_{\text{arc3}} = (4 + 3\sqrt{2})/(4\pi)$ and $K_{C_4} = 8/\pi^{3/2}$, with $\det \Sigma = 1/512$ for C_4 . So by Theorem 4.6 every crossing of the pair and C_4 satisfies $T + \log T = 2L - \log(8\sqrt{2}/\pi) + O(1/L)$. The 3-arc never holds the mode at first order, but Theorem 4.6 still applies to its face maximum. At third order, $c = -\frac{1}{8}$ for the pair by Proposition 4.13(iii). For the 3-arc and the whole cycle the formula of Proposition 4.13(i) is proved, and its value was evaluated exactly by computer (Section 9). This gives $c_{\text{arc3}} = \frac{5}{4}\sqrt{2} - 2$ and $c_{C_4} = -\frac{9}{16}$. So by Proposition 4.13(ii), T_N times the error in the pair-to-cycle equation tends to $-\frac{9}{16} + \frac{1}{8} = -\frac{7}{16}$.

Remark 5.7 (paths). The two-thirds rule also holds on the path P_d ($d \geq 3$, unit rates). The winners are the 2 end-arcs of k states for $k = 1, \dots, \lfloor 2d/3 \rfloor$ in turn, and then P_d . Reflecting an end-arc of k states at its leaf gives a path of $2k$ states, whose Perron vector is symmetric. So the end-arc has exit rate $\lambda'_k = \lambda_{2k} = 2 - 2\cos(\pi/(2k+1))$, below the rate λ_k of an inner arc, and it is strictly decreasing and convex in k by Lemma 5.3(b). The proof of Theorem 5.4 then goes through with λ'_k , because $D'_k = k + \lambda'_k/(\lambda'_{k-1} - \lambda'_k)$ satisfies $\frac{3}{2}k - \frac{1}{2} < D'_k < \frac{3}{2}k$ for all $k \geq 2$. As in Lemma 5.3(c), the lower bound follows from $(2k-1)^2(k+1) - (k-1)(2k+1)^2 = 2$. For the upper bound put $y = (8k-2)/((k+2)(2k-1)^2) \leq 3$, so that $\frac{k}{k+2} = (1+y)(\frac{2k-1}{2k+1})^2$. Using $\sqrt{1+y} \geq 1+y/3$, the sinc bounds and $1 - \pi^2/216 > 0.954$, it reduces to $\frac{\pi^2}{12k} + \frac{\pi^4}{7680k^2} < 0.318 \frac{8k-2}{k+2}$, which holds at $k = 2$ ($0.414 < 1.113$) and hence for all $k \geq 2$, since the left side decreases and the right side increases. The first switch is at the golden ratio $1/(\lambda'_1 - \lambda'_2) = (1 + \sqrt{5})/2$.

The lower bound is nearly attained, and this is visible at finite N . On P_4 , $D'_3 = 4.077 > 4$, so the end-arc of 3 states never wins at first order, but in the plot of Lemma 3.7(c) its point $(2, \underline{\lambda}_3)$ lies only 0.0071 above the lower hull. Numerically, the exact law on P_4 has this end-arc as its mode for T in $(9.381, 12.502)$, $(12.176, 15.142)$, $(15.804, 18.443)$ and $(19.292, 21.590)$ at $N = 40, 72, 150$ and 300, a window that shrinks only slowly. We also computed C_4, C_5, C_6 and the paw of Remark 5.9 exactly at some N between 40 and 300. On these graphs every size that never wins has its point $(s-1, \underline{\lambda}_s)$ at least 0.077 above the lower hull (on the paw every size wins), and the exact phases came in the order that first order predicts. So at feasible N the order of the phases can differ from first order when such a point lies close to the hull.

5.3 Two triangles

Let Θ_ε have states $1, \dots, 6$, unit rates on the triangles $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, and one more link 3–4 of rate $\varepsilon > 0$. We call 1, 2, 5, 6 the inner states and put $p_A(x) = x^2 - (3 + \varepsilon)x + \varepsilon$. For small ε each triangle is a well-separated community. It holds the mode from $\tau \approx 1$ until $\tau \approx 9/\varepsilon$ (Proposition 5.8).

Proposition 5.8 (two triangles joined by a weak link). *Let the start have full support.*

- (a) $\lambda_A = \lambda_B = \frac{1}{2}(3 + \varepsilon - \sqrt{\varepsilon^2 + 2\varepsilon + 9}) = \frac{\varepsilon}{3} - \frac{2\varepsilon^2}{27} + O(\varepsilon^3)$ is the smaller root of p_A , and $0 < \lambda_A < 1$.

(b) The least exit rates by size are $\underline{\lambda}_1 = 2$ (the inner states), $\underline{\lambda}_2 = 1$ (the inner edges $\{1, 2\}$ and $\{5, 6\}$), $\underline{\lambda}_3 = \lambda_A$ (the triangles), $\underline{\lambda}_4 = \lambda_{\{1, 2, 3, 4\}}$ (and its mirror image $\{3, 4, 5, 6\}$), $\underline{\lambda}_5 = \lambda_{\{2, \dots, 6\}}$ (and its images under the symmetries), and $\underline{\lambda}_6 = 0$. Here $\underline{\lambda}_4$ and $\underline{\lambda}_5$ are the least roots of $p_4(x) = x^3 - (5 + 2\varepsilon)x^2 + (6 + 6\varepsilon)x - 2\varepsilon$ and $p_5(x) = x^3 - (4 + 2\varepsilon)x^2 + (3 + 4\varepsilon)x - \varepsilon$.

(c) Let $\varepsilon_1 = (64 + \sqrt{46})/60 = 1.17971$, let $\varepsilon_2 = 4.48108$ be the positive root of $\varepsilon^3 + 2\varepsilon^2 - 19\varepsilon - 45$, and put

$$\tau_A = \frac{1}{1 - \lambda_A} = \frac{1 + \varepsilon + \sqrt{\varepsilon^2 + 2\varepsilon + 9}}{4}, \quad \tau^* = \frac{3}{\lambda_A} = \frac{3}{2\varepsilon} \left(3 + \varepsilon + \sqrt{\varepsilon^2 + 2\varepsilon + 9} \right).$$

For $\varepsilon < \varepsilon_1$ the winners are the inner states, then the inner edges from $\tau = 1$, then the triangles from τ_A , and then $[6]$ from τ^* . For $\varepsilon_1 < \varepsilon < \varepsilon_2$ a phase of $\{1, 2, 3, 4\}$ and its mirror image comes between the triangles and $[6]$, with switches at $1/(\lambda_A - \underline{\lambda}_4)$ and $2/\underline{\lambda}_4$. For $\varepsilon > \varepsilon_2$ the triangles never win, and the inner edges pass to $\{1, 2, 3, 4\}$ and its mirror image at $2/(1 - \underline{\lambda}_4)$ and then to $[6]$ at $2/\underline{\lambda}_4$. As $\varepsilon \rightarrow 0$, $\tau_A = 1 + \frac{\varepsilon}{3} + O(\varepsilon^2)$ and $\tau^* = \frac{9}{\varepsilon} + 2 + \frac{2\varepsilon}{9} + O(\varepsilon^2)$.

Proof. (a) On vectors (a, a, b) the matrix $-Q_A$ acts as $(a, b) \mapsto (a - b, -2a + (2 + \varepsilon)b)$, with characteristic polynomial p_A , and $(1, -1, 0)$ has eigenvalue 3. As $p_A(0) = \varepsilon > 0$ and $p_A(1) = -2 < 0$, the smaller root of p_A lies in $(0, 1)$, and it is λ_A . The expansion follows from $\sqrt{9 + 2\varepsilon + \varepsilon^2} = 3 + \frac{\varepsilon}{3} + \frac{4\varepsilon^2}{27} + O(\varepsilon^3)$.

(b) The admissible faces are the connected ones. For a face F , $-Q_F$ is the Laplacian of the graph on F , which is positive semidefinite, plus the diagonal matrix of the rates out of F . So $\lambda_F \geq 1$ whenever every state of F has rate at least 1 out of F . This covers $\{3, 4\}$, the pairs $\{1, 3\}$, $\{2, 3\}$ and their mirror images (for which in fact $\lambda = \frac{1}{2}(4 + \varepsilon - \sqrt{\varepsilon^2 + 4}) > 1$), every connected triple other than A and B , and the sets $\{a, 3, 4, b\}$ with $a \in \{1, 2\}$ and $b \in \{5, 6\}$. The other connected faces are the inner states (exit rate 2, against $2 + \varepsilon$ for 3 and 4), the inner edges ($\lambda = 1$), A and B , $\{1, 2, 3, 4\}$ and its mirror image, the complements of the inner states, which are all images of $\{2, \dots, 6\}$, and $[6]$. Since $\{2, \dots, 6\}$ contains the mirror image $\{3, 4, 5, 6\}$ of $\{1, 2, 3, 4\}$, Lemma 3.7(b) gives $\underline{\lambda}_5 < \lambda_{\{1, 2, 3, 4\}} < \lambda_A < 1$. Comparing within each size gives (b), once we know the polynomials. An exact computer calculation (Section 9) gives $\det(xI + Q_F) = (x - 3)p_4(x)$ for $F = \{1, 2, 3, 4\}$ and $(x - 3)^2 p_5(x)$ for $F = \{2, \dots, 6\}$. These are determinants of size 4 and 5, and they can also be checked by hand. So $\underline{\lambda}_4$ and $\underline{\lambda}_5$ are the least roots of p_4 and p_5 .

(c) Put $w_s = (s - 1, \underline{\lambda}_s)$. Suppose that a convex piecewise linear function through some of the points w_s lies strictly below the others. Then by Theorem 3.6(a) and Lemma 3.7(c) the sizes at its corners win in turn. The switch between 2 consecutive corners is at $-1/\sigma$, where σ is the slope between them. We need 3 comparisons, namely (i) $\underline{\lambda}_5 > \underline{\lambda}_4/2$ for every $\varepsilon > 0$, (ii) $\underline{\lambda}_4 > \frac{2}{3}\lambda_A$ if and only if $\varepsilon < \varepsilon_1$, and (iii) $\lambda_A < \frac{1}{2}(1 + \underline{\lambda}_4)$ if and only if $\varepsilon < \varepsilon_2$. The 3 differences are continuous in ε . If one of them vanishes, then $\underline{\lambda}_5$ is a common root of $p_5(x)$ and $p_4(2x)$, or $\underline{\lambda}_4$ is a common root of $p_4(x)$ and $p_A(\frac{3}{2}x)$, or $\underline{\lambda}_4$ is a common root of $p_4(x)$ and $p_A(\frac{1+x}{2})$. An exact computer calculation (Section 9) gives the resultants

$$\begin{aligned} \text{Res}_x(p_5(x), p_4(2x)) &= -8\varepsilon^3(16\varepsilon^2 + 12\varepsilon + 9), \\ \text{Res}_x(p_A(\tfrac{3}{2}x), p_4(x)) &= -\tfrac{1}{16}\varepsilon^2(120\varepsilon^2 - 256\varepsilon + 135), \\ \text{Res}_x(p_A(\tfrac{1+x}{2}), p_4(x)) &= \tfrac{1}{8}(\varepsilon^3 + 2\varepsilon^2 - 19\varepsilon - 45). \end{aligned}$$

So the difference in (i) never vanishes, the one in (ii) can vanish only at $\varepsilon_0 = (64 - \sqrt{46})/60 = 0.95363$ and at ε_1 , and the one in (iii) only at ε_2 , by Descartes' rule of signs.

We find the signs near 0 and ∞ . As $\varepsilon \rightarrow 0$, $\lambda_A \rightarrow 0$ and so $\underline{\lambda}_4, \underline{\lambda}_5 \rightarrow 0$. At $\varepsilon = 0$ the cubics are $x(x-2)(x-3)$ and $x(x-1)(x-3)$, so the implicit function theorem at the simple root 0 gives $\underline{\lambda}_4 = \frac{\varepsilon}{3} + O(\varepsilon^2)$ and $\underline{\lambda}_5 = \frac{\varepsilon}{3} + O(\varepsilon^2)$. With (a) the differences in (i), (ii) and (iii) are $\frac{\varepsilon}{6} + O(\varepsilon^2)$, $\frac{\varepsilon}{9} + O(\varepsilon^2)$ and $\frac{1}{2} + o(1)$, all positive. As $\varepsilon \rightarrow \infty$, $\lambda_A \rightarrow 1$, while $\underline{\lambda}_4 \leq (3 - \sqrt{5})/2 < \frac{1}{2}$ for every ε . This bound comes from Lemma 3.7(a) with the test vectors (a, a, c, c) on $\{1, 2, 3, 4\}$, whose quotient $((a-c)^2 + c^2)/(a^2 + c^2)$ does not involve ε and has least value $(3 - \sqrt{5})/2$. So the differences in (ii) and (iii) are negative for large ε . This proves (i) and (iii). It also proves (ii) once we know that the difference in (ii) is nonzero at ε_0 , since then ε_1 is its only possible zero. For that put $x_0 = 7/40$. Then

$$p_4(x_0) = \frac{57743}{64000} - \frac{809}{800}\varepsilon, \quad p_A(\frac{3}{2}x_0) = \frac{59}{80}\varepsilon - \frac{4599}{6400}$$

are both negative for $0.8923 < \varepsilon < 0.9743$, an interval that contains ε_0 . On $[0, \frac{1}{2}]$ we have $p'_4(x) \geq 1 + 4\varepsilon > 0$, and $p_4(0) < 0$, so p_4 has no root in $[0, x_0]$ and $\underline{\lambda}_4 > x_0$. Also $p_A(\frac{3}{2}x_0) < 0$ puts $\frac{3}{2}x_0$ between the roots of p_A , so $\lambda_A < \frac{3}{2}x_0$. Hence $\underline{\lambda}_4 > \frac{2}{3}\lambda_A$ at ε_0 .

Now we read off the hull. As $p_A(\frac{3}{4}) = \frac{\varepsilon}{4} - \frac{27}{16}$, we have $\lambda_A < \frac{3}{4}$ for $\varepsilon < \varepsilon_1 < \frac{27}{4}$. For $\varepsilon < \varepsilon_1$ the corners are w_1, w_2, w_3, w_6 , with slopes $-1, \lambda_A - 1$ and $-\lambda_A/3$, which increase because $0 < \lambda_A < \frac{3}{4}$. The segment w_3w_6 has height $\frac{2}{3}\lambda_A$ at $x = 3$ and $\frac{1}{3}\lambda_A$ at $x = 4$, so w_4 and w_5 lie above it by (i) and (ii). For $\varepsilon_1 < \varepsilon < \varepsilon_2$ the corners are w_1, w_2, w_3, w_4, w_6 , with slopes $-1, \lambda_A - 1, \underline{\lambda}_4 - \lambda_A$ and $-\underline{\lambda}_4/2$. These increase because $\lambda_A > 0$, then by (iii), then by (ii), and w_5 lies above w_4w_6 by (i). For $\varepsilon > \varepsilon_2$ the corners are w_1, w_2, w_4, w_6 , with slopes $-1, (\underline{\lambda}_4 - 1)/2$ and $-\underline{\lambda}_4/2$, which increase because $\underline{\lambda}_4 < \frac{1}{2}$. Here w_3 lies above w_2w_4 by (iii), and w_5 above w_4w_6 by (i). The switch values are minus the reciprocals of the slopes. The closed forms of τ_A and τ^* follow by rationalizing, and their expansions from (a). $\square$

So for small ε the inner edges win only on the short interval $(1, \tau_A)$, and a triangle then keeps the mode until τ^* . For $\varepsilon > \varepsilon_1$ the link is no longer weak enough, and the mode takes in the far end of the link before it spreads.

Remark 5.9 (the mode can leave a community). The winners need not grow around the first vertex. Take the paw, a triangle $\{1, 2, 3\}$ with a leaf 4 joined to 1, with unit rates and a start of full support. The single states have exit rates 3 for $\{1\}$, 2 for $\{2\}$ and $\{3\}$, and 1 for $\{4\}$. The connected pairs have $2 - \sqrt{2}$ for $\{1, 4\}$, 1 for $\{2, 3\}$ and $(5 - \sqrt{5})/2$ for $\{1, 2\}$ and $\{1, 3\}$. The triangle $\{1, 2, 3\}$ has $2 - \sqrt{3}$, from the vectors (c, e, e) with characteristic polynomial $x^2 - 4x + 1$. The triples $\{1, 2, 4\}$ and $\{1, 3, 4\}$ have $2 - 2\cos(2\pi/9) = 0.468$, the least root of $x^3 - 6x^2 + 9x - 3$. So $(\underline{\lambda}_1, \dots, \underline{\lambda}_4) = (1, 2 - \sqrt{2}, 2 - \sqrt{3}, 0)$. The slopes $1 - \sqrt{2} < \sqrt{2} - \sqrt{3} < \sqrt{3} - 2$ increase, since $(1 + \sqrt{3})^2 < 8$ and $(2 + \sqrt{2})^2 < 12$. So by Theorem 3.6(a) every size wins, in the order $\{4\}$, then $\{1, 4\}$ from $\tau = 1 + \sqrt{2}$, then $\{1, 2, 3\}$ from $\sqrt{2} + \sqrt{3}$, then all 4 states from $2 + \sqrt{3}$. The mode leaves the leaf, since $\{1, 4\} \not\subseteq \{1, 2, 3\}$.

6 Sticky priors for hidden Markov models

Fox, Sudderth, Jordan and Willsky [18] add a self-transition bias $\kappa \geq 0$ to the hierarchical Dirichlet process hidden Markov model (HDP-HMM), so that the hidden chain stays in a state for long stretches. With a finite vector β of base weights on d states, row j of the transition matrix has the Dirichlet law with parameters $\omega\beta_1, \dots, \omega\beta_j + \kappa, \dots, \omega\beta_d$ in the weak-limit approximation of [18]. Here $\omega > 0$ is the concentration. By the neutrality of the Dirichlet law (write it as independent gamma variables divided by their sum), the off-diagonal part of row j divided by $1 - \pi_{jj}$ is independent of π_{jj} and has the Dirichlet law with parameters $(\omega\beta_l)_{l \neq j}$. So κ only sets how sticky

the chain is, and the switching kernel is a random stochastic matrix with zero diagonal whose law does not involve κ . We idealize by fixing the exit probability of every state at $h = cL/N$, the scale of Kagey's crossing. In the prior itself $1 - \pi_{jj}$ has a Beta law and mixes all scales. That is a different regime, and we do not treat it.

Throughout this section $\Pi = (\pi_{ij})$ is a stochastic matrix on $[d]$ with zero diagonal, $c > 0$, and $h = cL/N$ with $0 < h < 1$. The transition matrix $(1 - h)I + h\Pi$ equals $I + hQ$ with $Q = \Pi - I$. So every exit rate is $q_i = 1$, $T = cL$ and $\tau = c$ (Example 2.5(iii)). We fix Π , which we think of as one draw from the prior, and we assume that the start μ has full support and that

$$\pi_{ij} > 0 \quad \text{for all } i \neq j. \quad (\text{A+})$$

Under the prior, (A+) holds with probability 1. For a face F let Π_F be the principal submatrix of Π on F and $\rho_F = \varrho(\Pi_F)$. Then $Q_F = \Pi_F - I$, so $\lambda_F = 1 - \rho_F$, and Q_F and Π_F have the same Perron vectors. A vertex has $\rho_F = 0$, and $\rho_{[d]} = 1$. If $|F| \geq 2$, then Π_F is irreducible by (A+), and $\rho_F > 0$ since ρ_F is at least the smallest row sum of Π_F . If also $F \neq [d]$, then $\rho_F < 1$, because the Perron root of an irreducible nonnegative matrix exceeds that of each proper principal submatrix [27]. With $\tau = c$ the exponent of a face is $\Phi_c(F) = c(1 - \rho_F) + |F| - 1$, and we put

$$c_- = \min_{|F| \geq 2} \frac{|F| - 1}{\rho_F}, \quad c_+ = \max_{F \neq [d]} \frac{d - |F|}{1 - \rho_F}.$$

We call a global mode of P_N a *MAP composition*. With no data the posterior is the prior, so a MAP composition is the most likely value of the occupation counts under the prior.

Corollary 6.1 (support and location of the MAP). *Assume (A+), let μ have full support, and fix $c > 0$. Let $\mathcal{F}_c^*$ be the set of faces that minimize Φ_c . There is $N_0(c)$ such that for $N \geq N_0(c)$ every MAP composition n^* satisfies $\text{supp } n^* \in \mathcal{F}_c^*$ and $n^*/N = \alpha_F^* + O(1/L)$ with $F = \text{supp } n^*$. If $\mathcal{F}_c^* = \{F^*\}$, then $n^*/N \rightarrow \alpha_{F^*}^* = l \circ r$, where l and r are the left and right Perron vectors of Π_{F^*} with $l \cdot r = 1$. No reversibility is needed.*

Proof. Every face is irreducible, since G_F is complete by (A+) when $|F| \geq 2$. Every face is also admissible, since a path that visits the states of F one after another has positive probability when μ has full support and $0 < h < 1$. So Theorem 3.3(iii) with $\tau = c$ gives $\text{supp } n^* \in \mathcal{F}_c^*$ for large N . Its assumption $hq_{\max} \leq \frac{1}{2}$ holds for large N . The faces of $\mathcal{F}_c^*$ are irreducible (as Theorem 3.6(c) also says), so Theorem 3.3(iv) gives $n^*/N = \alpha_F^* + O(1/L)$. For $|F| \geq 2$ the Perron vectors of Q_F are those of Π_F , so $\alpha_F^* = l \circ r$. For a vertex $\{i\}$, $n^* = Ne_i$. If $\mathcal{F}_c^* = \{F^*\}$, then $\text{supp } n^* = F^*$ and the error tends to 0. $\square$

The corollary is pointwise in c . At $c = c_-$ the vertices tie with the faces that attain the minimum in c_- . Every face is irreducible, so Theorem 4.7 applies. Part (iii) of that theorem, with a fixed large $t = K$, shows that for large N the MAP is not a single state at $c = c_- (1 - \frac{\log L - 2K}{2L})$, which is below c_- . So $N_0(c) \rightarrow \infty$ as $c \uparrow c_-$. Theorem 4.7 also describes the window at c_+ , with the coefficient $\frac{1}{2}$ of $\log \log N$.

Proposition 6.2 (thresholds and the direct jump). *Assume (A+) and $d \geq 3$.*

- (a) *If $0 < c < c'$, F minimizes Φ_c and G minimizes $\Phi_{c'}$, then $|F| \leq |G|$ and $\lambda_F \geq \lambda_G$.*
- (b) *The minimizers of Φ_c are exactly the vertices if and only if $c < c_-$, and no vertex minimizes Φ_c if $c > c_-$. The face $[d]$ is the unique minimizer if and only if $c > c_+$, and $[d]$ does not minimize Φ_c if $c < c_+$.*

(c) $1 < c_- \leq d - 1 \leq c_+$.

(d) $c_- = d - 1$ if and only if $c_+ = d - 1$, and both hold if and only if

$$\rho_F \leq \frac{|F| - 1}{d - 1} \quad \text{for every face } F. \quad (7)$$

If (7) fails, then $c_- < d - 1 < c_+$, and for $c \in (c_-, c_+)$ every minimizer F of Φ_c has $2 \leq |F| \leq d - 1$.

Consequently, let μ have full support, fix c and let $N \geq N_0(c)$. If $c < c_-$, the MAP compositions are the Ne_i with $\mu_i = \max_l \mu_l$. If $c > c_+$, every MAP composition has full support. If $c_- < c < c_+$, every MAP composition has a support F with $2 \leq |F| \leq d - 1$, and $F = F^*$ if F^* is the unique minimizer of Φ_c .

Proof. (a) Every face is admissible, so this is Theorem 3.6(b).

(b) A vertex has $\Phi_c = c$. For $|F| \geq 2$, $\Phi_c(F) - c = \rho_F((|F| - 1)/\rho_F - c)$ with $\rho_F > 0$. So if $c < c_-$, every face of size at least 2 is worse than the vertices, which all tie. The face attaining the minimum in c_- ties with the vertices at $c = c_-$ and beats them for $c > c_-$. In the same way, for $F \neq [d]$ we have $\Phi_c(F) - (d - 1) = (1 - \rho_F)(c - (d - |F|)/(1 - \rho_F))$ with $1 - \rho_F > 0$, and $\Phi_c([d]) = d - 1$. This gives the claims about $[d]$, using the face that attains the maximum in c_+ .

(c) If $2 \leq |F| \leq d - 1$, then $\rho_F < 1$ gives $(|F| - 1)/\rho_F > |F| - 1 \geq 1$, and $F = [d]$ gives $d - 1 \geq 2$. So $c_- > 1$. The face $[d]$ gives $c_- \leq d - 1$, and a vertex gives $c_+ \geq d - 1$.

(d) By (c), $c_+ = d - 1$ if and only if $(d - |F|)/(1 - \rho_F) \leq d - 1$ for all $F \neq [d]$, and $c_- = d - 1$ if and only if $(|F| - 1)/\rho_F \geq d - 1$ for all F with $|F| \geq 2$. Both inequalities say $(d - 1)\rho_F \leq |F| - 1$, which holds for every vertex ($\rho_F = 0$) and for $[d]$ ($\rho_F = 1$). This proves the equivalences. If (7) fails, (c) gives $c_- < d - 1 < c_+$, and for $c \in (c_-, c_+)$ part (b) excludes the vertices and $[d]$.

The consequences follow from Corollary 6.1, since $P_N(Ne_i) = \mu_i(1 - h)^{N-1}$. $\square$

We say that Π has a *direct jump* when (7) holds. Then at first order the MAP goes from one state straight to all d states at $c = d - 1$. The chance that m switches in a row from inside F all land in F is of order ρ_F^m . For the uniform kernel $\pi_{ij} = 1/(d - 1)$ every row sum of Π_F , and hence ρ_F , equals $(|F| - 1)/(d - 1)$. So (7) says that no set of states is more closed than under the uniform kernel. For that kernel all faces tie at $c = d - 1$, which is the degeneracy of Proposition 5.1(a), since $Q = (J - dI)/(d - 1)$ is the generator of Paper B [3] divided by $d - 1$ (Example 2.5(ii)).

Proposition 6.3 (when the direct jump happens). (a) Let $d \geq 3$, assume (A+), and let Π be symmetric. Then Π has a direct jump if and only if $\pi_{ij} = 1/(d - 1)$ for all $i \neq j$.

(b) Let $d = 3$ and assume (A+). Then Π has a direct jump if and only if $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$ for all 3 pairs. For example, the cyclic kernel $\pi_{i,i+1} = a$, $\pi_{i,i-1} = 1 - a$ (indices mod 3) has a direct jump for every $0 < a < 1$.

(c) Let $d = 3$ and let the base weights be uniform. In the weak-limit sticky HDP-HMM the rows of Π are then independent, and the first off-diagonal entry of each row has the law $\text{Beta}(\omega/3, \omega/3)$. Write $\mathbb{P}_\omega$ for this law of Π and D for the event that Π has a direct jump. Then $\mathbb{P}_\omega(D) \rightarrow \frac{1}{4}$ as $\omega \rightarrow 0$.

(d) In the setting of (c), $\mathbb{P}_\omega(D) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$ as $\omega \rightarrow \infty$.

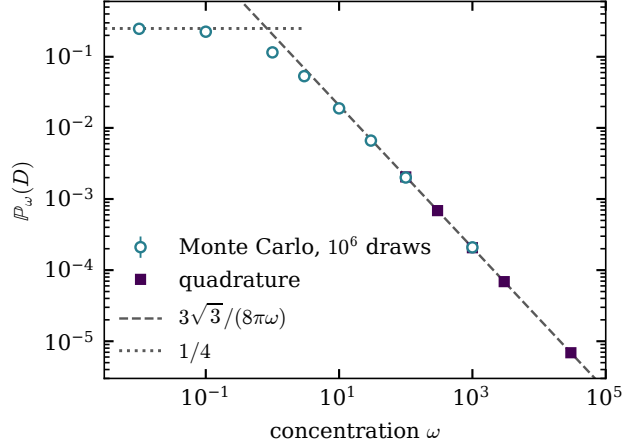


Figure 3: The probability $\mathbb{P}_\omega(D)$ that a random sticky kernel on 3 states with uniform base weights has a direct jump (Proposition 6.3), against the concentration ω . The open circles are Monte Carlo estimates with 10^6 draws, and their error bars of one standard error are smaller than the circles. The filled squares are quadrature values for $\omega \geq 100$. The dotted line is the limit $\frac{1}{4}$ as $\omega \rightarrow 0$, and the dashed line is the asymptote $3\sqrt{3}/(8\pi\omega)$ of Proposition 6.3(d).

Proof. For a pair $\{i, j\}$ the eigenvalues of $\Pi_{\{i, j\}}$ are $\pm\sqrt{\pi_{ij}\pi_{ji}}$, so $\rho_{\{i, j\}} = \sqrt{\pi_{ij}\pi_{ji}}$.

(a) Here $\rho_{\{i, j\}} = \pi_{ij}$. The rows sum to 1, so $\sum_{i < j} \pi_{ij} = d/2$, and the average of π_{ij} over the $\binom{d}{2}$ pairs is $1/(d-1)$. Unless all π_{ij} are equal, some pair has $\rho_{\{i, j\}} > 1/(d-1)$, and (7) fails. The uniform kernel satisfies (7) with equality, as shown above.

(b) For $d = 3$ the condition (7) is automatic for vertices and for [3], and for a pair it reads $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$. The cyclic kernel satisfies (A+) since $0 < a < 1$, and every pair has $\pi_{i, i+1}\pi_{i+1, i} = a(1-a) \leq \frac{1}{4}$.

(c) The law of Π comes from the neutrality property above. Let B_1, B_2, B_3 be independent with law Beta($\omega/3, \omega/3$) and put $\pi_{12} = B_1$, $\pi_{13} = 1 - B_1$, $\pi_{21} = B_2$, $\pi_{23} = 1 - B_2$, $\pi_{31} = B_3$ and $\pi_{32} = 1 - B_3$. Then (A+) holds with probability 1, and by (b) the event D is $\{B_1B_2 \leq \frac{1}{4}, (1-B_1)B_3 \leq \frac{1}{4}, (1-B_2)(1-B_3) \leq \frac{1}{4}\}$. As $\omega \rightarrow 0$ each B_j converges in law to a fair coin on $\{0, 1\}$. Under the limit law the 3 products take only the values 0 and 1, so the boundary of D has limit probability 0, and $\mathbb{P}_\omega(D)$ converges to the limit probability of D . Of the 8 points of $\{0, 1\}^3$, only (1, 0, 1) and (0, 1, 0) make all 3 products 0. So the limit is $\frac{1}{4}$.

(d) This is proved in Appendix C. $\square$

The 2 points in (c) are the directed 3-cycles, the only deterministic kernels in which no pair of states traps the chain. By (a) and (d), the tie of all faces that comes from the uniform kernel is not typical. Numerically $\mathbb{P}_\omega(D) = 0.115$ at $\omega = 1$ (Monte Carlo with 10^6 draws). So with probability about 0.88 a random kernel has a window of c in which the MAP sits on a pair of states (Proposition 6.2(d)). Figure 3 shows $\mathbb{P}_\omega(D)$ against ω . Quadrature values suggest $\mathbb{P}_\omega(D) = \frac{3\sqrt{3}}{8\pi\omega}(1 - \frac{9}{8\omega} + O(\omega^{-2}))$, but we have not proved the second term.

When the first switch of the MAP goes from one state to a symmetric pair, it has an exact description at every N . We use $S_{a,b}$, S_N , u_N , p_N and c_0 from Section 4.4.

Proposition 6.4 (an exact link to Kagey's crossing). *Let $i \neq j$ satisfy $\pi_{ij} = \pi_{ji} = \pi_0 > 0$ and*

$\mu_i = \mu_j$. We do not assume (A+). For $1 \leq m \leq N-1$,

$$\frac{P_N(me_i + (N-m)e_j)}{P_N(Ne_i)} = S_{m,N-m}(u'), \quad u' = \frac{h\pi_0}{1-h}.$$

- (a) Among the compositions with support exactly $\{i, j\}$, the most likely ones are the balanced ones, with $m = \lfloor N/2 \rfloor$ or $m = \lceil N/2 \rceil$.
- (b) The balanced composition $\lfloor N/2 \rfloor e_i + \lceil N/2 \rceil e_j$ is more likely than Ne_i if and only if $u' > u_N$, that is, if and only if $c > c^{\text{pair}}(N) := Nu_N/((\pi_0 + u_N)L)$.
- (c) If also $\mu_i = \mu_j = \max_l \mu_l$, then for every $N \geq 2$ and every c with $c^{\text{pair}}(N) < c < N/L$, no MAP composition is a single state.

Moreover,

$$c^{\text{pair}}(N) = \frac{1}{\pi_0} \left[1 - \frac{\frac{1}{2} \log L - c_0}{L} + O\left(\frac{\log L}{L^2}\right) \right],$$

which tends to the first-order value $1/\rho_{\{i,j\}} = 1/\pi_0$.

Proof. Every exit rate is 1, so the identity is Proposition 4.11 with $a = \pi_0$ and $q = 1$, whose condition $qh < 1$ is $h < 1$. Note that $P_N(Ne_i) = \mu_i(1-h)^{N-1} > 0$ and $u' > 0$.

(a) By [2, Proposition 6.1], if $a < b-1$, then $S_{a+1,b-1}$ dominates $S_{a,b}$ coefficientwise with at least one strict coefficient. So $S_{m+1,N-m-1}(u') > S_{m,N-m}(u')$ whenever $2m < N-1$, and $S_{m,N-m}(u')$ strictly increases in m up to $m = \lfloor N/2 \rfloor$. The run formula is symmetric, so $S_{a,b} = S_{b,a}$, and $S_{m,N-m}(u')$ strictly decreases from $m = \lceil N/2 \rceil$ on.

(b) For $m = \lfloor N/2 \rfloor$ the ratio is $S_N(u')$. By [2, Theorem 5.1] and its proof, S_N is strictly increasing on $[0, \infty)$ with the unique root u_N of $S_N(u) = 1$. So the balanced composition wins if and only if $u' > u_N$, that is $h\pi_0 > u_N(1-h)$, which is $c > c^{\text{pair}}(N)$.

(c) Every vertex has $P_N(Ne_l) = \mu_l(1-h)^{N-1} \leq P_N(Ne_i)$, and by (b) $P_N(Ne_i)$ is less than the probability of the balanced composition. The range of c is not empty since $u_N/(\pi_0 + u_N) < 1$.

For the expansion, [2, Theorem 5.2] gives $N(1-p_N) = L - \frac{1}{2} \log L + c_0 + O(\log L/L)$. Since $u_N = (1-p_N)/p_N$ and $1-p_N = O(L/N)$, also $Nu_N = L - \frac{1}{2} \log L + c_0 + O(\log L/L)$ and $u_N = O(L/N)$. Then $c^{\text{pair}}(N) = \frac{Nu_N}{\pi_0 L} (1+u_N/\pi_0)^{-1}$ gives the claim, since $L/N = O(\log L/L^2)$. $\square$

So for a sticky prior, one state against one pair with $\pi_{ij} = \pi_{ji}$ is exactly Kagey's crossing, at every N . With a uniform start, part (c) says that the MAP stops being a single state no later than $c^{\text{pair}}(N)$.

The last result says where the prior mass is. Put $\Lambda_F = 1 - h\lambda_F$.

Proposition 6.5 (face masses). *Assume (A+) and $0 < h < 1$. For every face F with $|F| \geq 2$,*

$$\frac{\min r_F}{\max r_F} \mu(F) \Lambda_F^{N-1} \leq \mathbb{P}(\text{supp } n \subseteq F) \leq \frac{\max r_F}{\min r_F} \mu(F) \Lambda_F^{N-1},$$

and for fixed c , $\Lambda_F^{N-1} = N^{-c\lambda_F} e^{O(L^2/N)}$. Also $\mathbb{P}(\text{one state}) = (1-h)^{N-1} \sim N^{-c}$ and $\mathbb{P}(\text{supp } n = [d]) \rightarrow 1$.

Proof. Let $U_F = (1-h)I + h\Pi_F$, which is nonnegative since $h < 1$. Then $\mathbb{P}(\text{supp } n \subseteq F) = \mu_F^\top U_F^{N-1} \mathbf{1}$, where μ_F is the restriction of μ to F . The Perron vector $r_F > 0$ of Π_F satisfies $U_F r_F = \Lambda_F r_F$. Entrywise $r_F / \max r_F \leq \mathbf{1} \leq r_F / \min r_F$, and U_F^{N-1} preserves these inequalities. So $U_F^{N-1} \mathbf{1}$ lies between $\Lambda_F^{N-1} r_F / \max r_F$ and $\Lambda_F^{N-1} r_F / \min r_F$, and pairing with μ_F gives the

(a) Switches of the MAP support along c					
N		60	120	240	480
first switch, vertex to $\{1, 2\}$		0.8248	0.8477	0.8673	0.8838
$c^{\text{pair}}(N)$		0.8248	0.8477	0.8673	0.8838
second switch, $\{1, 2\}$ to $[3]$		4.804	5.340	5.779	6.139

(b) The MAP and the prior mass at $N = 240$					
c	MAP composition	one state	$\{1, 2\}$	other pairs	$[3]$
0.5	one state (3-way tie)	0.0643	0.465	0.092	0.379
0.8	one state (3-way tie)	0.0122	0.423	0.034	0.531
3	(120, 120, 0)	4.3×10^{-8}	0.129	3.3×10^{-6}	0.871
8	(111, 111, 18)	1.2×10^{-21}	0.0081	2.1×10^{-16}	0.992

Table 2: The sticky chain with kernel Π_1 and the uniform start, for which $c_- = 10/9$ and $c_+ = 10$. (a) The 2 switches of the MAP support along c , from exact dynamic programming, and the exact value $c^{\text{pair}}(N)$ of Proposition 6.4. In these computations the faces $\{1, 3\}$ and $\{2, 3\}$ never hold the MAP. (b) At $N = 240$, the MAP composition and the prior probabilities of the possible supports. The column for the other pairs gives the total for $\{1, 3\}$ and $\{2, 3\}$.

bounds. Next, $(N-1)\log(1-h\lambda_F) = -(N-1)h\lambda_F + O(Nh^2) = -c\lambda_F L + O(L^2/N)$. A vertex composition is a single path, so $\mathbb{P}(\text{one state}) = \sum_i \mu_i (1-h)^{N-1} = (1-h)^{N-1}$. Finally, every $F \neq [d]$ has $\lambda_F > 0$, so $\mathbb{P}(\text{supp } n \neq [d])$ is at most a finite sum of terms $O(N^{-c\lambda_F})$. $\square$

Remark 6.6 (MAP versus mass). Let $d \geq 3$. For fixed $c < c_+$ the face $[d]$ does not minimize Φ_c , so for large N every MAP composition has a proper support, while $\mathbb{P}(\text{supp } n = [d]) \rightarrow 1$. For $c < c_-$ the most likely composition uses one state, and yet that event has probability about N^{-c} . The reason is that a composition on a face of size k is one of about N^{k-1} lattice points. So a face can carry most of the mass while each of its compositions is less likely than a vertex. Table 2 shows this for the kernel Π_1 with rows $(0, 0.9, 0.1)$, $(0.9, 0, 0.1)$, $(0.5, 0.5, 0)$ and the uniform start. Its pair $\{1, 2\}$ is a well-separated community with $\rho_{\{1,2\}} = 0.9$. In part (a) the first switch of the MAP agrees with $c^{\text{pair}}(N)$ to the digits shown. In part (b), at $c = 0.8$, the MAP is a single state (all 3 vertices tie), yet $\mathbb{P}(\text{one state}) = 0.0122$. At $c = 8$ the first order puts the MAP on $\{1, 2\}$, but at $N = 240$ the second switch has already happened. So the MAP is a poor summary of a sticky prior, and face selection says when it misleads.

7 The planar persistent walk

Kagey's Galton board has a natural version in 2 dimensions. A ball moves on $\mathbb{Z}^2$ in the directions east, north, west and south, numbered 0, 1, 2, 3 modulo 4, with unit vectors $e_0 = (1, 0)$, $e_1 = (0, 1)$, $e_2 = -e_0$ and $e_3 = -e_1$. Let $r, s \geq 0$ with $\beta = 2r + s > 0$, and let $0 < h < 1/\beta$. The first direction is uniform. After that the ball keeps its direction with probability $p_0 = 1 - \beta h$, turns left or right with probability rh each, and reverses with probability sh . If D_t is the direction of step t , the position is $Y_N = \sum_{t=1}^N e_{D_t}$. This is the chain of Example 2.5(iv), with $q_{j,j\pm 1} = r$ and $q_{j,j+2} = s$, and $Y_N = (n_0 - n_2, n_1 - n_3)$. In this section we allow every $h < 1/\beta$, and N is even. Only the path that always goes east reaches the corner Ne_0 , so $c_N = \mathbb{P}(Y_N = Ne_0) = \frac{1}{4}p_0^{N-1}$. We compare it with $o_N = \mathbb{P}(Y_N = 0)$. We write $u = sh/p_0$, $v = rh/p_0$ and $w = \beta h/p_0 = 2v + u$, so $1 + w = 1/p_0$. We use S_N , u_N , p_N and c_0 from Section 4.4. So $S_N(u)$ is the sum of u^j over the words of length

N with $N/2$ letters of each of 2 kinds, where j is the number of switches. It has nonnegative coefficients and no constant term, so it increases strictly on $[0, \infty)$.

We first guess the answer from the face exponents of Section 3, up to powers of $\log N$. Let $h = \tau \log N/N$. The corner is the vertex $\{0\}$, which has exit rate β , so c_N is about $N^{-\beta\tau}$. A vertex, a face with 3 directions, or a face with 2 adjacent ones misses the opposite of one of its directions, so it cannot reach the origin. On the axis $\{0, 2\}$, which has exit rate $2r$, the origin is the single composition $(N/2, 0, N/2, 0)$, of probability about $N^{-(2r\tau+1)}$. On the full face a single composition has probability about N^{-3} , but about $N/2$ compositions (a, b, a, b) end at the origin, and together they give about N^{-2} . So the origin should beat the corner when $\beta\tau > \min(2r\tau + 1, 2)$, that is when $\tau > \tau^* = \min(2/\beta, 1/s)$. At τ^* the axes win if $s > 2r$ and the full face wins if $s < 2r$. This applies Theorem 3.3 to a linear function of the composition, which we have not proved in general, so it is only a heuristic. Theorem 7.3 proves the conclusion directly.

Lemma 7.1 (three levels). *Write a sequence of directions as its number m of switches, its skeleton $Z = (Z_0, \dots, Z_m)$, which lists the directions of the runs so that $Z_t \neq Z_{t-1}$, and its run lengths $\ell = (\ell_0, \dots, \ell_m)$, which form a composition of N into $m+1$ positive parts. This is a bijection, and $Y_N = \sum_{t=0}^m \ell_t e_{Z_t}$. Under the walk, (a) m has the law $\text{Bin}(N-1, \beta h)$. Given m , (b) Z and ℓ are independent, (c) Z is the random walk on $\mathbb{Z}_4$ with uniform Z_0 whose independent increments are ± 1 with probability r/β each and 2 with probability s/β , and (d) ℓ is uniform on the $\binom{N-1}{m}$ compositions.*

Proof. Put $q(\pm 1) = r$ and $q(2) = s$. The sequence with data (m, Z, ℓ) has probability

$$\frac{1}{4} p_0^{N-1-m} \prod_{t=1}^m h q(Z_t - Z_{t-1}) = \binom{N-1}{m} (\beta h)^m p_0^{N-1-m} \cdot \frac{1}{4} \prod_{t=1}^m \frac{q(Z_t - Z_{t-1})}{\beta} \cdot \binom{N-1}{m}^{-1}.$$

For fixed m the middle factor is the probability of Z under the walk in (c), so it sums to 1 over the skeletons, and the last factor sums to 1 over the compositions. So the 3 factors are the laws in (a), (c) and (d), and the product form gives (b). $\square$

Proposition 7.2 (exact decomposition and a unique crossing). *Let $N \geq 2$ be even and $0 < h < 1/\beta$. Let $o_N^{(0)}$ and $o_N^{(4)}$ be the probabilities that $Y_N = 0$ and the walk never turns, and that $Y_N = 0$ and the walk uses all 4 directions. Then $o_N = o_N^{(0)} + o_N^{(4)}$,*

$$o_N^{(0)} = 2S_N(u) c_N, \quad 0 \leq o_N^{(4)} \leq \frac{w^2}{2p_0} (1 - p_0^N).$$

Equivalently, $o_N/c_N = 2S_N(u) + R_N$ with $0 \leq R_N \leq 2w^2((1+w)^N - 1)$. Moreover o_N/c_N is a polynomial in v and u with nonnegative coefficients and no constant term. If $s > 0$, or if $s = 0$ and $N \geq 4$, it increases strictly in h from 0 at $h = 0$ to ∞ as $h \uparrow 1/\beta$. So there is exactly one $h_N^ \in (0, 1/\beta)$ with $o_N = c_N$, and the corner is more likely than the origin exactly when $h < h_N^*$.*

Proof. An origin path with no turn stays on the axis of its first step. An origin path that turns uses both axes, and since $n_0 = n_2$ and $n_1 = n_3$ at the origin, it uses all 4 directions. So every origin path is of exactly one of the 2 kinds. A horizontal origin path with j switches, all of them reversals, has probability $\frac{1}{4} p_0^{N-1-j} (sh)^j = c_N u^j$. So these paths give $c_N S_N(u)$, and the vertical ones give the same. This is also Proposition 4.11 for the pairs $\{0, 2\}$ and $\{1, 3\}$. The bound on $o_N^{(4)}$ follows from Lemma 7.1 and a count of the run lengths that bring a skeleton back to the origin. We prove it in Appendix D. Dividing by $c_N = \frac{1}{4} p_0^{N-1}$ gives the form with R_N .

A path with a turns and b reversals has probability $\frac{1}{4}p_0^{N-1-a-b}(rh)^a(sh)^b = c_N v^a u^b$. So o_N/c_N is the sum of $v^a u^b$ over the origin paths, and $a + b \geq 1$ for each of them. Both u and v are nondecreasing in h , and each of them with a positive rate increases strictly to ∞ as $h \uparrow 1/\beta$. If $s > 0$, the path that goes east $N/2$ times and then west $N/2$ times gives the term u . If $s = 0$ and $N \geq 4$, then $r > 0$, and a path with 4 runs going east, north, west and south, of lengths $a, b, a, b \geq 1$, gives the term v^3 . This proves the claims on monotonicity and the limits. For $s = 0$ and $N = 2$ the origin cannot be reached. $\square$

We write $T_N^* = Nh_N^*$ and $u_N^* = sh_N^*/(1 - \beta h_N^*)$. Since $o_N = c_N$ at the crossing, $2S_N(u_N^*) = o_N^{(0)}/o_N$ there. We call it the zero-turn share of the origin mass.

Theorem 7.3 (first order). *Fix $r > 0$ and $s \geq 0$, put $\tau^* = \min(2/\beta, 1/s)$ with $1/0 = \infty$, and let $N \rightarrow \infty$ through even integers.*

- (a) *Fix $\tau > 0$ and put $h = \tau \log N/N$. If $\tau < \tau^*$, then $o_N/c_N \leq N^{-\gamma}$ for some $\gamma > 0$ and all large N . If $\tau > \tau^*$, then $o_N/c_N \geq N^\gamma$ for some $\gamma > 0$ and all large N .*
- (b) *$T_N^*/\log N \rightarrow \tau^*$.*
- (c) *If $s > 2r$, then $2S_N(u_N^*) \rightarrow 1$, so the origin mass at the crossing comes from the 2 axes, half from each. If $s < 2r$, then $2S_N(u_N^*) \leq N^{-\gamma}$ for some $\gamma > 0$ and all large N , so it comes from the paths that use all 4 directions.*

The proof is in Appendix D. Part (b) follows from (a) and the monotonicity in Proposition 7.2. The upper bound in (a) comes from [2, Eq. (6)] and Proposition 7.2, and the lower bound also uses Lemma 7.1 for the paths that turn.

When reversals dominate, the crossing is Paper A's with one change. Put

$$\mathcal{A}(\Lambda) = \Lambda - \frac{1}{2} \log \Lambda + c_0 + \frac{\frac{1}{4} \log \Lambda - \frac{1}{2} c_0 + \frac{1}{8}}{\Lambda},$$

so that [2, Eq. (8)] reads $N(1 - p_N) = \mathcal{A}(L) + O((\log L)^2/L^2)$.

Theorem 7.4 (reversals dominate). (a) *Let $s > 0$. For all even $N \geq 2$ and $0 < h < 1/\beta$,*

$$\frac{o_N}{c_N} = 2S_N(u)(1 + \varepsilon_N(h)), \quad 0 \leq \varepsilon_N(h) \leq \frac{w^2(1+w)^N}{S_N(u)}.$$

- (b) *Let $r \geq 0$ and $s > 2r$, and let $N \rightarrow \infty$ through even integers. Let z_N be the root of $z(I_0(z) + I_1(z)) = N/4$. Then $Nu_N^* = z_N + O(N^{-\gamma})$ for some $\gamma > 0$, and*

$$\varepsilon_N(h_N^*) = O((\log N)^{3/2-r/s} N^{-(s-2r)/s}), \quad z_N + \frac{1}{2} \log z_N = (L - \log 2) + c_0 + \frac{1}{8z_N} + O(z_N^{-2}).$$

- (c) *Under the hypotheses of (b),*

$$sT_N^* = \mathcal{A}(L - \log 2) + O\left(\frac{(\log L)^2}{L^2}\right), \quad sT_N^* - N(1 - p_N) = -\log 2 + \frac{\log 2}{2L} + O\left(\frac{(\log L)^2}{L^2}\right).$$

The equation in (b) is the one that Paper A solves for its crossing [2, proof of Thm. 5.2], with L replaced by $L - \log 2$. The reason is that the 2 axes double the mass at the origin, so e^{sT} must be smaller by a factor of about 2. Paper A also finds a shift of $\log 2$, between its periodic and free crossings [2, Thm. 7.2], but it has the opposite sign and a different cause. There a cyclic word must switch an even number of times, which makes the periodic ratio about half the free one near the crossing.

Proof. (a) Since $s > 0$, $S_N(u) \geq 2u > 0$. Divide Proposition 7.2 by $2S_N(u)$. Then $\varepsilon_N(h) = R_N/(2S_N(u))$ and $R_N \leq 2w^2(1+w)^N$.

(b) Put $\psi(z) = z(I_0(z) + I_1(z))$. Since $I'_0 = I_1$ and $(zI_1)' = zI_0$, we have $\psi' = \psi + I_0$. So $(\log \psi)' \geq 1$, ψ increases from 0 to ∞ , and z_N is well defined. Paper A writes h for $N/2$ in [2, Eq. (6)], while $h = T/N$ here. With $x = z/2$ that bound reads

$$\left(1 - \frac{z(z+2)}{2N}\right)\psi(z) \leq \frac{N}{2} S_N\left(\frac{z}{N}\right) \leq \psi(z) \quad (z > 0). \quad (8)$$

Write $u^* = u_N^*$, $w^* = \beta u^*/s$ and $R^* = R_N(h_N^*)$. At the crossing Proposition 7.2 gives $2S_N(u^*) + R^* = 1$. So $S_N(u^*) \leq \frac{1}{2} < S_N(u_N)$, and $u^* < u_N$. Put $w_N = \beta u_N/s$. Since $w \mapsto 2w^2(1+w)^N$ is increasing, $R^* \leq \eta_N := 2w_N^2(1+w_N)^N$. By [2, Eq. (7)], $Nu_N = N(1-p_N)/p_N = L - \frac{1}{2} \log L + O(1)$. So $(1+w_N)^N \leq e^{Nw_N} = O(N^{\beta/s} L^{-\beta/(2s)})$ and $w_N^2 = O(L^2/N^2)$. As $\beta/s - 2 = -(s-2r)/s$ and $2 - \beta/(2s) = \frac{3}{2} - \frac{r}{s}$, this gives $\eta_N = O(L^{3/2-r/s} N^{-(s-2r)/s})$, which tends to 0. Then $\varepsilon_N(h_N^*) = R^*/(1-R^*) \leq \eta_N/(1-\eta_N)$.

Next let $z^* = Nu^*$ and $\delta_N = Nu_N(Nu_N + 2)/(2N) = O(L^2/N)$. Since $z^* < Nu_N$ and $\frac{1}{2}(1-\eta_N) \leq S_N(u^*) \leq \frac{1}{2}$, (8) gives $\frac{N}{4}(1-\eta_N) \leq \psi(z^*) \leq \frac{N}{4}(1-\delta_N)^{-1}$. As $\psi(z_N) = N/4$ and $(\log \psi)' \geq 1$, we get $z_N + \log(1-\eta_N) \leq z^* \leq z_N - \log(1-\delta_N)$. So $z^* - z_N = O(\eta_N + \delta_N)$, which is $O(N^{-\gamma})$ for every $\gamma < \min(1, (s-2r)/s)$. Finally $I_0(z) + I_1(z) = \frac{2e^z}{\sqrt{2\pi z}}(1 - \frac{1}{8z} + O(z^{-2}))$ [29, Eq. 10.40.1], so

$$\log \psi(z) = z + \frac{1}{2} \log z + \frac{1}{2} \log \frac{2}{\pi} - \frac{1}{8z} + O(z^{-2}). \quad (9)$$

Since $c_0 = \frac{1}{2} \log \pi - \frac{3}{2} \log 2$, we have $\log(N/4) - \frac{1}{2} \log(2/\pi) = (L - \log 2) + c_0$. So by (9) the identity $\log \psi(z_N) = \log(N/4)$ is the equation in (b).

(c) Put $\Lambda = L - \log 2$, $d = -\frac{1}{2} \log \Lambda + c_0$ and $\tilde{z} = \mathcal{A}(\Lambda) = \Lambda + d + (\frac{1}{8} - \frac{d}{2})/\Lambda$. As in Paper A we use $\frac{1}{2} \log \tilde{z} = \frac{1}{2} \log \Lambda + \frac{d}{2\Lambda} + O((\log L)^2/L^2)$ and $1/(8\tilde{z}) = 1/(8\Lambda) + O(\log L/L^2)$. By (9) and the identity above, the terms of order 1 and $1/\Lambda$ in $\log \psi(\tilde{z}) - \log(N/4)$ are $d + \frac{1}{2} \log \Lambda - c_0 = 0$ and $(\frac{1}{8} - \frac{d}{2} + \frac{d}{2} - \frac{1}{8})/\Lambda = 0$. So $\log \psi(\tilde{z}) = \log(N/4) + O((\log L)^2/L^2)$, and $(\log \psi)' \geq 1$ gives $z_N = \tilde{z} + O((\log L)^2/L^2)$. Also $h_N^* = u^*/(s + \beta u^*)$, so $sT_N^* = Nu^*/(1 + \beta u^*/s) = Nu^* + O(L^2/N)$ since $u^* < u_N = O(L/N)$. With (b) this proves the first formula. For the second we subtract [2, Eq. (8)]. The last term of $\mathcal{A}$ has derivative $O(\log L/L^2)$ near L , so

$$\mathcal{A}(L - \log 2) - \mathcal{A}(L) = -\log 2 - \frac{1}{2} \log\left(1 - \frac{\log 2}{L}\right) + O\left(\frac{\log L}{L^2}\right) = -\log 2 + \frac{\log 2}{2L} + O\left(\frac{\log L}{L^2}\right). \quad \square$$

Numerically, for $(r, s) = (0.1, 1)$ the difference $sT_N^* - N(1-p_N)$ is -0.6317 at $N = 640$ and -0.6493 at $N = 10240$, while the first 2 terms in (c) give -0.6395 and -0.6556 (Figure 4(a)). When turns dominate, the origin mass comes from the paths with all 4 directions, and we need their local limit.

Theorem 7.5 (four-direction local limit). *Fix $r > 0$ and $s \geq 0$. Let $N \rightarrow \infty$ through even integers, let $T = T_N \rightarrow \infty$ with $T_N \leq N^{1/8}$, and put $h = T/N$. Then*

$$N^2 o_N^{(4)} = \frac{2(r+s)}{\pi} T (1 + o(1)).$$

The proof is in Appendix D. It uses Lemma 7.1, a local limit for the run lengths given the skeleton, and a central limit theorem for the skeleton. The functions $\cos(\pi j/2)$ and $\sin(\pi j/2)$ are eigenvectors of Q with eigenvalue $-2(r+s)$, so in this range the 2 coordinates of Y_N are uncorrelated with variance close to $N^2/(2(r+s)T)$. So Theorem 7.5 says that $o_N^{(4)}$ is the Gaussian density at 0

with this variance, times 2 because Y_N lies on a sublattice of index 2. It also gives the true size of the correction in Theorem 7.4(b). If $r > 0$ and $s > 2r$, then $T_N^* \sim L/s$ and $c_N^{-1} = 4e^{\beta T_N^*}(1 + o(1))$, so $\varepsilon_N(h_N^*) \sim o_N^{(4)}/c_N \sim \frac{8(r+s)}{\pi} T_N^* e^{\beta T_N^*} N^{-2}$. Since $e^{sT_N^*} = \frac{N}{2} \sqrt{\pi/(8sT_N^*)}(1 + o(1))$ by (9) and the proof above, this is of exact order $(\log N)^{1/2-r/s} N^{-(s-2r)/s}$. So the bound in Theorem 7.4(b) is too large only by a factor of order $\log N$.

Corollary 7.6 (turns dominate, and the switch). *Let $r > 0$ and let $N \rightarrow \infty$ through even integers.*

(a) *If $0 \leq s < 2r$, then $2S_N(u_N^*) \leq N^{-(2r-s)/\beta+o(1)}$, and the equivalent formulas*

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + o(1), \quad \beta T_N^* = 2N(1 - p_N) + \log \frac{\beta}{2(r+s)} + o(1)$$

hold.

(b) *If $s = 2r$, then $2S_N(u_N^*) \rightarrow \frac{2\sqrt{2}}{\sqrt{2}+\sqrt{5}} = 0.774852\dots$ and*

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + \log \frac{\sqrt{5} - \sqrt{2}}{\sqrt{5} + \sqrt{2}} + o(1).$$

Proof. Let $0 \leq s \leq 2r$. Then $\tau^* = 2/\beta$, so $T_N^*/L \rightarrow 2/\beta$ by Theorem 7.3(b), and Theorem 7.5 applies with $T_N = T_N^*$. At the crossing Proposition 7.2 gives $o_N^{(4)} = c_N(1 - 2S_N(u_N^*))$, and both sides are positive. Also $\log c_N = -\log 4 + (N-1)\log(1 - \beta h_N^*) = -\log 4 - \beta T_N^* + O(T_N^{*2}/N)$. Taking logarithms and using Theorem 7.5,

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + \log(1 - 2S_N(u_N^*)) + o(1). \quad (10)$$

(a) Since $I_1 \leq I_0 \leq e^z$ for $z \geq 0$, we have $\psi(z) \leq 2ze^z$, and (8) gives $2S_N(u) \leq 8ue^{Nu}$. At the crossing $Nu_N^* = sT_N^*/(1 - \beta h_N^*) = \frac{2s}{\beta}L(1 + o(1))$ and $u_N^* = O(L/N)$. So $2S_N(u_N^*) \leq N^{2s/\beta-1+o(1)}$, and $1 - 2s/\beta = (2r-s)/\beta$. Hence the last logarithm in (10) is $o(1)$. The equivalent form follows from $\log T_N^* = \log(2L/\beta) + o(1)$ and $2N(1 - p_N) = 2L - \log L + \log(\pi/8) + o(1)$ [2, Eq. (7)].

(b) Now $\beta = 2s$ and $2(r+s) = 3s$. Put $z^* = Nu_N^*$, so $z^* = sT_N^* + o(1) \rightarrow \infty$. By (8) and (9), $2S_N(u_N^*) = \frac{4}{N}\psi(z^*)(1 + o(1)) = \frac{8}{N}\sqrt{z^*/(2\pi)}e^{z^*}(1 + o(1))$. Let $q_N = Ne^{-sT_N^*}/\sqrt{T_N^*}$, $a = 8\sqrt{s/(2\pi)}$ and $b = 12s/\pi$. Then $2S_N(u_N^*) = (a/q_N)(1 + o(1))$, and Theorem 7.5 with $c_N^{-1} = 4e^{2sT_N^*}(1 + o(1))$ gives $o_N^{(4)}/c_N = (b/q_N^2)(1 + o(1))$. These 2 positive terms add up to 1. So q_N is bounded and bounded away from 0, and each limit point $\tilde{q}$ solves $\tilde{q}^2 - a\tilde{q} - b = 0$. As $a^2 = 32s/\pi$ and $a^2 + 4b = 80s/\pi$, the only positive root is $\tilde{q} = 2\sqrt{s/\pi}(\sqrt{2} + \sqrt{5})$. Hence $2S_N(u_N^*) \rightarrow a/\tilde{q} = 2\sqrt{2}/(\sqrt{2} + \sqrt{5})$, so $1 - 2S_N(u_N^*) \rightarrow (\sqrt{5} - \sqrt{2})/(\sqrt{5} + \sqrt{2})$, and (10) gives the display. $\square$

In (a) the coefficient of $\log T_N^*$ is 1, while Paper A has $\frac{1}{2}\log T$. The reason is that the origin is a point of a two-dimensional local limit. In 1 dimension the density at the center is of order $T^{1/2}/N$, and in 2 dimensions it is of order T/N^2 . Dividing by 2 gives $T_N^* = \frac{2}{\beta}[L - \frac{1}{2}\log L + O(1)]$. With Theorem 7.4(c) this shows that $T_N^* = \tau^*[L - \frac{1}{2}\log L + O(1)]$ for all $r > 0$ and $s \geq 0$. So the coefficient $\frac{1}{2}$ of $\log \log N$ in Theorem 4.6 appears here too, even when the origin mass comes from a line of compositions and not from a single one. At the switch $s = 2r$ neither kind of path wins, and the 2 shares tend to explicit constants. The approach is slow. For $(r, s) = (0.5, 1)$ the zero-turn share is 0.7650 at $N = 10240$, and for $(1, 1)$ it is still 0.1051 (Figure 4(b)). The errors in Theorem 7.5 and Corollary 7.6 are only $o(1)$. We state a heuristic next term in Section 9.

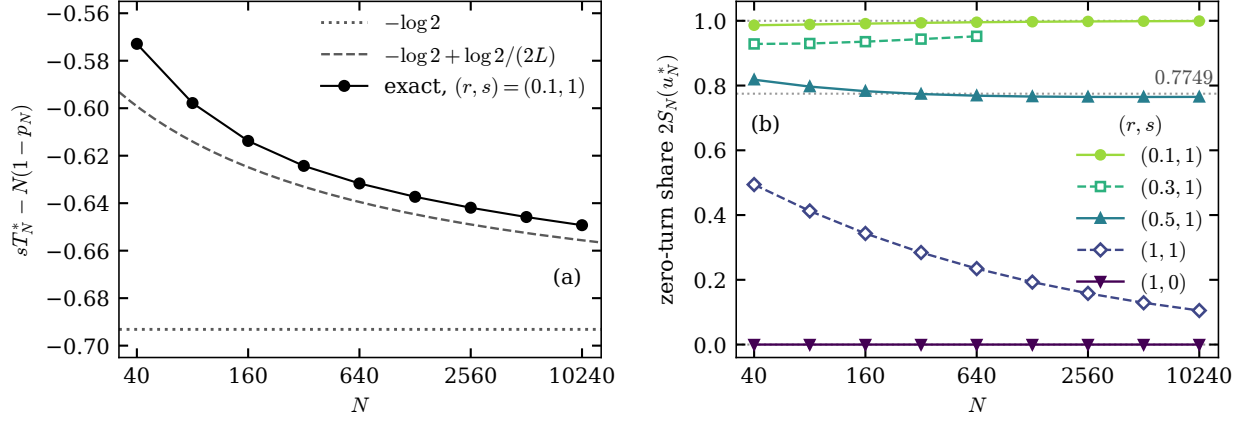


Figure 4: (a) The exact crossing of the planar walk with $(r, s) = (0.1, 1)$, written as $sT_N^* - N(1 - p_N)$, for $N = 40, 80, \dots, 10240$. Here p_N is the crossing of Paper A. The dashed curve is $-\log 2 + \log 2/(2L)$ from Theorem 7.4(c), and the dotted line is its limit $-\log 2$. (b) The zero-turn share $2S_N(u_N^*)$ of the origin mass at the crossing. It tends to 1 for $s > 2r$, to 0 for $s < 2r$, and to $2\sqrt{2}/(\sqrt{2} + \sqrt{5}) = 0.7749$ at the switch $s = 2r$ (Theorem 7.3 and Corollary 7.6). For $(r, s) = (1, 0)$ there are no reversals and the share is 0 for every N . The pair $(0.3, 1)$ was computed up to $N = 640$.

With turns only there is an exact reduction to Paper A. Rotate the lattice by 45° , that is, use $U = Y^{(1)} + Y^{(2)}$ and $V = Y^{(1)} - Y^{(2)}$, where $Y_N = (Y^{(1)}, Y^{(2)})$. Each step changes U and V by ± 1 . From any direction, a left turn changes the sign of the step of one of U and V and a right turn changes the other, while a reversal changes both. So when $s = 0$ the sign words of U and V are 2 copies of Paper A's walk that never switch at the same step, and $Y_N = 0$ exactly when both are balanced. This is the lattice form of the factorization in [24, Eq. (36)].

Proposition 7.7 (turns only). *Let $s = 0 < r$, let N be even, and put $x = rh/(1 - 2rh)$. For a word of length N in the letters $\pm$ with first letter σ , let $A \subseteq \{1, \dots, N-1\}$ be its set of switches, and call the word balanced if it has $N/2$ letters of each sign. Then $o_N/c_N = S_N(x)^2 - \Delta_N(x)$, where $\Delta_N(x) = \sum x^{|A|+|B|}$ over the pairs of balanced words (σ, A) and (σ', B) with $A \cap B \neq \emptyset$, and*

$$4x^2 \leq \Delta_N(x) \leq 4x^2 + 16x^3(1+x)^{N-2} + 16(N-1)x^4(1+x)^{2(N-2)}.$$

If x_N^* is the value of x at h_N^* ($N \geq 4$), then as $N \rightarrow \infty$

$$1 \leq \frac{x_N^*}{u_N} \leq 1 + \frac{\pi(\log N)^3}{N}(1 + o(1)), \quad rT_N^* = N(1 - p_N) + O\left(\frac{(\log N)^4}{N}\right).$$

The proof is in Appendix D. The term $4x^2$ comes from the 4 pairs with $A = B = \{N/2\}$, and the upper bound is a union bound over a common switch. At $s = 0$, Corollary 7.6(a) gives $rT_N^* = N(1 - p_N) + o(1)$, and Proposition 7.7 improves the error to $O((\log N)^4/N)$. Note that x_N^* and $rT_N^* = Nx_N^*/(1 + 2x_N^*)$ do not depend on r . Numerically the bound is far from sharp. At $N = 10240$, $x_N^*/u_N - 1 = 4.0 \times 10^{-4}$ and $rT_N^* - N(1 - p_N) = -0.0027$.

8 The information in the endpoint

In this section we return to the walk of Paper A, and the notation is local to the section. The first step is left or right with probability $\frac{1}{2}$ each, and each later step repeats the one before it with

probability $p = 1 - q$, where $0 < q < 1$. Put

$$\rho = 1 - 2q, \quad u = \frac{q}{p}, \quad x = Nq, \quad y = Nu = \frac{Nq}{p}.$$

Let J be the number of switches in the N steps, K the number of right steps, $X = 2K - N$ the endpoint and $W = X/N$. This is Example 2.5(i) with $h = q$, so x is the time horizon T . A path with J switches has probability $\frac{1}{2}p^{N-1-J}q^J$. So $J \sim \text{Bin}(N-1, q)$, the score of the path is $(J - (N-1)q)/(pq)$, and the path carries the Fisher information $\text{Var } J/(pq)^2 = (N-1)/(pq)$ about q . The endpoint carries $I_K(q) = \sum_{k=0}^N (\partial_q \mathbb{P}(K = k))^2 / \mathbb{P}(K = k)$, and we call $e_N(q) = I_K(q) pq / (N-1)$ the *efficiency* of the endpoint. It is the fraction of the information in the path that is left when we only see where the walk ends.

The endpoint sees the switches only through how far the walk spreads. If X were normal with mean 0 and variance Np/q , which is close to $\text{Var } X$ when x is large, it would carry the information $\frac{1}{2}(\partial_q \log(Np/q))^2 = 1/(2p^2q^2)$. This is the fraction $1/(2(N-1)pq) \approx 1/(2x)$ of the information in the path. It is only a heuristic, but Theorem 8.3 confirms the leading term $1/(2x)$. The leading term in the Gaussian approximation of Smiga, Radaelli, Binder and Landi [25] vanishes here, since the stationary law $(\frac{1}{2}, \frac{1}{2})$ does not depend on q . So in that approximation all of the endpoint information comes from the variance term that their leading-order formula drops.

Proposition 8.1 (a correlation ratio). *For $N \geq 2$ and $0 < q < 1$,*

$$e_N(q) = \frac{\text{Var}(\mathbb{E}[J \mid K])}{\text{Var } J}.$$

For $0 < k < N$ we have $\mathbb{E}[J \mid K = k] = \frac{d}{dt} \log S_{k, N-k}(e^t)$ at $t = \log(q/p)$, where $S_{a,b}$ is the polynomial of Section 4.4. For $k \in \{0, N\}$ we have $J = 0$.

Proof. For each k , $\mathbb{P}(K = k)$ is the sum of the probabilities of the paths with k right steps. Differentiating each term gives $\partial_q \log \mathbb{P}(K = k) = \mathbb{E}[(J - (N-1)q)/(pq) \mid K = k]$. Since $\mathbb{E}J = (N-1)q$, this gives $I_K(q) = \text{Var}(\mathbb{E}[J \mid K])/(pq)^2$, and we divide by $(N-1)/(pq) = \text{Var } J/(pq)^2$. By [2, Thm. 2.1], for $0 < k < N$ we have $\mathbb{P}(K = k, J = j) = \frac{1}{2}p^{N-1}W(N, k, j)u^j$, where $W(N, k, j)$ is the number of words with k letters R , $N-k$ letters L and j switches. Summing over j gives $\mathbb{P}(K = k) = \frac{1}{2}p^{N-1}S_{k, N-k}(u)$. Hence $\mathbb{E}[J \mid K = k] = uS'_{k, N-k}(u)/S_{k, N-k}(u)$, which is the stated derivative. The only paths with $K \in \{0, N\}$ are the 2 constant paths. $\square$

So e_N is the correlation ratio of J on K , and $e_N \leq 1$. Since $\mathbb{E}[J \mid K]$ is the orthogonal projection of J onto the functions of K in L^2 , we have $\text{Cov}(J, g(K)) = \text{Cov}(\mathbb{E}[J \mid K], g(K))$ for every function g . Then the Cauchy-Schwarz inequality gives

$$e_N(q) \geq \frac{\text{Cov}(J, g(K))^2}{\text{Var } J \text{Var } g(K)} \quad \text{whenever } \text{Var } g(K) > 0. \quad (11)$$

Proposition 8.2 (few switches). *For $N \geq 2$ and $0 < q < 1$,*

$$1 - (N-2)q \leq p^{N-2} \leq \frac{(N-1)qp^{N-2}}{1 - p^{N-1}} \leq e_N(q) \leq 1 - \frac{(N-2)qp^{N-3}}{2p + (N-2)q}.$$

Hence $e_N(q) = 1 - \frac{N-2}{2}q + O_N(q^2)$ as $q \rightarrow 0$ with N fixed. If $q \rightarrow 0$ and Nq stays bounded, then $e_N(q) \rightarrow 1$ if and only if $Nq \rightarrow 0$.

Proof. The first 2 inequalities are Bernoulli's inequality, in the forms $p^{N-2} \geq 1 - (N-2)q$ and $p^{N-1} \geq 1 - (N-1)q$. For the third, let $A = \mathbf{1}\{K \in \{0, N\}\}$. Only the constant paths have $K \in \{0, N\}$, so $A = \mathbf{1}\{J = 0\}$. Then $\text{Cov}(J, A) = -\mathbb{E}J\mathbb{E}A = -(N-1)qp^{N-1}$ and $\text{Var} A = p^{N-1}(1 - p^{N-1})$, and (11) with $g(K) = A$ gives the third inequality.

For the upper bound, $1 - e_N = \mathbb{E}[\text{Var}(J \mid K)] / \text{Var} J$ by the law of total variance, and $\mathbb{P}(K = k) \text{Var}(J \mid K = k)$ is the least value of $\mathbb{E}[(J - c)^2 \mathbf{1}\{K = k\}]$ over real c . For $0 < k < N$ there are 2 words with 1 switch and $N - 2$ words with 2 switches [2, Thm. 2.1]. So $a = \mathbb{P}(K = k, J = 1) = qp^{N-2}$ and $b = \mathbb{P}(K = k, J = 2) = \frac{1}{2}(N-2)q^2p^{N-3}$ do not depend on k . For every c we have $\mathbb{E}[(J - c)^2 \mathbf{1}\{K = k\}] \geq a(1 - c)^2 + b(2 - c)^2 \geq ab/(a + b)$, and summing over the $N - 1$ interior bins gives

$$\mathbb{E}[\text{Var}(J \mid K)] \geq \frac{(N-1)ab}{a+b} = \frac{(N-1)(N-2)q^2p^{N-2}}{2p + (N-2)q},$$

and we divide by $\text{Var} J = (N-1)pq$.

For fixed N , $1 - p^{N-1} = (N-1)q(1 - \frac{N-2}{2}q + O(q^2))$, so both bounds are $1 - \frac{N-2}{2}q + O(q^2)$. If $Nq \rightarrow 0$, then $e_N \geq 1 - (N-2)q \rightarrow 1$. Otherwise some subsequence has $Nq \rightarrow x \in (0, \infty)$ and $N \rightarrow \infty$, and along it the upper bound tends to $1 - xe^{-x}/(2+x) < 1$. $\square$

The reason is that 1 switch and 2 switches both leave the endpoint uniform on the interior bins. So the endpoint cannot tell them apart, and some information is lost as soon as 2 switches are likely.

Theorem 8.3 (many switches). *Let $R_N^2(q) = \text{Cov}(J, X^2)^2 / (\text{Var} J \text{Var} X^2)$ be the squared correlation of J and X^2 . For $N \geq 2$ and $0 < q < 1$,*

$$R_N^2(q) = \frac{1}{2(N-1)pq} \cdot \frac{[1 + \rho^N - p(1 - \rho^N)/x]^2}{V},$$

where

$$V = 1 - \frac{1 + 8\rho + \rho^2 + 8\rho^{N+1}}{4px} + \frac{\rho(1 - \rho^N)[2(1 + 2\rho)(2 + \rho) - \rho(1 - \rho^N)]}{8p^2x^2}.$$

There are absolute constants C and N_0 such that for $N \geq N_0$, $0 < q \leq \frac{1}{2}$ and $1 \leq x \leq N^{1/4}$,

$$0 \leq e_N(q) - R_N^2(q) \leq C\left(\frac{1}{x^3} + \frac{x^3}{N}\right).$$

Consequently, in the same range and with an absolute implied constant,

$$e_N(q) = \frac{1}{2x} + \frac{1}{4x^2} + O\left(\frac{1}{x^3} + \frac{x^3}{N}\right).$$

The lower bound $e_N \geq R_N^2$ is (11) with $g(K) = X^2$, and it holds for all $N \geq 2$ and $0 < q < 1$. The closed form and the upper bound are proved in Appendix E. The quadratic does well because Proposition 8.1 and Paper A's uniform Bessel estimate for every bin [2, Thm. 6.2] make $\mathbb{E}[J \mid K]$ close to $y\sqrt{1 - W^2} + \frac{1}{2} \approx y(1 - W^2/2) + \frac{1}{2}$. We do not compute C , and the error term we prove is small only for very large N , but numerically $e_N - R_N^2$ is already of order x^{-3} at $N = 50$ (Table 3).

Proof of the expansion in Theorem 8.3. If $x \leq 18$, then $|e_N - \frac{1}{2x} - \frac{1}{4x^2}| \leq 1 \leq 18^3 x^{-3}$, since e_N and $\frac{1}{2x} + \frac{1}{4x^2}$ lie in $[0, 1]$. Let $x \geq 18$. Then $p \geq \frac{1}{2}$, $0 \leq \rho < 1$ and $\rho^N \leq e^{-2x} \leq x^{-2}$. The middle term of

V lies in $[0, 9/x]$, and since $1 + 8\rho + \rho^2 = 10 - 20q + 4q^2$ it equals $\frac{5}{2x} + O(q/x + x^{-2})$. The last term lies in $[0, 9/x^2]$. So $V \geq \frac{1}{2}$, $1/V = 1 + \frac{5}{2x} + O(q/x + x^{-2})$, the squared bracket is $1 - \frac{2}{x} + O(q/x + x^{-2})$, and $1/(2(N-1)pq) = (1 + O(q + 1/N))/(2x)$. Multiplying, $R_N^2 = \frac{1}{2x} + \frac{1}{4x^2} + O(x^{-3} + q/x + 1/(Nx))$, and every implied constant is absolute. Since $q/x = 1/N \leq x^3/N$, the bound on $e_N - R_N^2$ gives the expansion. $\square$

Corollary 8.4 (at Kagey's crossing). *Let p_N be the value of p at which the middle bin and the endpoints are equally likely (Section 4.4), and put $q_N = 1 - p_N$ and $L = \log N$. Then*

$$e_N(q_N) = \frac{1}{2L} + \frac{\log L + 1 + \log(8/\pi)}{4L^2} + O\left(\frac{(\log L)^2}{L^3}\right).$$

Proof. By [2, Thm. 5.1], $p_N \geq \frac{2}{3}$, so $q_N \leq \frac{1}{3}$. By [2, Thm. 5.2, Eq. (7)], $x_N = Nq_N = L - \frac{1}{2} \log L + c_0 + O(\log L/L)$. So $1 \leq x_N \leq N^{1/4}$ for large N , and Theorem 8.3 gives $e_N(q_N) = \frac{1}{2x_N} + \frac{1}{4x_N^2} + O(L^{-3})$. Write $x_N = L(1 - \varepsilon)$ with $\varepsilon = (\frac{1}{2} \log L - c_0)/L + O(\log L/L^2)$. Then $\frac{1}{2x_N} = \frac{1}{2L} + \frac{\log L - 2c_0}{4L^2} + O((\log L)^2/L^3)$ and $\frac{1}{4x_N^2} = \frac{1}{4L^2} + O(\log L/L^3)$. Since $-2c_0 = \log(8/\pi)$, adding the 2 gives the claim. $\square$

So $e_N \sim 1/(2 \log N)$ at Kagey's crossing, but the limit is approached slowly. The relative correction is of order $(\log \log N)/\log N$, and Table 3 shows that it is large at every N we computed. The product $2e_N \log N$ is 1.74 at $N = 100$ and still 1.35 at $N = 3200$. Both R_N^2 and $\frac{1}{2x_N} + \frac{1}{4x_N^2}$ lie below e_N by an amount comparable to x_N^{-3} .

N	x_N	$e_N(q_N)$	$2x_N e_N$	$2Le_N$	$R_N^2(q_N)$	$\frac{1}{2x_N} + \frac{1}{4x_N^2}$
50	2.8998	0.24117	1.3987	1.8869	0.20873	0.20216
100	3.5034	0.18869	1.3221	1.7379	0.16642	0.16309
400	4.7501	0.12681	1.2047	1.5196	0.11715	0.11634
1600	6.0246	0.09433	1.1366	1.3919	0.09007	0.08988
3200	6.6684	0.08361	1.1151	1.3496	0.08070	0.08060

Table 3: The efficiency of the endpoint at Kagey's crossing $q_N = 1 - p_N$, computed from the exact law, with $x_N = Nq_N$ and $L = \log N$. The last 2 columns are the squared correlation of Theorem 8.3 and its two-term expansion.

Remark 8.5 (fixed q and fixed x). Let $Nq \rightarrow \infty$ with $q \leq q_0$ for a fixed $q_0 < 1$. Then $\liminf 2(N-1)pq e_N(q) \geq 1$. Indeed $e_N \geq R_N^2$, and $2(N-1)pq R_N^2$ is the squared bracket over V . Here $px \geq (1 - q_0)x \rightarrow \infty$, and $|\rho|^N \rightarrow 0$ because $\rho^N \leq e^{-2x}$ when $q \leq \frac{1}{2}$ and $|\rho| \leq 2q_0 - 1$ when $q > \frac{1}{2}$. The middle and last terms of V are at most $18/(4px)$ and $40/(8p^2x^2)$ in absolute value, so $V \rightarrow 1$. The bracket tends to 1 since $p(1 - \rho^N)/x \leq 2/x$. We have not proved the matching upper bound, but numerically the ratio tends to 1. At $q = 0.1$ it is 1.0113, 1.0055 and 1.0027 for $N = 400, 800$ and 1600. So at fixed q the efficiency is about $1/(2(N-1)pq)$, and $1/(2Nq)$ is too small by the factor $1/p$.

If instead $x = Nq$ is fixed and $N \rightarrow \infty$, Proposition 8.2 gives $x/(e^x - 1) \leq \liminf e_N \leq \limsup e_N \leq 1 - xe^{-x}/(2+x)$. We do not prove that e_N converges. Numerically it is close to the same efficiency $e_\infty(x)$ for the telegraph process on $[0, 1]$ that changes direction at rate x and is observed through the time it spends moving right. Along the crossing, $e_N/e_\infty(x_N)$ is 1.025 at $N = 100$ and 1.002 at $N = 3200$. The product $2xe_\infty(x)$ is 1.18, 1.40 and 1.34 at $x = 1, 2$ and 3, so for moderate x neither 1 nor $1/(2x)$ is a good approximation.

9 Verification and open questions

Three steps of the proofs rest on exact computer calculations. Apart from these and from decimal values of closed forms, the proofs are done by hand. In the proof of Lemma E.1, each moment that we need lies in a space of sequences in N of dimension at most 10, so its closed form follows from its values at $N = 1, \dots, 10$. The script `verify_sticky_fisher.py` computes these moments exactly for $N = 1, \dots, 12$, with q as a symbol, and checks with `sympy` the identities for $\text{Var } X^2$ and R_N^2 that follow from them. The proof of Proposition 5.8 uses 2 characteristic polynomials and 3 resultants of such polynomials, which `verify_examples.py` computes with `sympy`. The formula of Proposition 4.13(i) is proved by hand, and `verify_second_order.py` evaluates it exactly on the arc of 3 states and on the whole 4-cycle, which gives the values $\frac{5}{4}\sqrt{2} - 2$ and $-\frac{9}{16}$ in Remark 5.6.

All other computations are checks at finite size, in Python and in double precision unless we say otherwise. None of the proofs use them, and the statements that we call numerical rest on them alone. There is one script for each part of the paper. The script `verify_first_order.py` covers Sections 2 and 3. The script `verify_second_order.py` covers Section 4, Proposition 5.2 and Appendices A and B, and `verify_examples.py` covers the rest of Section 5. Sections 6 and 8 and Appendices C and E are checked by `verify_sticky_fisher.py`, and Section 7 and Appendix D by `verify_planar.py`. Each script computes the numbers in the paper in 2 independent ways where it can, and compares both with the printed values. For example, the run expansion of Lemma 2.3 agrees with a list of all words and with a dynamic program of the chain in exact rational arithmetic, and the torus representation behind Theorem 4.3 agrees with the run sum to 7×10^{-13} . On faces with 2, 3 and 4 states of a random generator the local law of Theorem 4.3 is accurate to 1% by $T = 40$, and the ratio in Lemma 4.4 is $1 - 2.7 \times 10^{-5}$ at $T = 10$ and $N = 10^6$. The constants K_F are computed from the tilted generator in double precision and from the Hessian of the Perron root in 30-digit arithmetic. We located the crossings $M_F = M_G$ on the 4-cycle from the exact face maxima up to $N = 10^{14}$, and they agree with Theorem 4.6 and with the next term in Proposition 4.13. For a vertex against an arc of 3 states at $N = 10^{14}$, the remainder in the crossing equation times T_N is -0.2333 , while $c_F - c_G = \frac{5}{4}\sqrt{2} - 2 = -0.2322$. The window of Proposition 5.2 was computed exactly up to $N = 10^{20}$, and its left end agrees to 3×10^{-5} with the second-order equation. For the directed 3-cycle of Proposition 4.12 the script follows the first crossing up to $N = 10^{12}$ and the second up to $N = 10^8$, and an exact dynamic program at $N = 300$ shows the 3 successive modes.

On the graphs of Section 5 the exact law of the composition comes from a dynamic program of the whole chain, which agrees with an enumeration of all d^N words in exact arithmetic to about 10^{-15} , and the face maxima come from the run expansion. For the planar walk the three-level form of Lemma 7.1, Fourier inversion, a dynamic program and an enumeration of all 4^N words agree to 2×10^{-13} for $N \leq 10$, and the crossings up to $N = 10240$ come from the three-level form. The sticky dynamic program agrees with an enumeration of all paths to 10^{-16} , and the probabilities of the direct jump come from Monte Carlo with 10^6 draws and from quadrature in 2 coordinate systems. The Fisher information is computed from the run formula of Paper A and from its recurrence with a complex step, and the 2 agree with exact rational arithmetic for $N \leq 14$ and with 30-digit arithmetic for $N \leq 400$. The script `verify_all.py` runs all 5 scripts in a few minutes, and with the option `--full` it also runs the largest cells. The scripts are in the folder `paperC` of the Problem 131 folder, and their output is in `paperC/data`. Before each new computation we wrote a prediction, and a condition that would refute it, in the ledgers `ledger_C1.md` to `ledger_C5.md` in `paperC/data`. The outcome is recorded below each prediction, also when it refuted the prediction. None of the refuted predictions concerns a statement that we

call a theorem.

Some of the finite and algebraic steps are also proved in Lean 4, in the folder `lean/PaperC` of the Problem 131 folder. For Sections 2 and 3 these are the run expansion of Lemma 2.3, the zero set in Example 3.5, the lower bound and the value at α_F^* in Proposition 3.1(a), the form in part (c) for symmetric Q at interior points, and the finite statements behind Theorem 3.6(b) and Lemma 3.7(c). For Section 4 they are the identity of Proposition 4.11, the exact parts of Proposition 4.12(i) and (ii), and the offsets b_{FG} in Corollaries 4.9 and 4.10. For Section 5 they are the exit rates of the faces of K_d , of the arcs of cycles and of the end-arcs of paths, Proposition 5.1(b)–(d), the window of Proposition 5.2(a) and (b), Lemma 5.3(b) and (c), the cascade of Theorem 5.4(b) and (c) among the arcs and the whole cycle, the constants K_{pair} and K_{arc3} of Remark 5.6, parts of Proposition 5.8, and the winners on the paw in Remark 5.9. For Sections 6 to 8 they are Propositions 6.2(b)–(d) and 6.3(a) and (b), the finite bounds of Proposition 6.5, the identity $o_N^{(0)} = 2S_N(u)c_N$ of Proposition 7.2, and Proposition 8.1. The header of each Lean file lists what it proves. The local laws, the crossings and the other asymptotic results are not in Lean.

Several questions remain open. Theorem 4.3 holds on compact subsets of the open face. When some $T\alpha_i$ is $O(1)$ there is a boundary layer, and the rare-coordinate crossover of Paper B is the case of the complete graph. A local law that holds uniformly up to the boundary of a face is open (Example 3.5). For a reducible face, Theorem 3.3(ii) only places the power of T in M_F between $k_F - k^*$ and $k_F - 1$. Heuristically it is $(k^* - 1)/2 + k_F - k^*$, and Proposition 4.12 shows that the coefficient of $\log \log N$ in a crossing then changes. We have no formula for c_F in terms of trees or cycles of the graph, and for non-reversible faces we have no tree formula for K_F like Proposition 4.2(b). Theorem 4.6 puts all zeros in an interval of length $O(1/L)$ but does not show that there is only one. Papers A and B and Proposition 7.2 prove uniqueness with polynomial arguments that do not extend to general faces. For linear functions of the composition, such as the other endpoints of the planar walk, face selection is only heuristic here. At finite N the order of the phases can differ from first order when a face that never wins lies close to the hull, as on the path with 4 states (Remark 5.7). We do not know a condition on the hull that rules this out. For the planar walk a heuristic expansion suggests that the next term of Theorem 7.5 comes from replacing T by $T + \frac{1}{4r} - \frac{1}{2(r+s)}$, and the crossover near $s = 2r$, where $|s - 2r| \log N = O(1)$, is open. For sticky priors we do not treat exit probabilities drawn at random as in [18], or the probability of the direct jump for $d \geq 4$, where triples of states enter the criterion. Finally, we do not compute the constant in Theorem 8.3, and at fixed q we prove only a lower bound for $2(N - 1)pq e_N(q)$.

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Data and code availability. The manuscript source, verification scripts, and recorded results are in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>.

A The torus representation and the local law

This appendix proves Lemma 3.2, Theorem 4.3 and Lemma 4.4. All 3 rest on a formula that writes the run sum $H(t)$ of Lemma 2.4 as an integral over a torus.

Throughout, F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$, and $\mathbf{1}$ is the all-ones vector of the appropriate size. We write i for the imaginary

unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \quad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j \geq 2}$, $b_1 = (q_{i1})_{i \geq 2}$ and $A' = (q_{ij})_{i,j \geq 2}$. For $z' \in \mathbb{C}^{k-1}$ let $Z' = \text{diag}(z')$. Where $Z' - A'$ is invertible we put

$$X(z') = (Z' - A')^{-1}, \quad g(z') = a_1^\top X(z') b_1, \quad b(z') = (\mu_1 + \mu'^\top X(z') b_1) (1 + a_1^\top X(z') \mathbf{1}).$$

We use 2 standard facts. First, let $B \geq 0$ be a square matrix and S a diagonal matrix with positive diagonal. Then $\varrho(B - S) < 0$ holds if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some vector $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. These are among the standard characterizations of nonsingular M-matrices [27, Ch. 6]. Second, if M is an irreducible Metzler matrix, then $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M , and every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$ [27].

Lemma A.1 (torus representation). *Let $s' \in (0, \infty)^{k-1}$ satisfy $\varrho(A' - \text{diag } s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \geq s_i$ for every $i \geq 2$, the matrix $Z' - A'$ is invertible, and X , g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. In particular $|g(z')| \leq g(s')$ and $|b(z')| \leq b(s')$ there. For every $t \in (0, \infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\left(t_1 g(z') + \sum_{i \geq 2} t_i z_i\right) b(z') \prod_{i \geq 2} z_i d\varphi, \quad z_i = s_i e^{i\varphi_i}.$$

Proof. Put $S' = \text{diag}(s')$. By the first fact, $\varrho(S'^{-1}A') < 1$. So the series

$$X(z') = \sum_{j \geq 0} (Z'^{-1}A')^j Z'^{-1}$$

converges absolutely for $|z_i| \geq s_i$, since its terms are bounded entrywise in modulus by those of the same series at $z' = s'$. This gives the claims about X , g and b , because a_1 , b_1 and μ are nonnegative.

Choose $s_1 > g(s')$, put $s = (s_1, s')$ and $S = \text{diag}(s)$. We show that $\varrho(S^{-1}A_F) < 1$. This is the Schur complement criterion for M-matrices, and we check it directly. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon \mathbf{1})$. Then $v' \geq S'^{-1}(b_1 + \varepsilon \mathbf{1}) > 0$, and

$$((S - A_F)v)_i = ((S' - A')v')_i - (b_1)_i = \varepsilon \quad (i \geq 2), \quad ((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^\top X(s') \mathbf{1}.$$

For small ε both are positive, so the first fact gives the claim.

Now let $Z = \text{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. Expanding $(Z - A_F)^{-1} = Z^{-1}(I - A_F Z^{-1})^{-1}$ in powers of $A_F Z^{-1}$ gives

$$\mu_F^\top (Z - A_F)^{-1} \mathbf{1} = \sum_c \mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}} \prod_{i \in F} z_i^{-m_i(c)},$$

where $c = c_1 \cdots c_M$ runs over all words in F with $M \geq 1$ and $c_j \neq c_{j+1}$, and $m_i(c)$ is the number of letters i in c . Since $\varrho(A_F S^{-1}) = \varrho(S^{-1}A_F) < 1$, the expansion is valid and the sum converges absolutely and uniformly on the torus. For an integer m , $t_i > 0$ and $z_i = s_i e^{i\varphi_i}$,

$$\frac{1}{2\pi} \int_0^{2\pi} e^{t_i z_i} z_i^{1-m} d\varphi_i = \frac{1}{2\pi i} \oint_{|z_i|=s_i} e^{t_i z_i} z_i^{-m} dz_i = \begin{cases} t_i^{m-1}/(m-1)!, & m \geq 1, \\ 0, & m \leq 0. \end{cases}$$

We integrate term by term. The words that miss a state of F give 0, and grouping the others by $m = (m_i(c))_{i \in F}$ gives $W(m) \prod_i t_i^{m_i-1}/(m_i-1)!$. Hence

$$H(t) = \frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} e^{\langle t, z \rangle} \mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i \in F} z_i d\varphi. \quad (12)$$

We do the integral over φ_1 first. Put $\zeta = z_1 - g(z')$, the Schur complement of $Z' - A'$ in $\mathcal{Z} - A_F$. It is not 0, because $|g(z')| \leq g(s') < s_1$. With $X = X(z')$, block inversion gives $(\mathcal{Z} - A_F)^{-1}$ with first row $\zeta^{-1}(1, a_1^\top X)$, first column $\zeta^{-1}(1, Xb_1)^\top$ and lower right block $X + \zeta^{-1}Xb_1a_1^\top X$. So

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^\top X(z') \mathbf{1}.$$

The second term does not depend on z_1 . So by Cauchy's integral formula the integral over φ_1 in (12), divided by 2π , equals $e^{t_1 g(z')} b(z')$ times the factors that do not involve z_1 . In other words $b(z')$ is the residue at the pole $z_1 = g(z')$. This proves the formula. $\square$

The reason for this form is positivity. All coefficients are nonnegative, so on the torus the modulus of the integrand is at most $\exp(t_1 g(s') + \sum_{i \geq 2} t_i s_i) b(s') \prod_{i \geq 2} s_i$, which is its value at $\varphi = 0$. This bound gives Lemma 3.2, and a Laplace expansion around $\varphi = 0$ gives Theorem 4.3.

A.1 Proof of Lemma 3.2

Let F be irreducible and $\theta \in V_F$. Then $A_F - \text{diag } \theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for the eigenvalue 0, and $\theta_i = (A_F u)_i / u_i$. Since G_F is strongly connected and $k \geq 2$, each row of A_F has a positive entry, so $\theta > 0$. By the second fact, $\varrho(A' - \text{diag } \theta') < 0$. The rows $i \geq 2$ of $(A_F - \text{diag } \theta)u = 0$ read $(\text{diag } \theta' - A')u' = u_1 b_1$, so $u' = u_1 X(\theta') b_1$. Then row 1 gives $\theta_1 u_1 = a_1^\top u' = u_1 g(\theta')$. Hence

$$\theta > 0, \quad \varrho(A' - \text{diag } \theta') < 0, \quad g(\theta') = \theta_1 \quad \text{for every } \theta \in V_F. \quad (13)$$

Now fix n with support F and h with $h q_{\max} \leq \frac{1}{2}$, and put $|n| = \sum_i n_i$ and $t = hn$. Lemma 2.3, applied with $|n|$ in place of N and the same h , gives the probability $P_{|n|}(n)$ that the first $|n|$ visits have composition n . Only terms with $m_i \leq n_i$ for every i are nonzero. In them we use $h^{m_i-1} \binom{n_i-1}{m_i-1} \leq t_i^{m_i-1}/(m_i-1)!$ and $(1 - h q_i)^{n_i-m_i} \leq e^{-q_i t_i} \sigma^{m_i}$, where $\sigma = e^{h q_{\max}}$. Note that $h^{M-1} = h^{k-1} \prod_i h^{m_i-1}$, and that $\sigma^M W(m) = \sigma W_\sigma(m)$, where W_σ is W computed with A_F replaced by σA_F . So

$$P_{|n|}(n) \leq \sigma h^{k-1} e^{-\langle q, t \rangle} H_\sigma(t), \quad (14)$$

where H_σ is H computed with σA_F in place of A_F . We bound $H_\sigma(t)$ by Lemma A.1 for σA_F with the radii $\sigma \theta'$. They satisfy its hypothesis, since $\varrho(\sigma A' - \text{diag } \sigma \theta') = \sigma \varrho(A' - \text{diag } \theta') < 0$. The functions of Lemma A.1 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. By (13) and the bounds in Lemma A.1, the integrand for σA_F has modulus at most $e^{\sigma \langle t, \theta \rangle} b(\theta') \sigma^{k-1} \prod_{i \geq 2} \theta_i$. With (14) this gives

$$P_{|n|}(n) \leq \sigma^k b(\theta') \prod_{i \geq 2} \theta_i h^{k-1} \exp(-h \langle n, q - \theta \rangle + (\sigma - 1) h \langle n, \theta \rangle).$$

Finally $\sigma \leq e^{1/2}$ and $\sigma - 1 \leq 2h q_{\max}$, so $(\sigma - 1) h \langle n, \theta \rangle \leq 2q_{\max} (\max_i \theta_i) h^2 |n|$. This proves the lemma with $C_\theta = \max(e^{k/2} b(\theta') \prod_{i \geq 2} \theta_i, 2q_{\max} \max_i \theta_i)$. For the bound stated after the lemma take $\theta = q - \lambda_F \mathbf{1}$. It lies in V_F because $\varrho(A_F - \text{diag } \theta) = \varrho(Q_F) + \lambda_F = 0$. Then $\langle n, q - \theta \rangle = \lambda_F |n|$, and for $|n| = N$ we have $h|n| = T$ and $h^2|n| = T^2/N$. $\square$

A.2 Proof of Theorem 4.3

For $k = 1$ the formula is part of Lemma 2.4. Let $k \geq 2$, fix $\alpha \in \Delta_F^\circ$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$ and $l = l_\alpha$ as in Definition 4.1, and $\Theta' = \text{diag}(\theta')$. By (13) we may use Lemma A.1 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1}e^{-T\langle q, \alpha \rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha, \varphi)} \chi(\alpha, \varphi) d\varphi, \quad (15)$$

where $z_i = \theta_i e^{i\varphi_i}$ and

$$\Psi(\alpha, \varphi) = \alpha_1 g(z') + \sum_{i \geq 2} \alpha_i z_i, \quad \chi(\alpha, \varphi) = b(z') \prod_{i \geq 2} z_i.$$

We check the hypotheses of Lemma B.1 with $m = k - 1$, phase Ψ and amplitude χ , and then compute the constant.

We first show that $\varphi = 0$ is the only maximum of $\text{Re } \Psi(\alpha, \cdot)$ on $\mathbb{T}^{k-1}$. By (13), $\Psi(\alpha, 0) = \alpha_1 \theta_1 + \sum_{i \geq 2} \alpha_i \theta_i = \langle \alpha, \theta \rangle$, which is real. By Lemma A.1, $|g(z')| \leq g(\theta') = \theta_1$ on the torus, so

$$\text{Re } \Psi(\alpha, \varphi) \leq \alpha_1 \theta_1 + \sum_{i \geq 2} \alpha_i \theta_i \cos \varphi_i = \Psi(\alpha, 0) - \sum_{i \geq 2} \alpha_i \theta_i (1 - \cos \varphi_i).$$

Every α_i is positive, and every θ_i is positive by (13). So the last sum is positive unless $\varphi = 0$ in $\mathbb{T}^{k-1}$, which is hypothesis (iii) of Lemma B.1. Note that the gain comes from the angles and not from g . For $k = 2$, $g(z_2) = q_{12}q_{21}/z_2$ has constant modulus on the circle. No aperiodicity condition is needed.

Next we compute the derivatives at $\varphi = 0$. For $z \in \mathbb{R}^F$ near θ let $\lambda(z) = \varrho(A_F - \text{diag } z)$, and for $\eta \in \mathbb{R}^F$ near 0 let $\Lambda(\eta) = \varrho(\hat{Q}_\alpha + \text{diag } \eta)$. Since $A_F - \text{diag}(\theta - \eta) = R(\hat{Q}_\alpha + \text{diag } \eta)R^{-1}$ with $R = \text{diag } r$, we have $\lambda(\theta - \eta) = \Lambda(\eta)$. The Perron root of the irreducible matrix $\hat{Q}_\alpha + \text{diag } \eta$ is a simple eigenvalue. So by the implicit function theorem, applied to the characteristic polynomial, Λ and the right eigenvector $v(\eta)$ with $\alpha^\top v(\eta) = 1$ are real-analytic near 0, with $\Lambda(0) = 0$ and $v(0) = \mathbf{1}$. Recall $Z\mathbf{1} = 0$, $\alpha^\top Z = 0$ and $-\hat{Q}_\alpha Z = I - \mathbf{1}\alpha^\top$ for the matrix Z of Definition 4.1. These follow from $(\mathbf{1}\alpha^\top - \hat{Q}_\alpha)\mathbf{1} = \mathbf{1}$ and $\alpha^\top(\mathbf{1}\alpha^\top - \hat{Q}_\alpha) = \alpha^\top$. Differentiating $(\hat{Q}_\alpha + \text{diag } \eta)v = \Lambda v$ at $\eta = 0$ gives $e_j + \hat{Q}_\alpha \partial_j v = (\partial_j \Lambda)\mathbf{1}$. Multiplying by $\alpha^\top$ gives $\partial_j \Lambda = \alpha_j$. Then $\partial_j v = Z e_j$, because both vectors solve $\hat{Q}_\alpha x = \alpha_j \mathbf{1} - e_j$ with $\alpha^\top x = 0$, and the kernel of $\hat{Q}_\alpha$ is spanned by $\mathbf{1}$. Differentiating twice and multiplying by $\alpha^\top$ gives $\partial_i \partial_j \Lambda(0) = \alpha_i Z_{ij} + \alpha_j Z_{ji}$. Hence

$$\nabla \lambda(\theta) = -\alpha, \quad \nabla^2 \lambda(\theta) = \Gamma_\alpha.$$

For real z' near θ' we have $\varrho(A' - \text{diag } z') < 0$ by continuity, so the vector $(1, X(z')b_1)$ is nonnegative by the first fact. The definition of g shows that this vector lies in the kernel of $A_F - \text{diag}(g(z'), z')$, so $\lambda(g(z'), z') = 0$ by the second fact. Write g_j and g_{ij} for the first and second partial derivatives of g at θ' , and put $w_j = e_j - (\alpha_j/\alpha_1)e_1 \in \mathbb{R}^F$ for $j \geq 2$. Differentiating $\lambda(g(z'), z') = 0$ once and twice at $z' = \theta'$ gives

$$g_j = -\frac{\alpha_j}{\alpha_1}, \quad \alpha_1 g_{ij} = w_i^\top \Gamma_\alpha w_j.$$

So $\partial \Psi / \partial z_j = \alpha_1 g_j + \alpha_j = 0$ at θ' , and the matrix $\mathcal{H} = (\partial^2 \Psi / \partial z_i \partial z_j)_{i,j \geq 2}$ at θ' equals $V^\top \Gamma_\alpha V$, where V has columns $w_2, \dots, w_k$. Since $\partial / \partial \varphi_j = iz_j \partial / \partial z_j$, at $\varphi = 0$ we get $\nabla_\varphi \Psi = 0$ and $\nabla_\varphi^2 \Psi = -\Theta' \mathcal{H} \Theta'$, because the extra term involves only first derivatives, which vanish there. So hypothesis (i) of Lemma B.1 holds, and $-\nabla_\varphi^2 \Psi(\alpha, 0) = \Theta' \mathcal{H} \Theta'$ is real and symmetric.

Now we show that $\mathcal{H}$ is positive definite and find its determinant. Let Σ_α be Γ_α without the row and column of state 1, and $B = I + \mathbf{1}\alpha'^\top/\alpha_1$. For $y \in \mathbb{R}^{k-1}$ we have $Vy + (\alpha'^\top y/\alpha_1)\mathbf{1} = (0, By)$. Since $\Gamma_\alpha \mathbf{1} = 0$, adding a multiple of $\mathbf{1}$ does not change the quadratic form, so $y^\top \mathcal{H}y = (By)^\top \Sigma_\alpha (By)$. The matrix determinant lemma gives $\det B = 1 + (1 - \alpha_1)/\alpha_1 = 1/\alpha_1$. Hence

$$\det \mathcal{H} = \frac{\det \Sigma_\alpha}{\alpha_1^2},$$

and $\mathcal{H}$ is positive definite if Σ_α is. Let $y \in \mathbb{R}^{k-1}$ be nonzero and $x = (0, y) \in \mathbb{R}^F$. Put $f = x - (\alpha^\top x)\mathbf{1}$ and $u = Zf$. Then $\alpha^\top f = 0$, $\alpha^\top u = 0$ and $-\widehat{Q}_\alpha u = f$. With $D = \text{diag } \alpha$, using $\Gamma_\alpha \mathbf{1} = 0$ again,

$$y^\top \Sigma_\alpha y = f^\top \Gamma_\alpha f = 2f^\top DZf = 2 \sum_i \alpha_i u_i (-\widehat{Q}_\alpha u)_i = \sum_{i \neq j} \alpha_i (\widehat{Q}_\alpha)_{ij} (u_i - u_j)^2.$$

The last step uses $\widehat{Q}_\alpha \mathbf{1} = 0$ and $\alpha^\top \widehat{Q}_\alpha = 0$. For $i \neq j$, $(\widehat{Q}_\alpha)_{ij} = q_{ij}r_j/r_i$ is positive exactly on the edges of G_F , which is strongly connected. So the sum vanishes only if u is constant. Then $u = 0$ since $\alpha^\top u = 0$, so $f = 0$ and x is a multiple of $\mathbf{1}$. This is impossible, since $x_1 = 0$ and $x \neq 0$. Hence Σ_α and $\mathcal{H}$ are positive definite, and hypothesis (ii) of Lemma B.1 holds.

Next we check the regularity. First, $\alpha \mapsto \theta(\alpha)$ is real-analytic on Δ_F° . Indeed, since $I_F(\alpha) = \langle q, \alpha \rangle - \inf_{u>0} \sum_i \alpha_i (A_F u)_i / u_i$, Proposition 3.1(b) says that r_α is the unique minimizer of this function up to scaling. With $u = e^w$ and $w_1 = 0$ the function is $\mathcal{U}(\alpha, w) = \sum_{i \neq j} \alpha_i q_{ij} e^{w_j - w_i}$. Its Hessian in $(w_2, \dots, w_k)$ is the quadratic form $y \mapsto \sum_{i \neq j} \alpha_i q_{ij} e^{w_j - w_i} (y_j - y_i)^2$ with $y_1 = 0$, which is positive definite because G_F is strongly connected. So the analytic implicit function theorem, applied to $\nabla_w \mathcal{U}(\alpha, w) = 0$, shows that r_α and $\theta(\alpha) = A_F r_\alpha / r_\alpha$ are real-analytic in α . Now fix $\alpha_0 \in \mathcal{K}$. By (13) and continuity of the Perron root there is $\delta \in (0, \min_i \theta_i(\alpha_0))$ with $\varrho(A' - \text{diag}(\theta'(\alpha_0) - \delta \mathbf{1})) < 0$. By Lemma A.1, X , g and b are holomorphic on the set where $|z_i| > \theta_i(\alpha_0) - \delta$ for every $i \geq 2$, as locally uniform limits of their series. For α near α_0 we have $\theta_i(\alpha) > \theta_i(\alpha_0) - \delta$. So Ψ and χ are real-analytic in (α, φ) on a neighborhood of $\{\alpha_0\} \times \mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such neighborhoods, we see that Ψ and χ are jointly real-analytic near $\mathcal{K} \times \mathbb{T}^{k-1}$, and all their φ -derivatives are jointly continuous on $\mathcal{K} \times \mathbb{T}^{k-1}$.

Lemma B.1 now gives, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi} \chi d\varphi = e^{T\langle \alpha, \theta \rangle} \left(\frac{2\pi}{T} \right)^{\frac{k-1}{2}} \det(\Theta' \mathcal{H} \Theta')^{-1/2} \left[\chi(\alpha, 0) + \frac{c_1(\alpha)}{T} + O(T^{-3/2}) \right],$$

where c_1 is the coefficient given there. Here $\det(\Theta' \mathcal{H} \Theta') = (\prod_{i \geq 2} \theta_i)^2 \det \Sigma_\alpha / \alpha_1^2$ and $\chi(\alpha, 0) = b(\theta') \prod_{i \geq 2} \theta_i$. To evaluate $b(\theta')$, which is the residue of the generating function at $z_1 = g(\theta')$, we use the Perron vectors. As in the derivation of (13), $r' = r_1 X(\theta') b_1$. In the same way the columns $j \geq 2$ of $l^\top (A_F - \text{diag } \theta) = 0$ give $l' = l_1 X(\theta')^\top a_1$. Hence $\mu \cdot r = r_1(\mu_1 + \mu'^\top X(\theta') b_1)$ and $l \cdot \mathbf{1} = l_1(1 + a_1^\top X(\theta') \mathbf{1})$. Since $l_1 r_1 = \alpha_1$,

$$b(\theta') = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{\alpha_1}, \quad \text{so} \quad \frac{\chi(\alpha, 0)}{\det(\Theta' \mathcal{H} \Theta')^{1/2}} = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(\det \Sigma_\alpha)^{1/2}}.$$

Here we use $\mu(F) > 0$. Since $r_\alpha, l_\alpha > 0$, the value $\chi(\alpha, 0)$ is positive exactly when $\mu(F) > 0$, which holds by (H). It is continuous in α , so it is bounded below on $\mathcal{K}$, and we may divide by it. (If $\mu(F) = 0$, then $f_T = 0$ and the relative form makes no sense.) With $c(\alpha) = c_1(\alpha)/\chi(\alpha, 0)$, equation (15) becomes

$$f_T(\alpha) = T^{\frac{k-1}{2}} (2\pi)^{-\frac{k-1}{2}} \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(\det \Sigma_\alpha)^{1/2}} e^{-T\langle q - \theta, \alpha \rangle} \left(1 + \frac{c(\alpha)}{T} + O(T^{-3/2}) \right),$$

uniformly on $\mathcal{K}$. Since $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$, this is the claim. The function c is real-analytic, since c_1 is by Lemma B.1 and $\chi(\alpha, 0) > 0$ is. It is real, because $f_T(\alpha)$ and the main term are real, so $\text{Im } c(\alpha) = O(T^{-1/2})$ for every $T \geq 1$, which forces $\text{Im } c(\alpha) = 0$. $\square$

A.3 Proof of Lemma 4.4

We compare Lemma 2.3 with its continuum version term by term. Under the continuum weights a path has at most of order T runs, and each run after the first in state i changes the number of placements by a factor $1 - O(m_i/n_i)$. So the relative error is of order T^2/N .

For $k = 1$ we have $P_N(Ne_i) = \mu_i(1 - hq_i)^{N-1}$ and $f_T = \mu_i e^{-q_i T}$. Since $hq_i \leq \frac{1}{2}$, $(N-1) \log(1 - hq_i) + q_i T = O(T/N + T^2/N) = O(T^2/N)$, and $T^2/N \leq 1$, so the ratio is $1 + O(T^2/N)$. Let $k \geq 2$, and write $t = hn$ and $\alpha = n/N \in \mathcal{K}$. By Lemma 2.4, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t \rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m) h^{k-1} \prod_{i \in F} e^{-q_i t_i} \frac{t_i^{m_i-1}}{(m_i-1)!}, \quad m \in \mathbb{Z}_{\geq 1}^F.$$

Since $h^{m_i-1} \binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!} \prod_{1 \leq j < m_i} (1 - j/n_i)$ and $h^{M-1} = h^{k-1} \prod_i h^{m_i-1}$, the m -th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i \in F} \left[\prod_{1 \leq j < m_i} \left(1 - \frac{j}{n_i} \right) \right] (1 - hq_i)^{n_i - m_i} e^{hq_i n_i}.$$

We bound κ_m . Since $(1 - hq_i)^{n_i - m_i} \leq e^{-hq_i(n_i - m_i)}$, we get $\kappa_m \leq e^{hq_{\max} M}$. For the lower bound, $\log(1 - x) \geq -x - x^2$ for $0 \leq x \leq \frac{1}{2}$ gives $(1 - hq_i)^{n_i - m_i} e^{hq_i n_i} \geq (1 - hq_i)^{n_i} e^{hq_i n_i} \geq e^{-n_i h^2 q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2 T^2/N}$. If $m_i \leq n_i$ for every i , then all factors $1 - j/n_i$ lie in $[0, 1]$, and their product is at least $1 - \sum_i m_i^2/(2n_i) \geq 1 - M^2/(2n_{\min})$, where $n_{\min} = \min_i n_i$. If some $m_i > n_i$, then $\kappa_m = 0$, while $M \geq n_{\min} + 1$ gives $M^2/(2n_{\min}) \geq 2$. In all cases

$$\left(1 - \frac{M^2}{2n_{\min}} \right) e^{-q_{\max}^2 T^2/N} \leq \kappa_m \leq e^{hq_{\max} M}. \quad (16)$$

Let $w_m = v_m / \sum_{m'} v_{m'}$. This is a probability on $\mathbb{Z}_{\geq 1}^F$, since $H(t) > 0$ under (H). By Lemma 2.3 the ratio in the lemma is $\mathbb{E}_w \kappa_m$. So by (16) we need $\mathbb{E}_w e^{hq_{\max} M}$ and $\mathbb{E}_w M^2$. Since W for $e^c A_F$ is $e^{c(M-1)} W(m)$, we have, in the notation of the proof of Lemma 3.2,

$$\psi(c) := \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t).$$

The function ψ is convex with $\psi(0) = 0$, so $\psi(c) \leq c\psi(1)$ for $0 \leq c \leq 1$. Also $\psi(1) \geq 1$, since $M \geq k \geq 2$. We show that $\psi(1) \leq C_{\mathcal{K}} T$. The bound for H_{σ} in the proof of Lemma 3.2 holds for every $\sigma > 0$, and with $\sigma = e$ it gives $H_e(T\alpha) \leq e^{k-1} b(\theta') \prod_{i \geq 2} \theta_i e^{eT\langle \alpha, \theta \rangle}$ with $\theta = \theta(\alpha)$. For the lower bound, Theorem 4.3 gives T_1 such that, for $T \geq T_1$ and $\alpha \in \mathcal{K}$,

$$H(T\alpha) = T^{-(k-1)} e^{T\langle q, \alpha \rangle} f_T(\alpha) \geq \frac{1}{2} (2\pi T)^{-\frac{k-1}{2}} (\mu \cdot r_{\alpha}) (l_{\alpha} \cdot \mathbf{1}) (\det \Sigma_{\alpha})^{-1/2} e^{T\langle \alpha, \theta \rangle}.$$

For $1 \leq T \leq T_1$, the function $H(T\alpha) e^{-T\langle \alpha, \theta(\alpha) \rangle}$ is continuous and positive on the compact set $[1, T_1] \times \mathcal{K}$, so it is bounded below there. The prefactor in the display is also continuous and positive on $\mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}} T^{-(k-1)/2} e^{T\langle \alpha, \theta \rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, this gives $\psi(1) \leq C_{\mathcal{K}} T$ for $T \geq 1$.

Now let $x = hq_{\max} \leq \frac{1}{2}$. Then $\mathbb{E}_w e^{xM} = e^x e^{\psi(x)} \leq e^{x(1+C_K T)} = 1 + O(T^2/N)$, because $x(1+C_K T) = O(T^2/N)$ and $T^2/N \leq 1$. The Chernoff bound $\mathbb{P}_w(M-1 \geq y) \leq e^{\psi(1)-y}$ gives

$$\mathbb{E}_w(M-1)^2 = \int_0^\infty 2y \mathbb{P}_w(M-1 \geq y) dy \leq \psi(1)^2 + \int_{\psi(1)}^\infty 2ye^{\psi(1)-y} dy = \psi(1)^2 + 2\psi(1) + 2,$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \geq \varepsilon_K N$ with $\varepsilon_K = \min_{\alpha \in \mathcal{K}} \min_i \alpha_i > 0$. By (16),

$$1 - O\left(\frac{T^2}{N}\right) \leq e^{-q_{\max}^2 T^2/N} \left(1 - \frac{\mathbb{E}_w M^2}{2\varepsilon_K N}\right) \leq \mathbb{E}_w \kappa_m \leq \mathbb{E}_w e^{hq_{\max} M} \leq 1 + O\left(\frac{T^2}{N}\right),$$

uniformly for $\alpha \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$. $\square$

B A Laplace lemma with a parameter, and the next term

This appendix proves the Laplace lemma that Appendix A uses for Theorem 4.3, and then Proposition 4.13. Let $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. For $\varphi \in \mathbb{T}^m$ we use the max-norm $\|\varphi\| = \max_i \text{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm, so that $\|x\| \leq |x|$. Repeated indices are summed from 1 to m .

Lemma B.1 (Laplace integrals with a parameter). *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \rightarrow \mathbb{C}$ be such that the φ -derivatives of Ψ up to order 5 and of a up to order 3 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

- (i) $\Psi(\alpha, 0) \in \mathbb{R}$ and $\nabla_\varphi \Psi(\alpha, 0) = 0$,
- (ii) $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$ is real, symmetric and positive definite, and
- (iii) $\text{Re } \Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha, \varphi)} a(\alpha, \varphi) d\varphi = e^{T\Psi(\alpha, 0)} \left(\frac{2\pi}{T}\right)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \left(a_0 + \frac{c_1}{T} + O(T^{-3/2})\right).$$

Here $a_0 = a(\alpha, 0)$. Write $\mathcal{J}^{ij}$ for the entries of $\mathcal{J}(\alpha)^{-1}$, and $\Psi_{ijk}, \Psi_{ijkl}, a_i, a_{ij}$ for the φ -derivatives at $(\alpha, 0)$. Then

$$c_1 = \frac{1}{2} a_{ij} \mathcal{J}^{ij} + \frac{1}{2} a_i \Psi_{jkl} \mathcal{J}^{ij} \mathcal{J}^{kl} + a_0 \left[\frac{1}{8} \Psi_{ijkl} \mathcal{J}^{ij} \mathcal{J}^{kl} + \frac{1}{12} \Psi_{ijk} \Psi_{lrs} \mathcal{J}^{il} \mathcal{J}^{jr} \mathcal{J}^{ks} + \frac{1}{8} \Psi_{ijk} \Psi_{lrs} \mathcal{J}^{ij} \mathcal{J}^{kl} \mathcal{J}^{rs} \right].$$

It is continuous on $\mathcal{K}$, and it is real-analytic in α when Ψ and a are real-analytic in (α, φ) .

Proof. Constants. The derivatives are jointly continuous on the compact set $\mathcal{K} \times \mathbb{T}^m$. So $B_\Psi = \max_{|\beta| \leq 5} \sup |\partial_\varphi^\beta \Psi|$ and $B_a = \max_{|\beta| \leq 3} \sup |\partial_\varphi^\beta a|$ are finite, and $\mathcal{J}(\alpha)$ is continuous in α . Hence the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $h_{\min}$ over $\mathcal{K}$. Also $\det \mathcal{J}(\alpha) \leq (mB_\Psi)^m$, since every eigenvalue is at most the largest absolute row sum. Below, C is a constant that depends only on $m, h_{\min}, B_\Psi, B_a$ and the numbers ε and η chosen below. It may change from line to line.

The chart. Let $C_3 = m^3 B_\Psi / 6$ and choose ε with $0 < \varepsilon < 1$ and $C_3 \varepsilon \leq h_{\min} / 4$. Since $\varepsilon < \pi$, reduction modulo 2π is a measure-preserving bijection from the cube $\{x \in \mathbb{R}^m : \|x\| \leq \varepsilon\}$ onto $\{\varphi \in \mathbb{T}^m : \|\varphi\| \leq \varepsilon\}$, and we identify the two. The cube is convex, so Taylor's theorem with integral remainder applies to $\Psi(\alpha, \cdot)$ and $a(\alpha, \cdot)$ on it. It holds for complex values, since the remainder is

an integral. Put $R_3(\alpha, \varphi) = \Psi(\alpha, \varphi) - \Psi(\alpha, 0) + \frac{1}{2}\varphi^\top \mathcal{J}(\alpha)\varphi$. By (i) and (ii), R_3 and its derivatives of order 1 and 2 vanish at $\varphi = 0$. The remainder formula of order 3 and $\sum_{|\beta|=3} 1/\beta! = m^3/6$ give $|R_3(\alpha, \varphi)| \leq C_3 \|\varphi\|^3$. Since $\|\varphi\|^3 \leq \|\varphi\| |\varphi|^2$, we get

$$|R_3(\alpha, \varphi)| \leq C_3 \varepsilon |\varphi|^2 \leq \frac{1}{4} h_{\min} |\varphi|^2 \quad \text{for } \|\varphi\| \leq \varepsilon. \quad (17)$$

Step 1 (localization). The set of (α, φ) with $\|\varphi\| \geq \varepsilon$ is compact, and $\text{Re } \Psi(\alpha, \varphi) - \Psi(\alpha, 0)$ is continuous and, by (iii), negative on it. So it is at most $-\eta$ there, for some $\eta > 0$. Hence the integral over $\|\varphi\| \geq \varepsilon$ has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha, 0) - \eta)}$. Since $T^{m/2+3/2} e^{-\eta T} \leq C$, this is at most $CT^{-3/2} \cdot e^{T\Psi(\alpha, 0)} T^{-m/2}$.

Step 2 (scaling). On $\|\varphi\| \leq \varepsilon$ we substitute $\varphi = x/\sqrt{T}$. With $w(x) = TR_3(\alpha, x/\sqrt{T})$, the integral over this set is

$$T^{-m/2} e^{T\Psi(\alpha, 0)} \int_{\|x\| \leq \varepsilon\sqrt{T}} e^{-\frac{1}{2}x^\top \mathcal{J}x + w(x)} a(\alpha, x/\sqrt{T}) dx.$$

By (17), $|w(x)| \leq \frac{1}{4} h_{\min} |x|^2$. Since $x^\top \mathcal{J}x \geq h_{\min} |x|^2$, we get $|e^{-\frac{1}{2}x^\top \mathcal{J}x + w(x)}| \leq e^{-h_{\min} |x|^2/4}$.

Step 3 (expansion). Put $\mathcal{P}_3(x) = \frac{1}{6}\Psi_{ijk}x^i x^j x^k$ and $\mathcal{P}_4(x) = \frac{1}{24}\Psi_{ijkl}x^i x^j x^k x^l$. For $\|x\| \leq \varepsilon\sqrt{T}$, Taylor's theorem to order 5 for Ψ and to order 3 for a gives

$$w = T^{-1/2}\mathcal{P}_3 + T^{-1}\mathcal{P}_4 + \omega_1, \quad a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2}a_i x^i + \frac{1}{2T}a_{ij}x^i x^j + \omega_2,$$

with $|\omega_1| \leq CT^{-3/2}|x|^5$ and $|\omega_2| \leq CT^{-3/2}|x|^3$. Also $|w| \leq C_3 T^{-1/2}\|x\|^3$ and $e^{|w|} \leq e^{h_{\min}|x|^2/4}$. Let $\mathcal{Y} = 1 + T^{-1/2}\mathcal{P}_3 + T^{-1}(\mathcal{P}_4 + \frac{1}{2}\mathcal{P}_3^2)$. Then $e^w - \mathcal{Y}$ is the sum of ω_1 , of $\frac{1}{2}(w - T^{-1/2}\mathcal{P}_3)(w + T^{-1/2}\mathcal{P}_3)$ with $w - T^{-1/2}\mathcal{P}_3 = T^{-1}\mathcal{P}_4 + \omega_1$, and of $e^w - 1 - w - \frac{1}{2}w^2$, which is at most $\frac{1}{6}|w|^3 e^{|w|}$ in modulus. So $|e^w - \mathcal{Y}| \leq CT^{-3/2}(1 + |x|)^9 e^{h_{\min}|x|^2/4}$. We multiply $e^w = \mathcal{Y} + (e^w - \mathcal{Y})$ by the expansion of a , and use $T \geq 1$ for the products that carry $T^{-3/2}$ or T^{-2} . This gives

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2}\mathcal{Q}_1(x) + T^{-1}\mathcal{Q}_2(x) + \omega(x), \quad |\omega(x)| \leq CT^{-3/2}(1 + |x|)^{11} e^{h_{\min}|x|^2/4},$$

where $\mathcal{Q}_1 = a_0\mathcal{P}_3 + a_i x^i$ is odd and $\mathcal{Q}_2 = a_0(\mathcal{P}_4 + \frac{1}{2}\mathcal{P}_3^2) + a_i x^i \mathcal{P}_3 + \frac{1}{2}a_{ij}x^i x^j$.

Step 4 (integration). Since $\frac{1}{2}x^\top \mathcal{J}x \geq \frac{1}{2}h_{\min}|x|^2$, the remainder ω contributes at most $CT^{-3/2} \int_{\mathbb{R}^m} (1 + |x|)^{11} e^{-h_{\min}|x|^2/4} dx = CT^{-3/2}$ to the integral in Step 2. If $\|x\| > \varepsilon\sqrt{T}$ then $|x|^2 > \varepsilon^2 T$. So extending the integrals of the 3 polynomial terms from $\|x\| \leq \varepsilon\sqrt{T}$ to $\mathbb{R}^m$ changes them by at most $Ce^{-h_{\min}\varepsilon^2 T/4} \leq CT^{-3/2}$. On $\mathbb{R}^m$, $\int e^{-\frac{1}{2}x^\top \mathcal{J}x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd term $\mathcal{Q}_1$ integrates to 0. This is why the first correction is of order $1/T$ and not $T^{-1/2}$. For $\mathcal{Q}_2$ let x be a centered Gaussian vector with $\mathbb{E}[x^i x^j] = \mathcal{J}^{ij}$. By Isserlis' theorem, a moment of order 4 or 6 is the sum, over the 3 or 15 ways to pair the indices, of the products of the $\mathcal{J}^{ij}$ of the pairs. Since the derivatives of Ψ are symmetric in their indices, this gives $\mathbb{E}[a_i x^i \mathcal{P}_3] = \frac{1}{2}a_i \Psi_{jkl} \mathcal{J}^{ij} \mathcal{J}^{kl}$ and $\mathbb{E}[\mathcal{P}_4] = \frac{1}{8}\Psi_{ijkl} \mathcal{J}^{ij} \mathcal{J}^{kl}$. In $\mathbb{E}[\mathcal{P}_3^2] = \frac{1}{36}\Psi_{ijk} \Psi_{lrs} \mathbb{E}[x^i x^j x^k x^l x^r x^s]$, the 6 pairings that join each index of the first factor to an index of the second each give $\Psi_{ijk} \Psi_{lrs} \mathcal{J}^{il} \mathcal{J}^{jr} \mathcal{J}^{ks}$. The other 9 pair 2 indices inside each factor and join the remaining 2, and each gives $\Psi_{ijk} \Psi_{lrs} \mathcal{J}^{ij} \mathcal{J}^{kl} \mathcal{J}^{rs}$. So $\mathbb{E}[\mathcal{Q}_2] = c_1$.

By Steps 2 to 4 the integral over $\|\varphi\| \leq \varepsilon$ is $e^{T\Psi(\alpha, 0)} T^{-m/2} [(2\pi)^{m/2} \det \mathcal{J}^{-1/2} (a_0 + c_1/T) + O(T^{-3/2})]$. Since $\det \mathcal{J} \leq (mB_\Psi)^m$, this error and the one in Step 1 are $O(T^{-3/2})$ relative to $e^{T\Psi(\alpha, 0)} (2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. This proves the expansion. All constants depend only on m , $h_{\min}$, ε , η , B_Ψ and B_a , so it holds uniformly on $\mathcal{K}$. Finally c_1 is a polynomial in a_0 , the derivatives at $\varphi = 0$ and the entries of $\mathcal{J}^{-1}$, which gives its continuity and analyticity. $\square$

Proof of Proposition 4.13. If $k = 1$, then $f_T = \mu_i e^{-q_i T} = K_F e^{-T\lambda_F}$ exactly (Theorem 4.3), so the bracket in (i) vanishes and $c_F = 0$. This proves (i), and (iii) for single states. Until the proof of (iii) let $k \geq 2$. We use the coordinates α' and the Euclidean norm $|\cdot|$ of Section 4, and put $x = \alpha' - \alpha_F^*$, $c^* = c(\alpha_F^*)$ and $\bar{c} = c^* + \frac{1}{2}\gamma^\top \Sigma \gamma$.

Taylor bounds. By Proposition 4.2(a) and Theorem 4.3, the functions I_F , $\log C_F$ and c are real-analytic on Δ_F° . Here $C_F > 0$ because $\mu(F) > 0$. Also $\nabla I_F(\alpha_F^*) = 0$, since α_F^* is an interior minimum of I_F (Proposition 3.1(a)), and $\nabla^2 I_F(\alpha_F^*) = \Sigma^{-1}$. Let $4c_U > 0$ be the least eigenvalue of Σ^{-1} . Then there are a closed ball $U = \{|x| \leq \rho_0\} \subset \Delta_F^\circ$ and $L_1 > 0$ with $L_1 \rho_0 \leq c_U$ such that for all $\alpha, \beta \in U$

$$\begin{aligned} \left| \log \frac{C_F(\alpha)}{K_F} - \gamma \cdot x \right| &\leq L_1 |x|^2, & \left| I_F(\alpha) - \lambda_F - \frac{1}{2} x^\top \Sigma^{-1} x \right| &\leq L_1 |x|^3, \\ |\log C_F(\alpha) - \log C_F(\beta)| + |c(\alpha) - c(\beta)| + |\nabla I_F(\alpha) - \nabla I_F(\beta)| &\leq L_1 |\alpha' - \beta'|, \\ |I_F(\alpha) - I_F(\beta) - \nabla I_F(\beta) \cdot (\alpha' - \beta')| &\leq L_1 |\alpha' - \beta'|^2. \end{aligned}$$

In particular $I_F(\alpha) - \lambda_F \geq 2c_U |x|^2 - L_1 |x|^3 \geq c_U |x|^2$ on U .

(i), *outside* U . We first show that for $\theta \in V_F$, $T \geq 1$ and $\alpha \in \Delta_F^\circ$,

$$f_T(\alpha) \leq C_\theta T^{k-1} e^{-T\langle q - \theta, \alpha \rangle}, \quad (18)$$

with C_θ from Lemma 3.2. Fix T and α , and for large N choose n with support F and $|n_i - N\alpha_i| < 1$ for all i . By Lemma 4.4 on a closed ball around α in Δ_F° , and the continuity of f_T , we have $N^{k-1} P_N(n) \rightarrow f_T(\alpha)$ as $N \rightarrow \infty$. By Lemma 3.2 with $h = T/N$, $N^{k-1} P_N(n) \leq C_\theta T^{k-1} \exp(-T\langle q - \theta, n/N \rangle + C_\theta T^2/N)$, and the right side tends to that of (18). The argument for the region outside U in the proof of Theorem 4.5 gives $\eta > 0$ and $\theta_1, \dots, \theta_J \in V_F$ with $\max_j \langle q - \theta_j, \alpha \rangle > \lambda_F + 2\eta$ on $\Delta_F \setminus U^\circ$. So (18) gives $f_T(\alpha) \leq C T^{k-1} e^{-T(\lambda_F + 2\eta)}$ for $\alpha \in \Delta_F^\circ \setminus U$.

(i), *inside* U . Put $X_T = T^{(k-1)/2} K_F e^{-T\lambda_F}$ and $R_T = f_T/X_T$. By Theorem 4.3 with $\mathcal{K} = U$, for $\alpha \in U$,

$$R_T(\alpha) = \frac{C_F(\alpha)}{K_F} e^{-T(I_F(\alpha) - \lambda_F)} \left(1 + \frac{c(\alpha)}{T} + \epsilon_T(\alpha) \right), \quad |\epsilon_T(\alpha)| \leq C_1 T^{-3/2}. \quad (19)$$

Let $x_0 = \Sigma \gamma / T$, and for every x put $y = x - x_0$. Completing the square gives

$$\gamma \cdot x - \frac{T}{2} x^\top \Sigma^{-1} x = \frac{\gamma^\top \Sigma \gamma}{2T} - \frac{T}{2} y^\top \Sigma^{-1} y. \quad (20)$$

For large T the point α_0 with $x = x_0$ lies in U . There the Taylor bounds give $\log(C_F/K_F) - T(I_F - \lambda_F) \geq \frac{\gamma^\top \Sigma \gamma}{2T} - L_1 |x_0|^2 - L_1 T |x_0|^3$ and $c(\alpha_0) \geq c^* - L_1 |x_0|$. Since $|x_0| = O(1/T)$,

$$R_T(\alpha_0) \geq 1 + \frac{\bar{c}}{T} - O(T^{-3/2}). \quad (21)$$

Now let $\alpha \in U$. If $|x| \geq T^{-1/3}$, then $\log(C_F/K_F) - T(I_F - \lambda_F) \leq |\gamma||x| + L_1 |x|^2 - c_U T |x|^2 \leq -\frac{c_U}{2} T^{1/3}$ for large T , so $R_T(\alpha) \leq 2e^{-c_U T^{1/3}/2}$. If $|x| \leq T^{-1/3}$, then $|y| \leq 2T^{-1/3}$ for large T . From $|x|^2 \leq 2|y|^2 + 2|x_0|^2$ and $|x|^3 \leq 4|y|^3 + 4|x_0|^3$ we get $L_1 |x|^2 + L_1 T |x|^3 \leq (2L_1 + 8L_1 T^{2/3})|y|^2 + O(T^{-2}) \leq c_U T |y|^2 + O(T^{-2})$ for large T . With (20) and $\frac{T}{2} y^\top \Sigma^{-1} y \geq 2c_U T |y|^2$, this gives

$$\log \frac{C_F(\alpha)}{K_F} - T(I_F(\alpha) - \lambda_F) \leq \frac{\gamma^\top \Sigma \gamma}{2T} - c_U T |y|^2 + O(T^{-2}).$$

Also $c(\alpha) \leq c^* + L_1|y| + L_1|x_0|$ and $\sup_{u \geq 0} ue^{-c_U T u^2} = O(T^{-1/2})$. Since $1 + c^*/T > 0$ for large T , (19) gives

$$R_T(\alpha) \leq e^{\gamma^\top \Sigma \gamma / (2T) + O(T^{-2})} \left[e^{-c_U T |y|^2} \left(1 + \frac{c^*}{T} \right) + O(T^{-3/2}) \right] \leq 1 + \frac{\bar{c}}{T} + O(T^{-3/2}). \quad (22)$$

Off U we showed $R_T = O(T^{(k-1)/2} e^{-2\eta T})$. Together with (21) and (22), this shows that as $T \rightarrow \infty$

$$\sup_{\alpha \in \Delta_F^\circ} f_T(\alpha) = T^{(k-1)/2} K_F e^{-T\lambda_F} \left(1 + \frac{\bar{c}}{T} + O(T^{-3/2}) \right). \quad (23)$$

So the limit in (i) exists and equals $\bar{c}$.

We also locate the maximizer. For large T , the value $R_T(\alpha_0)$ exceeds the bounds above off U and for $|x| \geq T^{-1/3}$. Since f_T is continuous on U , it has a maximizer $\hat{\alpha}$ over Δ_F° , with $\hat{\alpha} \in U$ and $|\hat{x}| \leq T^{-1/3}$. Comparing the middle of (22) at $\hat{\alpha}$ with (21) gives $e^{-c_U T |\hat{y}|^2} \geq 1 - O(T^{-3/2})$. So $|\hat{y}| = O(T^{-5/4})$ and

$$|\hat{\alpha}' - \alpha_F^*| \leq |x_0| + |\hat{y}| = O(1/T). \quad (24)$$

(ii). We show that for every F that satisfies (H) there are T_0 and N_0 such that, for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \frac{k-1}{2} \log T + \log K_F - T\lambda_F + \frac{c_F}{T} + O\left(T^{-3/2} + \frac{T^2}{N}\right). \quad (25)$$

For $k = 1$ this is Theorem 4.5, since $c_F = 0$. Let $k \geq 2$ and $X = N^{-(k-1)} X_T$. For $n/N \in U$, Lemma 4.4 and (23) give $P_N(n) \leq N^{-(k-1)} \sup f_T(1 + O(T^2/N)) = X(1 + c_F/T + O(T^{-3/2} + T^2/N))$. For $n/N \notin U$, Lemma 3.2 with the θ_j above and $T^2/N \leq 1$ gives $P_N(n) \leq C(T/N)^{k-1} e^{-T(\lambda_F + 2\eta)} = X \cdot O(T^{(k-1)/2} e^{-2\eta T})$.

For the lower bound let $\hat{\alpha}$ be the maximizer in (24). Choose n with support F and $|n_i - N\hat{\alpha}_i| < 1$ for all i , and put $\beta = n/N$, so that $|\beta' - \hat{\alpha}'| \leq k/N$. For $T \geq T_0$ and $N \geq N_0$ both $\hat{\alpha}$ and β lie in U , and we apply Theorem 4.3 at both points. By the Taylor bounds, $|\log C_F(\beta) - \log C_F(\hat{\alpha})|$ and $|c(\beta) - c(\hat{\alpha})|$ are at most $L_1 k/N$. Also $|\nabla I_F(\hat{\alpha})| = |\nabla I_F(\hat{\alpha}) - \nabla I_F(\alpha_F^*)| \leq L_1 |\hat{x}| = O(1/T)$ by (24), so

$$T|I_F(\beta) - I_F(\hat{\alpha})| \leq T(|\nabla I_F(\hat{\alpha})| |\beta' - \hat{\alpha}'| + L_1 |\beta' - \hat{\alpha}'|^2) = O(1/N),$$

since $T \leq N$. Hence $f_T(\beta) = f_T(\hat{\alpha})(1 + O(1/N + T^{-3/2}))$. Note that this step uses Theorem 4.3 at the 2 points, and no bound on the derivatives of f_T . By Lemma 4.4 and (23),

$$M_F \geq P_N(n) = N^{-(k-1)} f_T(\beta) (1 + O(T^2/N)) = X(1 + c_F/T - O(T^{-3/2} + T^2/N)),$$

since $1/N \leq T^2/N$. So $M_F = X(1 + c_F/T + O(T^{-3/2} + T^2/N))$. Taking logarithms gives (25), because $T^2/N \leq N^{-1/3}$.

Now let $T_N \in [1, N^{1/3}]$ be a zero of $\log M_F - \log M_G$. By (5), $T_N \rightarrow \infty$, so $T_N \geq T_0$ for large N . Subtracting (25) for G from (25) for F at $T = T_N$ gives

$$\Delta L - \frac{\Delta}{2} \log T_N - \delta T_N - \log \frac{K_F}{K_G} = \frac{c_F - c_G}{T_N} + O\left(T_N^{-3/2} + \frac{T_N^2}{N}\right),$$

and multiplying by T_N gives (ii).

(iii). Single states were treated at the start. In the 2 other cases a group S of permutations of F acts transitively on F and preserves A_F , q and the restriction of μ to F . We first show

that then α_F^* is uniform on F and $\gamma = 0$, so that $c_F = c(\alpha_F^*)$. For $\sigma \in S$ and $v \in \mathbb{R}^F$ let σv be v with its coordinates permuted by σ . A permutation in S maps each word in the definition of W to a word with the same weight, so $H(\sigma t) = H(t)$ and $f_T(\sigma \alpha) = f_T(\alpha)$. Substituting $u \mapsto \sigma u$ in the definition of I_F gives $I_F(\sigma \alpha) = I_F(\alpha)$. So $\sigma \alpha_F^*$ also minimizes I_F , and $\sigma \alpha_F^* = \alpha_F^*$ by Proposition 3.1(a). Since S is transitive, α_F^* is uniform on F . By Theorem 4.3, $C_F(\alpha) = \lim_{T \rightarrow \infty} T^{-(k-1)/2} e^{T I_F(\alpha)} f_T(\alpha)$, so $C_F(\sigma \alpha) = C_F(\alpha)$ as well. Hence the derivative of $\log C_F$ at α_F^* is a linear form on $\{v \in \mathbb{R}^F : \sum_i v_i = 0\}$ that takes the same value at v and at σv . The average of σv over $\sigma \in S$ is the constant vector with entries $\frac{1}{k} \sum_i v_i = 0$. So the form vanishes and $\gamma = 0$.

A symmetric pair. Let $F = \{i, j\}$ with $q_{ij} = q_{ji} = \kappa$, $q_i = q_j = q$ and $\mu_i = \mu_j = \mu_0$. By (H), $\kappa > 0$ and $\mu_0 > 0$. The swap of i and j gives S , so $\alpha_F^* = (\frac{1}{2}, \frac{1}{2})$ and $c_F = c(\alpha_F^*)$. Also $Q_F(1, 1)^\top = (\kappa - q)(1, 1)^\top$, so $\lambda_F = q - \kappa$. The words in F alternate. A word that starts at i and contains j has either $2l$ letters and run counts (l, l) , or $2l + 1$ letters and run counts $(l + 1, l)$, with $l \geq 1$, and its weight is $\mu_0 \kappa^{M-1}$. The words that start at j give the same terms at $t_i = t_j$. With $I_\nu(\xi) = \sum_{l \geq 0} (\xi/2)^{2l+\nu} / (l!(l+\nu)!)$ [29, Eq. 10.25.2] we get

$$H\left(\frac{T}{2}, \frac{T}{2}\right) = 2\mu_0 \sum_{l \geq 1} \left(\frac{\kappa^{2l-1} (T/2)^{2l-2}}{((l-1)!)^2} + \frac{\kappa^{2l} (T/2)^{2l-1}}{l!(l-1)!} \right) = 2\mu_0 \kappa (I_0(\kappa T) + I_1(\kappa T)).$$

By Lemma 2.4, $f_T(\alpha_F^*) = T e^{-qT} H(T/2, T/2)$. As $\xi \rightarrow \infty$, $I_\nu(\xi) = (2\pi\xi)^{-1/2} e^\xi (1 - \frac{4\nu^2-1}{8\xi} + O(\xi^{-2}))$ [29, Eq. 10.40.1], so $I_0(\xi) + I_1(\xi) = 2(2\pi\xi)^{-1/2} e^\xi (1 - \frac{1}{8\xi} + O(\xi^{-2}))$. Hence

$$f_T(\alpha_F^*) = \frac{4\mu_0 \sqrt{\kappa}}{\sqrt{2\pi}} T^{1/2} e^{-(q-\kappa)T} \left(1 - \frac{1}{8\kappa T} + O(T^{-2}) \right).$$

Theorem 4.3 at α_F^* gives $f_T(\alpha_F^*) = T^{1/2} K_F e^{-(q-\kappa)T} (1 + c(\alpha_F^*)/T + O(T^{-3/2}))$. Letting $T \rightarrow \infty$ gives first $K_F = 4\mu_0 \sqrt{\kappa} / \sqrt{2\pi}$ and then $c(\alpha_F^*) = -1/(8\kappa)$. So $c_F = -1/(8\kappa)$, which is $-\frac{1}{8}$ when $\kappa = 1$. For $\kappa = 1$ and $\mu_0 = \frac{1}{2}$ we recover $K_F = \sqrt{2/\pi}$ from Corollary 4.9.

A face of the complete graph. Let $Q = J - dI$, so that $q_{ij} = 1$ for $i \neq j$ and $q_i = d - 1$. Let $k = |F| \geq 2$ and $\mu_i = \mu_0$ for $i \in F$, where $\mu_0 > 0$ by (H). All permutations of F form a group S as above, so $c_F = c(\alpha_F^*)$, and $\lambda_F = d - k$ (Example 2.5). Both f_T and C_F are linear in the restriction of μ to F . So replacing μ by the uniform law on $[d]$ multiplies both by $1/(d\mu_0)$, and by Theorem 4.3 it does not change c . So we may take μ uniform. Then the chain is the model of [3] with $p = 1 - (d-1)h$ (Example 2.5), and a vertex bin has probability p^{N-1}/d . Put $s = k - 1$, $z = Nh/p$ and $\mathcal{D}_k = k^{k+1/2} (4\pi)^{-s/2}$, and let n be a balanced bin with support F , so that $|n_i - N/k| < 1$ for $i \in F$. By [3, Eq. (7)], which gives the ratio of $P_N(n)$ to p^{N-1}/d , uniformly for $\frac{1}{2} \log N \leq z \leq 2 \log N$,

$$P_N(n) = \frac{p^{N-1}}{d} \mathcal{D}_k \frac{z^{s/2} e^{sz}}{N^s} \left(1 - \frac{s}{8z} + O\left(\frac{1}{z^2} + \frac{z^2}{N}\right) \right).$$

Given a large T , let $N = \lceil e^T \rceil$. Then $T \leq \log N < T + 1$, $T^2/N \leq T^2 e^{-T}$ and $z - T = (d-1)hz = O(T^2/N)$, so z lies in the range above. Also $(N-1) \log p = -(d-1)T + O(T^2/N)$. Substituting, and using $d-1-s = d-k$, we get

$$P_N(n) = N^{-s} \frac{\mathcal{D}_k}{d} T^{s/2} e^{-(d-k)T} \left(1 - \frac{s}{8T} + O(T^{-2}) \right).$$

On the other hand $|n'/N - \alpha_F^*| \leq k/N$, so $n/N \in U$ for large T . We use Lemma 4.4, Theorem 4.3 at n/N and the Taylor bounds, where $\nabla I_F(\alpha_F^*) = 0$ gives $T(I_F(n/N) - \lambda_F) = O(T/N^2)$. This gives

$$P_N(n) = N^{-s} T^{s/2} K_F e^{-(d-k)T} \left(1 + \frac{c(\alpha_F^*)}{T} + O(T^{-3/2}) \right).$$

Letting $T \rightarrow \infty$ in the ratio of the 2 expressions gives first $K_F = \mathcal{D}_k/d$, in agreement with Corollary 4.10, and then $c(\alpha_F^*) = -s/8$. So $c_F = -(k-1)/8$. $\square$

C The large-concentration limit of the direct jump

In this appendix we prove Proposition 6.3(d). As in the proof of part (c), B_1, B_2, B_3 are independent with the law $\text{Beta}(a, a)$, where $a = \omega/3$ throughout, and $\mathbb{P}_\omega(D)$ is the probability of the event D defined there. We let $a \rightarrow \infty$, so we may assume $a \geq 1$.

For large a each B_j lies within about $a^{-1/2}$ of $\frac{1}{2}$. At the uniform kernel $B_1 = B_2 = B_3 = \frac{1}{2}$ all 3 constraints are equalities. To first order they allow only the cyclic kernels of Proposition 6.3(b), so D is a thin tube around this curve of kernels. Its cross-section at distance δ from the uniform kernel has area of order δ^4 . The density of (B_1, B_2, B_3) near the uniform kernel is of order $a^{3/2}$, so $\mathbb{P}_\omega(D)$ is of order $a^{3/2} \int \delta^4 e^{-ca\delta^2} d\delta$, that is of order $1/a$.

Logits. Put $\ell_j = \log(B_j/(1-B_j))$ and $g(y) = 2 \log \cosh(y/2)$. The function g is even, convex and nonnegative, with $g(0) = g'(0) = 0$. Since $B_j = 1/(1 + e^{-\ell_j})$, we have $2B_j = e^{\ell_j/2} / \cosh(\ell_j/2)$, $2(1-B_j) = e^{-\ell_j/2} / \cosh(\ell_j/2)$ and $B_j(1-B_j) = \frac{1}{4} e^{-g(\ell_j)}$. As $dB_j/d\ell_j = B_j(1-B_j)$, the logit ℓ_j has the density $(B_j(1-B_j))^a \Gamma(2a)/\Gamma(a)^2 = e^{-ag(\ell_j)}/Z$ with $Z = 4^a \Gamma(a)^2/\Gamma(2a) = 2\sqrt{\pi} \Gamma(a)/\Gamma(a + \frac{1}{2})$ by the duplication formula. Write $g_j = g(\ell_j)$. Then $\log(2B_j) = \frac{1}{2}(\ell_j - g_j)$ and $\log(2(1-B_j)) = \frac{1}{2}(-\ell_j - g_j)$. So after taking logarithms the 3 constraints of D read

$$M_1 \leq g_1 + g_2, \quad M_2 \leq g_1 + g_3, \quad M_3 \leq g_2 + g_3, \quad (26)$$

where $M_1 = \ell_1 + \ell_2$, $M_2 = \ell_3 - \ell_1$ and $M_3 = -\ell_2 - \ell_3$. Note that $M_1 + M_2 + M_3 = 0$. From now on we regard D as the set of $\ell \in \mathbb{R}^3$ that satisfy (26). With $\psi(\ell) = g_1 + g_2 + g_3$ we get

$$\mathbb{P}_\omega(D) = Z^{-3} \int_D e^{-a\psi(\ell)} d\ell. \quad (27)$$

Two bounds on g . For all real y ,

$$\frac{y^2}{4} - \frac{y^4}{96} \leq g(y) \leq \frac{y^2}{4}, \quad (g1)$$

$$g(y) \geq \frac{1}{10} \min(y^2, |y|). \quad (g2)$$

Put $u = y/2$, so that $g(y) = 2 \log \cosh u$. The upper bound in (g1) is $\cosh u \leq e^{u^2/2}$, which holds term by term in the power series since $(2m)! \geq 2^m m!$. For the lower bound let $\chi(u) = \log \cosh u - \frac{u^2}{2} + \frac{u^4}{12}$. Then $\chi(0) = \chi'(0) = 0$ and $\chi''(u) = u^2 - \tanh^2 u \geq 0$, so $\chi \geq 0$. Since $2(\frac{u^2}{2} - \frac{u^4}{12}) = \frac{y^2}{4} - \frac{y^4}{96}$, this is the lower bound in (g1). For (g2), note that $g''(y) = \frac{1}{2} \text{sech}^2(y/2)$ is at least $\frac{1}{2} \text{sech}^2 1 > 0.2$ on $|y| \leq 2$. With $g(0) = g'(0) = 0$ this gives $g(y) \geq 0.1y^2$ there. For $|y| \geq 2$, convexity and $g(0) = 0$ give $g(y) \geq \frac{1}{2}|y|g(2)$, and $\frac{1}{2}g(2) = \log \cosh 1 > 0.43$.

Coordinates. Let $v = (1, -1, 1)$ and $t = \langle \ell, v \rangle / 3$. The linear map $\ell \mapsto (t, M_1, M_2)$ has determinant $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$, as we see by expanding along the row of t . Its inverse is $\ell = tv + w$ with $w = \frac{1}{3}(M_1 - M_2, 2M_1 + M_2, M_1 + 2M_2)$, and w is orthogonal to v . Hence $|\ell|^2 = 3t^2 + |w|^2$, and a direct computation gives $|w|^2 = \frac{1}{3}(M_1^2 + M_2^2 + M_3^2)$. We write $|M|$ for the Euclidean norm of (M_1, M_2) and $|M|_\infty = \max(|M_1|, |M_2|)$. Since $M_3 = -(M_1 + M_2)$,

$$\frac{1}{3}|M|^2 \leq |w|^2 \leq |M|^2 \leq 2|M|_\infty^2. \quad (28)$$

The line $M = 0$ is $\ell = tv$, that is $B_1 = B_3 = 1 - B_2$. These are the cyclic kernels $\pi_{i,i+1} = B_1$ of Proposition 6.3(b).

Proof of Proposition 6.3(d). Step 1. We bound M on D . By (26) and $g \geq 0$ we have $M_1 \leq g_1 + g_2 \leq \psi$ and $M_1 = -M_2 - M_3 \geq -(g_1 + g_3) - (g_2 + g_3) \geq -2\psi$. The same argument applies to M_2 . With (g1) we get, on D ,

$$|M|_\infty \leq 2\psi(\ell) \leq \frac{|\ell|^2}{2} = \frac{3t^2 + |w|^2}{2}. \quad (29)$$

Step 2. We show that the far region is negligible. Fix $\varepsilon = \frac{1}{12}$ and let $U = \{|t| \leq \varepsilon, |M|_\infty \leq \varepsilon\}$. Off U we have $|\ell| \geq \varepsilon/\sqrt{3}$. Indeed, either $|t| > \varepsilon$ and $|\ell| \geq \sqrt{3}|t|$, or $|M|_\infty > \varepsilon$ and $|\ell| \geq |w| \geq |M|/\sqrt{3}$ by (28). Some $|\ell_j|$ is at least $|\ell|/\sqrt{3}$, and $\min(y^2, |y|)$ increases with $|y|$. So (g2) gives $\psi(\ell) \geq \gamma(\ell)$ with $\gamma(\ell) = 0.1 \min(|\ell|^2/3, |\ell|/\sqrt{3})$. As γ increases with $|\ell|$, it is at least $\gamma_0 = \varepsilon^2/90$ off U . Hence $e^{-a\psi} \leq e^{-(a-1)\gamma_0} e^{-\gamma}$ off U . Since γ grows linearly, $e^{-\gamma}$ is integrable over $\mathbb{R}^3$, and so $\int_{D \setminus U} e^{-a\psi(\ell)} d\ell = O(e^{-\gamma_0 a})$.

Step 3. We show that the sections of $D \cap U$ are close to triangles. Let $(t, M) \in D \cap U$. By (29) and (28), $|M|_\infty \leq \frac{3}{2}t^2 + |M|_\infty^2 \leq \frac{3}{2}t^2 + \varepsilon|M|_\infty$, so $|M|_\infty \leq 2t^2$.

Next let E be the set of (t, M) with $|t| \leq \varepsilon$ and $|M|_\infty \leq 2t^2$. On E , (28) gives $|w| \leq 3t^2$, so $\ell_j = \pm t + w_j$ with $|w_j| \leq 3t^2$. As $|t| \leq \frac{1}{12}$, this gives $|\ell_j| \leq 2|t|$ and $|\ell_j^2 - t^2| \leq 6|t|^3 + 9t^4 \leq 7|t|^3$. Then (g1) gives $|g_j - \frac{1}{4}t^2| \leq \frac{1}{96}\ell_j^4 + \frac{7}{4}|t|^3 \leq 2|t|^3$. So on E

$$|g_i + g_j - \frac{1}{2}t^2| \leq 4|t|^3 \quad (i \neq j). \quad (30)$$

Also on E , (g1) gives $\frac{1}{4}|\ell|^2 - \frac{1}{2}t^4 \leq \psi(\ell) \leq \frac{1}{4}|\ell|^2$, and $|\ell|^2 = 3t^2 + |w|^2$ with $|w|^2 \leq 8t^4$. Hence

$$|\psi(\ell) - \frac{3}{4}t^2| \leq 2t^4. \quad (31)$$

For $h \geq 0$ let $\mathcal{T}(h) = \{M : M_1 \leq h, M_2 \leq h, M_1 + M_2 \geq -h\}$. This is the triangle with vertices (h, h) , $(h, -2h)$ and $(-2h, h)$. It has area $\frac{9}{2}h^2$ and lies in $\{|M|_\infty \leq 2h\}$. For $|t| \leq \varepsilon$ let D_t be the set of M with $(t, M) \in D \cap U$, and put $h_\pm = \frac{1}{2}t^2 \pm 4|t|^3$. Note that $h_- \geq 0$, since $8|t| < 1$. We claim that $\mathcal{T}(h_-) \subseteq D_t \subseteq \mathcal{T}(h_+)$. If $M \in D_t$, then $(t, M) \in E$ by the first part of this step. So (26) and (30) give $M_1 \leq h_+$, $M_2 \leq h_+$ and $-(M_1 + M_2) = M_3 \leq h_+$. If $M \in \mathcal{T}(h_-)$, then $|M|_\infty \leq 2h_- \leq t^2$, and $t^2 \leq \min(2t^2, \varepsilon)$. So $(t, M) \in E \cap U$. Each of M_1, M_2, M_3 is at most h_- , so (30) gives (26), and $M \in D_t$. Since $h_\pm = \frac{1}{2}t^2(1 \pm 8|t|)$, the area of D_t satisfies

$$\frac{9}{8}t^4(1 - 8|t|)^2 \leq |D_t| \leq \frac{9}{8}t^4(1 + 8|t|)^2. \quad (32)$$

Step 4. We apply Laplace's method. The coordinates (t, M_1, M_2) have Jacobian 1. So by Fubini's theorem, (31) and (32), the integral $J = \int_{D \cap U} e^{-a\psi} d\ell$ satisfies

$$\int_{-\varepsilon}^{\varepsilon} \frac{9}{8}t^4(1 - 8|t|)^2 e^{-a(\frac{3}{4}t^2 + 2t^4)} dt \leq J \leq \int_{-\varepsilon}^{\varepsilon} \frac{9}{8}t^4(1 + 8|t|)^2 e^{-a(\frac{3}{4}t^2 - 2t^4)} dt.$$

Substitute $t = s/\sqrt{a}$. Both bounds become $a^{-5/2}$ times an integral over $|s| \leq \varepsilon\sqrt{a}$, and both integrands tend to $\frac{9}{8}s^4 e^{-3s^2/4}$ for each s . On $|s| \leq \varepsilon\sqrt{a}$ we have $2s^4/a \leq 2\varepsilon^2 s^2 \leq s^2/4$ and $8|s|/\sqrt{a} \leq 8\varepsilon < 1$. So both integrands are at most $\frac{9}{2}s^4 e^{-s^2/2}$, which is integrable. By dominated convergence and $\int_{\mathbb{R}} s^4 e^{-bs^2} ds = \frac{3}{4}\sqrt{\pi} b^{-5/2}$ with $b = \frac{3}{4}$,

$$J = a^{-5/2} \left(\frac{9}{8} \cdot \frac{8\sqrt{\pi}}{3\sqrt{3}} + o(1) \right) = a^{-5/2} (\sqrt{3\pi} + o(1)).$$

By Step 2 the same holds for the integral over all of D .

It remains to estimate Z . The Gamma function is log-convex. So $\Gamma(a + \frac{1}{2})^2 \leq \Gamma(a)\Gamma(a+1) = a\Gamma(a)^2$ and $\Gamma(a+1)^2 \leq \Gamma(a + \frac{1}{2})\Gamma(a + \frac{3}{2}) = (a + \frac{1}{2})\Gamma(a + \frac{1}{2})^2$. Hence $a(a + \frac{1}{2})^{-1/2} \leq \Gamma(a + \frac{1}{2})/\Gamma(a) \leq \sqrt{a}$, and $Z = 2\sqrt{\pi/a} (1 + O(1/a))$. Now (27) gives

$$\mathbb{P}_\omega(D) = \frac{a^{3/2}}{8\pi^{3/2}} a^{-5/2} \sqrt{3\pi} (1 + o(1)) = \frac{\sqrt{3}}{8\pi a} (1 + o(1)) = \frac{3\sqrt{3}}{8\pi\omega} (1 + o(1)). \quad \square$$

D Proofs for the planar walk

This appendix proves the bound on $o_N^{(4)}$ in Proposition 7.2, Theorems 7.3 and 7.5, and Proposition 7.7. We identify $\mathbb{R}^2$ with $\mathbb{C}$ and write i for the imaginary unit, so that $e_z = i^z$ for $z \in \mathbb{Z}_4$. We use the notation of Lemma 7.1. So a path is given by its number m of switches, its skeleton Z and its run lengths ℓ , and $Y_N = \sum_{t=0}^m \ell_t i^{Z_t}$. Let $\text{bin}(m) = \binom{N-1}{m} (\beta h)^m p_0^{N-1-m}$ be the law of m , whose mean is $\bar{m} = (N-1)\beta h$. We write $\mathbb{P}_z$ and $\mathbb{E}_z$ for the skeleton walk of Lemma 7.1(c) started at z , $\mathbb{P}$ for its uniform start, and ξ for one of its increments. Put $\varkappa = s/\beta$, so $\varkappa < 1$ when $r > 0$. For a skeleton, let a, b, c, d be the numbers of t with $Z_t = 0, 2, 1, 3$, that is, the numbers of runs going east, west, north and south. So $a + b + c + d = m + 1$, and the path uses all 4 directions exactly when $a, b, c, d \geq 1$. Throughout, $N = 2\nu$ is even, and $[\cdot]$ is 1 if the statement inside holds and 0 otherwise. We also use $\psi(z) = z(I_0(z) + I_1(z))$ and the form (8) of [2, Thm. 5.2].

D.1 Return counts and skeletons

For $k \geq 1$ and an integer $X \geq 0$ let $\text{comp}_k(X) = \binom{X-1}{k-1} [X \geq 1]$, and let $\text{comp}_0(X) = [X = 0]$. Then $\text{comp}_k(X)$ is the number of compositions of X into k positive parts.

Lemma D.1 (return counts). *Let Z be a skeleton with m switches and counts a, b, c, d . The number of compositions ℓ of N into $m+1$ parts with $\sum_t \ell_t i^{Z_t} = 0$ is*

$$K_N(a, b, c, d) = \sum_{X=0}^{\nu} \text{comp}_a(X) \text{comp}_b(X) \text{comp}_c(\nu - X) \text{comp}_d(\nu - X).$$

If $a, b, c, d \geq 1$, then $m \geq 3$ and $K_N(a, b, c, d) \leq \frac{1}{2} \binom{N}{m-2}$.

Proof. The sum $\sum_t \ell_t i^{Z_t}$ vanishes exactly when the runs going east have the same total length X as the runs going west, and the runs going north have the same total length X' as the runs going south. Then $2X + 2X' = N$, so $X' = \nu - X$. The lengths of the a runs going east, in their order, form a composition of X into a parts, and similarly for the other 3 directions. These 4 compositions determine ℓ and can be chosen independently. This gives the formula.

Now let $a, b, c, d \geq 1$, so $m+1 = a+b+c+d \geq 4$. If $m = 3$, there is 1 run in each direction, and K_N counts the pairs $X, X' \geq 1$ with $X + X' = \nu$. So $K_N = \nu - 1 < \frac{1}{2} \binom{N}{1}$. Let $m \geq 4$. Pick

positions t_0, t_1, t_2, t_3 with $Z_{t_i} = z$ and fix the other $m - 3$ run lengths. If these add up to $N - j$, then $j \geq 4$, and there are $\binom{N-j-1}{m-4}$ ways to choose them. The conditions for $Y_N = 0$ now read $\ell_{t_0} - \ell_{t_2} = \delta_1$ and $\ell_{t_1} - \ell_{t_3} = \delta_2$, where δ_1 and δ_2 depend only on the fixed lengths, and the 4 free lengths add up to j . So ℓ_{t_2} determines the other 3. Moreover $\ell_{t_0} + \ell_{t_2} = 2\ell_{t_2} + \delta_1$ lies in $[2, j - 2]$ and has the parity of δ_1 . So there are at most $j/2$ solutions, and

$$K_N(a, b, c, d) \leq \sum_{j=0}^{N-1} \frac{j}{2} \binom{N-1-j}{m-4} = \frac{1}{2} \binom{N}{m-2}.$$

The identity counts the subsets of $\{0, \dots, N-1\}$ of size $m-2$ by their second smallest element j . $\square$

By Lemma 7.1, given m and Z the run lengths are uniform on the $\binom{N-1}{m}$ compositions. So $\varpi_N = K_N(a, b, c, d) / \binom{N-1}{m}$ is the probability that $Y_N = 0$ given m and Z , and

$$o_N^{(4)} = \sum_{m \geq 3} \text{bin}(m) \mathbb{E}[\varpi_N \mathbf{1}\{a, b, c, d \geq 1\} \mid m]. \quad (33)$$

Lemma D.2 (skeleton returns). *Let $W_m = \sum_{t=0}^m i^{Z_t}$, so $W_m = (a - b) + i(c - d)$.*

- (a) $\mathbb{E}_z |W_m|^2 \leq m + 1$ for every start z .
- (b) If $s > 0$ and m is odd, then $\mathbb{P}(W_m = 0) \geq \varkappa / (36(m + 1))$.
- (c) If $s = 0$ and $m \equiv 3 \pmod{4}$, then $\mathbb{P}(W_m = 0) \geq 1 / (m + 1)$.

Proof. (a) For $t > t'$ the difference $Z_t - Z_{t'}$ is a sum of $t - t'$ independent copies of ξ , and $\mathbb{E} i^\xi = \frac{r}{\beta} (i + i^{-1}) + \frac{s}{\beta} i^2 = -\varkappa$. So

$$\mathbb{E}_z |W_m|^2 = \sum_{t, t'=0}^m \mathbb{E}_z i^{Z_t - Z_{t'}} = m + 1 + 2 \sum_{k=1}^m (m + 1 - k) (-\varkappa)^k.$$

The last sum has terms of alternating sign, the first one negative, and their absolute values do not increase since $0 \leq \varkappa \leq 1$. So it is at most 0.

(b) Write $m = 2m_1 + 1$ and split $W_m = W' + W''$, where W' sums over $t \leq m_1$ and W'' over $t > m_1$. The law of ξ is symmetric on $\mathbb{Z}_4$ and the uniform law is stationary, so $(Z_{m_1}, Z_{m_1-1}, \dots, Z_0)$ is again the walk from a uniform start. Hence $\mathbb{P}(W' = w, Z_{m_1} = z) = \frac{1}{4} \mathbb{P}_z(W_{m_1} = w)$. The event $Z_{m_1+1} = Z_{m_1} + 2$ has probability $\varkappa$ and is independent of $Z_0, \dots, Z_{m_1}$. On this event, given $Z_{m_1} = z$, the second half is the walk from $z + 2$, and adding 2 to every entry changes the sign of W'' . So $\mathbb{P}_{z+2}(W_{m_1} = -w) = \mathbb{P}_z(W_{m_1} = w)$ and

$$\mathbb{P}(W_m = 0) \geq \frac{\varkappa}{4} \sum_{z \in \mathbb{Z}_4} \sum_w \mathbb{P}_z(W_{m_1} = w)^2.$$

Let $R^2 = m + 1 = 2(m_1 + 1)$. By (a) and Chebyshev's inequality, $\mathbb{P}_z(|W_{m_1}| \leq R) \geq \frac{1}{2}$. At most $(2R + 1)^2 \leq 9R^2$ Gaussian integers w have $|w| \leq R$, so by the Cauchy-Schwarz inequality the inner sum is at least $\frac{1}{4} / (9R^2)$. Summing over the 4 values of z gives $\mathbb{P}(W_m = 0) \geq \varkappa / (36(m + 1))$.

(c) Now every switch is a turn, so $i^{Z_t} = i^{Z_{t-1}} \cdot i\epsilon_t$ with independent fair signs ϵ_t . Hence $i^{Z_t} = i^{Z_0} i^t \sigma_t$, where $\sigma_0 = 1$ and $\sigma_t = \epsilon_1 \cdots \epsilon_t$, and $\sigma_1, \dots, \sigma_m$ are independent fair signs. Write

$m = 4k - 1$. Then $i^{-Z_0}W_m$ has real part $\sum_{t \text{ even}} (-1)^{t/2} \sigma_t$, which is 1 plus a sum of $2k - 1$ independent fair signs, and imaginary part $\sum_{t \text{ odd}} (-1)^{(t-1)/2} \sigma_t$, an independent sum of $2k$ fair signs. So

$$\mathbb{P}(W_m = 0) = \binom{2k-1}{k} 2^{-(2k-1)} \binom{2k}{k} 2^{-2k} = \left(\binom{2k}{k} 4^{-k} \right)^2 = \prod_{j=1}^k \frac{(2j-1)^2}{(2j)^2} \geq \frac{1}{4} \prod_{j=2}^k \frac{2j-2}{2j} = \frac{1}{4k},$$

using $(2j-1)^2 \geq (2j-2)2j$. This is $1/(m+1)$. $\square$

Lemma D.3 (balanced skeletons). *If $a = b \geq 1$, $c = d \geq 1$ and $m = 2(a+c) - 1 \leq N - 1$, then $K_N(a, b, c, d) / \binom{N-1}{m} \geq \sqrt{m+1}/(2N^2)$.*

Proof. Put $k = a + c = (m+1)/2$, so $2 \leq k \leq \nu$. Only $1 \leq X \leq \nu - 1$ contribute to K_N . So by the Cauchy-Schwarz inequality and Vandermonde's identity,

$$K_N = \sum_{X=1}^{\nu-1} \binom{X-1}{a-1}^2 \binom{\nu-X-1}{c-1}^2 \geq \frac{1}{\nu} \left(\sum_{X=1}^{\nu-1} \binom{X-1}{a-1} \binom{\nu-X-1}{c-1} \right)^2 = \frac{1}{\nu} \binom{\nu-1}{k-1}^2.$$

Next, $\binom{\nu-1}{k-1}^2 = \frac{k}{\nu} \binom{\nu-1}{k-1} \binom{\nu}{k}$, and $\binom{\nu-1}{k-1} \binom{\nu}{k} / \binom{2\nu-1}{2k-1} = \mathbb{P}(H = k-1)$, where H is the number of red balls among $2k-1$ balls drawn without replacement from $\nu-1$ red and ν blue ones. If $k = \nu$, all balls are drawn and $H = k-1$. Otherwise the ratio

$$\frac{\mathbb{P}(H = j+1)}{\mathbb{P}(H = j)} = \frac{(\nu-1-j)(2k-1-j)}{(j+1)(\nu-2k+2+j)}$$

decreases in j , equals $\frac{\nu-k}{\nu-k+1} < 1$ at $j = k-1$ and exceeds 1 at $j = k-2$. So $k-1$ is a mode of H . Since $\text{Var } H \leq (2k-1)/4$, Chebyshev's inequality puts mass at least $\frac{3}{4}$ on the at most $2\sqrt{2k-1} + 1 \leq 3\sqrt{2k}$ integers within $\sqrt{2k-1}$ of $\mathbb{E}H$. So $\mathbb{P}(H = k-1) \geq 1/(4\sqrt{2k})$. As $\binom{N-1}{m} = \binom{2\nu-1}{2k-1}$ and $\nu = N/2$,

$$\frac{K_N}{\binom{N-1}{m}} \geq \frac{k}{\nu^2} \cdot \frac{1}{4\sqrt{2k}} = \frac{\sqrt{m+1}}{2N^2}. \quad \square$$

D.2 The four-direction bound and the first order

Proof of the bound on $o_N^{(4)}$ in Proposition 7.2. Note that $(\beta h)^m p_0^{N-1-m} = p_0^{N-1} w^m$. By (33) and Lemma D.1,

$$o_N^{(4)} \leq \sum_{m=3}^{N-1} \binom{N-1}{m} (\beta h)^m p_0^{N-1-m} \frac{\binom{N}{m-2}}{2 \binom{N-1}{m}} = \frac{p_0^{N-1}}{2} \sum_{m=3}^{N-1} \binom{N}{m-2} w^m \leq \frac{p_0^{N-1} w^2}{2} ((1+w)^N - 1).$$

Since $1+w = 1/p_0$, the right side equals $\frac{w^2}{2p_0} (1 - p_0^N)$. Dividing by $c_N = \frac{1}{4} p_0^{N-1}$ gives $R_N = o_N^{(4)}/c_N \leq 2w^2((1+w)^N - 1)$. $\square$

Proof of Theorem 7.3. We first prove 3 bounds. Since $I_1 \leq I_0 \leq e^z$ for $z \geq 0$, we have $\psi(z) \leq 2ze^z$, and (8) gives $2S_N(u) \leq 8ue^{Nu}$. With Proposition 7.2 and $(1+w)^N \leq e^{Nw}$,

$$\frac{o_N}{c_N} \leq 8ue^{Nu} + 2w^2 e^{Nw}. \quad (34)$$

Since $o_N \geq o_N^{(0)}$, (8) with $z = Nu$ gives

$$\frac{o_N}{c_N} \geq 2S_N(u) \geq \frac{4}{N} \left(1 - \frac{z(z+2)}{2N}\right) \psi(z). \quad (35)$$

For the third bound we show that there are $c' > 0$ and T_0 , depending only on r and s , with

$$o_N^{(4)} \geq \frac{c'}{N^2 \sqrt{T}} \quad \text{whenever } T = Nh \geq T_0 \text{ and } \beta h \leq \frac{1}{2}. \quad (36)$$

Let $\mathcal{M}$ be the set of odd m if $s > 0$, and the set of $m \equiv 3 \pmod{4}$ if $s = 0$. Let $m \in \mathcal{M}$. The event $\{a = b \geq 1, c = d \geq 1\}$ is $\{W_m = 0\}$ minus the events $\{a = b = 0\}$ and $\{c = d = 0\}$. Each of these needs every switch to be a reversal, so it has probability at most $\frac{1}{2}\kappa^m$, and it is empty if $s = 0$. If $s > 0$, choose m_2 with $\kappa^m \leq \kappa/(72(m+1))$ for $m \geq m_2$, and put $c_2 = \kappa/72$. If $s = 0$, put $m_2 = 3$ and $c_2 = 1$. By Lemma D.2, $\mathbb{P}(a = b \geq 1, c = d \geq 1 \mid m) \geq c_2/(m+1)$ for $m \in \mathcal{M}$ with $m \geq m_2$. On this event Lemma D.3 gives $\varpi_N \geq \sqrt{m+1}/(2N^2)$. Let $J = [\bar{m}/2, 2\bar{m}]$. By (33), if $\bar{m} \geq 2m_2$ then

$$o_N^{(4)} \geq \sum_{m \in \mathcal{M} \cap J} \text{bin}(m) \frac{c_2}{2N^2 \sqrt{m+1}} \geq \frac{c_2}{2N^2 \sqrt{2\bar{m}+1}} \mathbb{P}(m \in \mathcal{M} \cap J).$$

By the Chernoff bound, $\mathbb{P}(m \notin J) \leq e^{-\bar{m}/8} + e^{-\bar{m}/3} \leq 2e^{-\bar{m}/8}$. Put $p = \beta h \leq \frac{1}{2}$. Then $\mathbb{P}(m \text{ odd}) = \frac{1}{2}(1 - (1-2p)^{N-1}) \geq \frac{1}{2}(1 - e^{-2\bar{m}})$. For $s = 0$ we use the roots of unity filter,

$$\mathbb{P}(m \equiv 3 \pmod{4}) = \frac{1}{4} \sum_{k=0}^3 i^{-3k} (1 - p + p i^k)^{N-1} \geq \frac{1}{4} - \frac{3}{4} e^{-\bar{m}/2},$$

since $|1 - p + p i^{\pm 1}|^2 = 1 - 2p(1-p) \leq e^{-p}$ and $|1 - 2p| \leq e^{-2p}$. So in both cases $\mathbb{P}(m \in \mathcal{M} \cap J) \geq \frac{1}{4} - \frac{3}{4} e^{-\bar{m}/2} - 2e^{-\bar{m}/8}$, which is at least $\frac{1}{8}$ when $\bar{m}$ is large. Since $\bar{m} \geq \beta T/2$, and $2\bar{m} + 1 \leq 3\beta T$ when $\beta T \geq 1$, this proves (36) with $c' = c_2/(16\sqrt{3}\beta)$.

(a) Let $h = \tau L/N$ and $T = \tau L$. Then $u, w = O(L/N)$, $Nu = sT/p_0 = s\tau L + O(L^2/N)$ and $Nw = \beta T/p_0 = \beta\tau L + O(L^2/N)$. So (34) gives $o_N/c_N = O(LN^{s\tau-1} + L^2N^{\beta\tau-2})$. If $\tau < \tau^*$, then $s\tau < 1$ and $\beta\tau < 2$, and this is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < \min(1-s\tau, 2-\beta\tau)$. Now let $\tau > \tau^*$, so that $s\tau > 1$ or $\beta\tau > 2$. If $s\tau > 1$, then $s > 0$, $z = Nu \rightarrow \infty$ and $z^2/N \rightarrow 0$, and (9) gives $\psi(z) = \sqrt{2z/\pi} e^z(1 + o(1))$. So (35) gives $o_N/c_N \geq (1 + o(1)) \frac{4}{N} \sqrt{2z/\pi} e^z$, which is at least $N^{s\tau-1}$ for large N . If $\beta\tau > 2$, we use (36) and $c_N^{-1} = 4p_0^{-(N-1)} \geq 4e^{\beta h(N-1)} = 4e^{\beta T - \beta h}$. They give $o_N/c_N \geq 4c' e^{-\beta h} N^{\beta\tau-2} (\tau L)^{-1/2}$, which is at least $N^{(\beta\tau-2)/2}$ for large N .

(b) Let $\tau < \tau^* < \tau'$. By (a), for large N the ratio o_N/c_N is less than 1 at $h = \tau L/N$ and greater than 1 at $h = \tau' L/N < 1/\beta$. By Proposition 7.2 it increases in h , so $\tau L < T_N^* < \tau' L$.

(c) Put $\tau_N = T_N^*/L$, so $\tau_N \rightarrow \tau^*$ by (b), and let $u^* = u_N^*$ and $w^* = \beta h_N^*/(1 - \beta h_N^*)$. As in (a), $u^*, w^* = O(L/N)$, $Nu^* = s\tau_N L + o(1)$ and $Nw^* = \beta\tau_N L + o(1)$. At the crossing Proposition 7.2 gives $2S_N(u^*) + R_N = 1$. If $s > 2r$, then $\tau^* = 1/s$ and $\beta\tau^* = \beta/s < 2$. So $R_N \leq 2w^{*2} e^{Nu^*} = N^{\beta/s-2+o(1)} \rightarrow 0$, and $2S_N(u^*) \rightarrow 1$. The 2 axes carry the same mass by symmetry. If $s < 2r$, then $\tau^* = 2/\beta$ and $s\tau^* = 2s/\beta < 1$. So $2S_N(u^*) \leq 8u^* e^{Nu^*} = N^{2s/\beta-1+o(1)}$, which is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < 1 - 2s/\beta$. $\square$

D.3 The four-direction local limit

Given the skeleton, the run lengths are close to the gaps between m uniform points in $[0, N]$. So the total lengths in the 4 directions are close to N times a Dirichlet vector (y_0, y_1, y_2, y_3) with

parameters (a, c, b, d) . Then $N^2\varpi_N$ should be close to twice the density of $(y_0 - y_2, y_1 - y_3)$ at 0, where the factor 2 is the parity of Y_N . This is the quantity $g(a, b, c, d)$ below. Divided by m , it is close to a Gaussian function of $W_m/\sqrt{m}$, and W_m satisfies a central limit theorem.

Proof of Theorem 7.5. Let $A = a + b$ and $B = c + d$, so $A + B = m + 1$, and put $\Gamma_4 = (a - 1)!(b - 1)!(c - 1)!(d - 1)!$ and

$$g(a, b, c, d) = \frac{m(m-1)}{2} \cdot \frac{\binom{A-2}{a-1}}{2^{A-2}} \cdot \frac{\binom{B-2}{c-1}}{2^{B-2}} = \frac{m!(A-2)!(B-2)!}{2^{m-2}(m-2)!\Gamma_4}.$$

Let $J_N = [\bar{m}/2, 2\bar{m}]$, $\mathcal{G}_m = \{a, b, c, d \geq m/8\}$ and $m_0 = 20$. Note that $\bar{m} = \beta T(1 - 1/N) \rightarrow \infty$ and $\bar{m} \leq \beta N^{1/8}$. Below, C is a constant that depends only on r and s and may change from line to line.

Step 1 (few or many switches). If $a, b, c, d \geq 1$ and $m \leq N/5$, then Lemma D.1 and $\binom{N}{m-2}/\binom{N-1}{m} = Nm(m-1)/((N-m+2)(N-m+1)(N-m))$ give $\varpi_N \leq m^2/N^2$. If $m > N/5$, then $\varpi_N \leq 1 < 25m^2/N^2$. So $N^2\varpi_N \leq 25m^2$ whenever the path uses all 4 directions. By the Chernoff bound, $\mathbb{P}(m \notin J_N) \leq 2e^{-\bar{m}/8}$. Also $\mathbb{E}m^4 = \sum_{j=1}^4 S(4, j)(N-1)_j(\beta h)^j \leq 15\bar{m}^4$, where $S(4, j) = 1, 7, 6, 1$ are Stirling numbers of the second kind, $(N-1)_j$ is a falling factorial, and $\bar{m} \geq 1$. By the Cauchy–Schwarz inequality, the terms of (33) with $m \notin J_N$ contribute at most $25\mathbb{E}[m^2\mathbf{1}\{m \notin J_N\}] \leq 25\sqrt{30}\bar{m}^2e^{-\bar{m}/16} = o(1)$ to $N^2o_N^{(4)}$.

Step 2 (unbalanced skeletons). For $k = 1, 2, 3$ let $V_k = \sum_{t=0}^m i^{kZ_t}$. The number of t with $Z_t = z$ is $\frac{m+1}{4} + \frac{1}{4}\sum_{k=1}^3 i^{-kz}V_k$. Let $\vartheta_k = \mathbb{E}i^{k\xi}$, so $\vartheta_1 = \vartheta_3 = -\kappa$ and $\vartheta_2 = 2\kappa - 1$. Then $\mathbb{E}[i^{kZ_t} \mid Z_0, \dots, Z_{t-1}] = \vartheta_k i^{kZ_{t-1}}$, so $M_m^{(k)} = \sum_{t=1}^m (i^{kZ_t} - \vartheta_k i^{kZ_{t-1}})$ is a martingale with increments of modulus at most 2, and

$$(1 - \vartheta_k)V_k = M_m^{(k)} + i^{kZ_0} - \vartheta_k i^{kZ_m}.$$

Here $1 - \vartheta_k \geq \min(1, 4r/\beta) > 0$. Burkholder's inequality [26, Ch. 2], applied to the real and imaginary parts, gives $\mathbb{E}|M_m^{(k)}|^4 \leq Cm^2$. So $\mathbb{E}|V_k|^4 \leq Cm^2$ and $\mathbb{E}(a - \frac{m+1}{4})^4 \leq Cm^2$, and the same holds for b, c, d . Since $\frac{m+1}{4} - \frac{m}{8} \geq \frac{m}{8}$, Markov's inequality gives $\mathbb{P}(\mathcal{G}_m^c \mid m) \leq C/m^2$. Every skeleton in $\mathcal{G}_m$ uses all 4 directions. By Step 1, the skeletons with all 4 directions that are not in $\mathcal{G}_m$, with $m \in J_N$, contribute at most $\sum_m \text{bin}(m) 25m^2 C/m^2 \leq 25C$ to $N^2o_N^{(4)}$.

Step 3 (run lengths given the skeleton). We show that $N^2\varpi_N = g(a, b, c, d)(1 + o(1))$ uniformly over the skeletons in $\mathcal{G}_m$ with $m \in J_N$. For large N such m satisfy $m_0 \leq m \leq 2\beta N^{1/8} \leq N^{1/6}$. On $\mathcal{G}_m$ we have $A, B \geq m/4$, so $A-1, B-1 \geq m/8$. Put $f(X) = X^{A-2}(\nu - X)^{B-2}$. For $X \geq 1$ we have $\text{comp}_a(X) = \frac{X^{a-1}}{(a-1)!} \prod_{j < a} (1 - j/X)_+$, and similarly for b, c, d . So

$$K_N(a, b, c, d) = \frac{1}{\Gamma_4} \sum_{X=1}^{\nu-1} f(X)\chi(X),$$

where $\chi(X) \in [0, 1]$ is the product of the 4 products. The function f vanishes at 0 and ν and is unimodal, and its mode $X^* = \nu(A-2)/(m-3)$ lies in $[\nu/6, 5\nu/6]$ since $A, B \geq m/4$ and $m \geq 18$. On each monotone piece of f every value $f(X)$ lies between the integrals of f over the 2 unit intervals next to X . So $|\sum_{j \leq X \leq k} f(X) - \int_j^k f| \leq 2 \max f$ for all integers $0 \leq j \leq k \leq \nu$. Moreover $f/\int_0^\nu f$ is the density of ν times a Beta variable with parameters $A-1$ and $B-1$, and $\int_0^\nu f = \nu^{m-2}B(A-1, B-1)$.

We need 3 estimates. First, this Beta law is the law of the $(A - 1)$ -th smallest of $m - 2$ independent uniform points in $[0, 1]$. A union bound over sets of $A - 1$ points gives, for $0 < t < 1$,

$$\mathbb{P}(\text{Beta} \leq t) \leq \binom{m-2}{A-1} t^{A-1} \leq \left(\frac{e(m-2)t}{A-1} \right)^{A-1} \leq (8et)^{A-1}.$$

We take $t = 2\nu^{-1/4}$. Since $A - 1 \geq 2$ and $16e\nu^{-1/4} < 1$ for large N , this gives $\int_0^{\nu^{3/4+1}} f \leq (16e)^2 \nu^{-1/2} \int_0^\nu f$, and the same bound holds near ν . Second, for $\nu^{3/4} \leq X \leq \nu - \nu^{3/4}$, the inequality $1 - \prod_j (1 - t_j)_+ \leq \sum_j t_j$ for $t_j \geq 0$ gives $1 - \chi(X) \leq (a^2 + b^2 + c^2 + d^2)/(2\nu^{3/4}) \leq m^2 \nu^{-3/4}$. Third, on $[\nu/12, 11\nu/12]$ we have $(\log f)'' = -(A-2)/X^2 - (B-2)/(\nu-X)^2 \geq -144m/\nu^2$. This interval contains every X with $|X - X^*| \leq \nu/(12\sqrt{m})$, and for these X Taylor's formula at X^* gives $f(X) \geq e^{-1/2} \max f$. So $\int_0^\nu f \geq e^{-1/2} \max f \cdot \nu/(6\sqrt{m})$, that is, $\max f \leq 6e^{1/2} \sqrt{m} \nu^{-1} \int_0^\nu f$.

Now $\sum_{X=1}^{\nu-1} f\chi \leq \sum_X f \leq \int_0^\nu f + 2 \max f$. Summing only over $\nu^{3/4} \leq X \leq \nu - \nu^{3/4}$ gives

$$\sum_{X=1}^{\nu-1} f\chi \geq (1 - m^2 \nu^{-3/4}) \left(\int_0^\nu f - 2(16e)^2 \nu^{-1/2} \int_0^\nu f - 2 \max f \right).$$

Since $m^2 \leq N^{1/3}$ and $\sqrt{m}/\nu \leq N^{-5/6}$, we get $\Gamma_4 K_N = \nu^{m-2} B(A-1, B-1)(1 + o(1))$, uniformly. Also $\binom{N-1}{m} = \frac{N^m}{m!} \prod_{j=1}^m (1 - \frac{j}{N}) = \frac{N^m}{m!} (1 + O(m^2/N))$. With $\nu^{m-2}/N^{m-2} = 2^{-(m-2)}$ and $B(A-1, B-1) = (A-2)!(B-2)!/(m-2)!$ this gives $N^2 \varpi_N = g(a, b, c, d)(1 + o(1))$.

Step 4 (average over the skeleton). We show that $\mathbb{E}[g \mathbf{1}_{\mathcal{G}_m} | m]/m \rightarrow c_* = \frac{2(r+s)}{\pi\beta}$ as $m \rightarrow \infty$. First we need a local limit for the fair binomial law. Let $0 \leq k \leq n$ and $y = (2k - n)/n$. If $|2k - n| \leq n^{3/5}$, Stirling's formula gives

$$\begin{aligned} \sqrt{n} \binom{n}{k} 2^{-n} &= \sqrt{\frac{2}{\pi(1-y^2)}} e^{-\frac{n}{2}[(1+y)\log(1+y) + (1-y)\log(1-y)]} (1 + O(\frac{1}{n})) \\ &= \sqrt{\frac{2}{\pi}} e^{-\frac{(2k-n)^2}{2n}} (1 + O(n^{-3/5})), \end{aligned}$$

since the bracket is $y^2 + O(y^4)$ and $ny^4 \leq n^{-3/5}$. Both sides decrease as $|2k - n|$ grows, so outside this range both are $O(e^{-n^{1/5}/3})$. Hence

$$\sup_{0 \leq k \leq n} \left| \sqrt{n} \binom{n}{k} 2^{-n} - \sqrt{\frac{2}{\pi}} e^{-(2k-n)^2/(2n)} \right| = O(n^{-1/2}). \quad (37)$$

On $\mathcal{G}_m$ we have $A-2, B-2 \geq m/8$. We apply (37) with $n = A-2$, $k = a-1$ and with $n = B-2$, $k = c-1$, and note that $2(a-1) - (A-2) = a-b$. Then, uniformly on $\mathcal{G}_m$,

$$\frac{g}{m} = \frac{m-1}{\pi \sqrt{(A-2)(B-2)}} \exp\left(-\frac{(a-b)^2}{2(A-2)} - \frac{(c-d)^2}{2(B-2)}\right) + O(m^{-1/2}).$$

Now we replace $m-1$ by m , then $A-2$ by A and $B-2$ by B , and finally B by $m(1-\theta)$, where $\theta = A/m$. The last step is needed because $B = m(1-\theta) + 1$. Each step changes the prefactor by a factor $1 + O(1/m)$ and each exponential by $O(1/m)$, since $0 \leq e^{-y/A_2} - e^{-y/A_1} \leq \frac{A_2 - A_1}{eA_1}$ for $y \geq 0$ and $0 < A_1 \leq A_2$. So, uniformly on $\mathcal{G}_m$,

$$\frac{g}{m} = \Psi\left(\frac{a-b}{\sqrt{m}}, \frac{c-d}{\sqrt{m}}, \frac{A}{m}\right) + O(m^{-1/2}), \quad \Psi(x_1, x_2, \theta) = \frac{e^{-x_1^2/(2\theta) - x_2^2/(2(1-\theta))}}{\pi \sqrt{\theta(1-\theta)}}.$$

On $\mathcal{G}_m$ we have $\theta \in [\frac{1}{4}, \frac{3}{4} + \frac{1}{m}] \subset [\frac{1}{5}, \frac{4}{5}]$, as $m \geq 20$. Let $\tilde{\Psi}$ be Ψ with θ replaced by the nearest point of $[\frac{1}{5}, \frac{4}{5}]$. It is bounded and continuous on $\mathbb{R}^3$ and equals Ψ on $\mathcal{G}_m$.

Next we prove a central limit theorem for $W_m = (a - b) + i(c - d)$. Put $\zeta_j = i\xi_j + \varkappa$, where $\xi_j = Z_j - Z_{j-1}$. The ζ_j are independent with $\mathbb{E}\zeta_j = 0$, $\mathbb{E}|\zeta_j|^2 = 1 - \varkappa^2$ and $\mathbb{E}\zeta_j^2 = \vartheta_2 - \varkappa^2 = -(1 - \varkappa)^2$. Since $i^{Z_j} = i^{Z_{j-1}}(\zeta_j - \varkappa)$,

$$(1 + \varkappa)W_m = M_m + i^{Z_0} + \varkappa i^{Z_m}, \quad M_m = \sum_{j=1}^m i^{Z_{j-1}} \zeta_j,$$

and M_m is a martingale with increments of modulus at most 2. We take the skeletons for all m to be the first steps of 1 infinite walk, so the σ -fields are nested. Fix $\varphi \in \mathbb{R}$. For a constant ω with $|\omega| = 1$ we have $\mathbb{E}(\text{Re } \omega \zeta_j)^2 = \frac{1}{2}(\mathbb{E}|\zeta_j|^2 + \text{Re } \omega^2 \mathbb{E}\zeta_j^2)$. So the j -th increment of $\text{Re}(e^{-i\varphi} M_m)$ has conditional variance $\frac{1}{2}[(1 - \varkappa^2) - (1 - \varkappa)^2 \cos(2\varphi)(-1)^{Z_{j-1}}]$. The average of these variances tends to $\frac{1}{2}(1 - \varkappa^2)$ in probability, because $\sum_{t=0}^m (-1)^{Z_t} = V_2$ and $\mathbb{E}|V_2/m|^4 \leq C/m^2$ by Step 2. The increments are bounded, so the martingale central limit theorem [26, Cor. 3.1] and the Cramér–Wold device give $M_m/\sqrt{m} \Rightarrow N(0, \frac{1}{2}(1 - \varkappa^2)I_2)$. Hence

$$\frac{W_m}{\sqrt{m}} \Rightarrow X \sim N(0, \sigma_*^2 I_2), \quad \sigma_*^2 = \frac{1 - \varkappa}{2(1 + \varkappa)} = \frac{r}{2(r + s)}.$$

Also $A = \frac{m+1}{2} + \frac{1}{2}V_2$, so $A/m \rightarrow \frac{1}{2}$ in probability, and $(W_m/\sqrt{m}, A/m) \Rightarrow (X, \frac{1}{2})$. By Step 2, $\mathbb{E}[\tilde{\Psi} \mathbf{1}_{\mathcal{G}_m^c} | m] \leq C/m^2$. So, since $\tilde{\Psi}$ is bounded and continuous,

$$\frac{\mathbb{E}[g \mathbf{1}_{\mathcal{G}_m} | m]}{m} = \mathbb{E} \tilde{\Psi} \left(\frac{W_m}{\sqrt{m}}, \frac{A}{m} \right) + O(m^{-1/2}) \rightarrow \mathbb{E} \Psi \left(X, \frac{1}{2} \right) = \frac{2}{\pi} \mathbb{E} e^{-|X|^2} = \frac{2}{\pi(1 + 2\sigma_*^2)} = c_*.$$

Step 5 (conclusion). By Steps 1 and 2, $N^2 o_N^{(4)} = \sum_{m \in J_N} \text{bin}(m) \mathbb{E}[N^2 \varpi_N \mathbf{1}_{\mathcal{G}_m} | m] + O(1)$. By Step 3 the expectation is $\mathbb{E}[g \mathbf{1}_{\mathcal{G}_m} | m](1 + o(1))$, with $o(1)$ uniform in $m \in J_N$. Since $\min J_N \rightarrow \infty$, Step 4 turns this into $N^2 o_N^{(4)} = c_*(1 + o(1)) \sum_{m \in J_N} \text{bin}(m) m + O(1)$. As in Step 1, $\mathbb{E}[m \mathbf{1}_{\{m \notin J_N\}}] \rightarrow 0$, and $\bar{m} = \beta T(1 + O(1/N))$. Hence $N^2 o_N^{(4)} = c_* \beta T(1 + o(1)) + O(1) = \frac{2(r+s)}{\pi} T(1 + o(1))$. $\square$

D.4 Turns only

Proof of Proposition 7.7. Let $s = 0$, so $\beta = 2r$ and $p_0 = 1 - 2rh$. Step t changes $U = Y^{(1)} + Y^{(2)}$ by a sign σ_t^U and $V = Y^{(1)} - Y^{(2)}$ by a sign σ_t^V . The directions east, north, west and south give the sign pairs $(\sigma_t^U, \sigma_t^V) = (+, +), (+, -), (-, -)$ and $(-, +)$. So directions and sign pairs correspond one to one, and $Y_N = 0$ exactly when $\sum_t \sigma_t^U = \sum_t \sigma_t^V = 0$. The 2 neighbors of a direction differ from it in exactly 1 sign each, and in different signs. So at each step the walk changes only σ^U with probability rh , only σ^V with probability rh , and neither with probability $1 - 2rh$. Hence a path is given by 2 words in $\pm$ with first letters σ, σ' and switch sets $A, B \subseteq \{1, \dots, N-1\}$ with $A \cap B = \emptyset$, and each such quadruple gives exactly 1 path. Its probability is $\frac{1}{4}(1 - 2rh)^{N-1-|A|-|B|}(rh)^{|A|+|B|} = c_N x^{|A|+|B|}$. So o_N/c_N is the sum of $x^{|A|+|B|}$ over the pairs of balanced words with disjoint switch sets. By the definition of S_N , the sum over all pairs of balanced words is $S_N(x)^2$. This proves $o_N/c_N = S_N(x)^2 - \Delta_N(x)$.

The 4 pairs with $A = B = \{\nu\}$ are balanced, since a single switch after step ν gives a balanced word for either first letter. So $\Delta_N(x) \geq 4x^2$. For the upper bound, a union bound over a common switch $i \in A \cap B$ gives $\Delta_N(x) \leq \sum_{i=1}^{N-1} S_{N,i}(x)^2$, where $S_{N,i}(x)$ is the sum of $x^{|A|}$ over the balanced

words (σ, A) with $i \in A$. If $A = \{t_1 < \dots < t_k\}$, the word has sum $\sigma(2 \sum_j (-1)^{j-1} t_j + (-1)^k N)$. So being balanced does not depend on σ , and it is a linear equation in $t_1, \dots, t_k$ with coefficients ± 2 . For $k = 1$ it gives $A = \{\nu\}$. Let $k \geq 2$ and $i \in A$, and let t' be the largest element of $A \setminus \{i\}$. There are at most $\binom{N-2}{k-2}$ choices for $A \setminus \{t', i\}$. Given this set and whether $t' > i$, the position in A of every element is known, since t' is the largest or the second largest element. So the equation determines t' . Hence at most $2 \binom{N-2}{k-2}$ sets $A \ni i$ of size k give balanced words, and

$$S_{N,i}(x) \leq 2x[i = \nu] + 4 \sum_{k \geq 2} \binom{N-2}{k-2} x^k = 2x[i = \nu] + 4x^2(1+x)^{N-2}.$$

Summing the squares gives $\Delta_N(x) \leq (2x + 4x^2(1+x)^{N-2})^2 + 16(N-2)x^4(1+x)^{2(N-2)}$, which is the stated upper bound.

Now let $N \geq 4$ and $x^* = x_N^*$. At the crossing, $S_N(x^*)^2 = 1 + \Delta_N(x^*) > 1 = S_N(u_N)^2$, so $x^* > u_N$. The polynomial S_N has nonnegative coefficients and no constant term, so $S_N(x)/x$ is nondecreasing, and

$$\frac{x^*}{u_N} \leq \frac{S_N(x^*)}{S_N(u_N)} = \sqrt{1 + \Delta_N(x^*)} \leq 1 + \frac{1}{2} \Delta_N(x^*).$$

By Theorem 7.3(b) with $\tau^* = 1/r$, $rT_N^* = L(1 + o(1))$. So $Nx^* = rT_N^*/(1 - 2rT_N^*/N) = L(1 + o(1))$, and with $(1+x)^n \leq e^{nx}$ the upper bound on Δ_N gives

$$\Delta_N(x^*) \leq 4x^{*2} + 16x^{*3}e^{Nx^*} + 16Nx^{*4}e^{2Nx^*} = N^{-1+o(1)}.$$

Hence $N(x^* - u_N) \leq Nu_N N^{-1+o(1)} \rightarrow 0$, since $Nu_N = O(L)$. Now we can be more precise. By [2, Eq. (7)], $Nu_N = N(1 - p_N)/p_N = L - \frac{1}{2} \log L + c_0 + o(1)$, and so also $Nx^* = L - \frac{1}{2} \log L + c_0 + o(1)$. Then $e^{2Nx^*} = \frac{\pi}{8} N^2 L^{-1} (1 + o(1))$ and $x^* = \frac{L}{N} (1 + o(1))$. So the last term of the bound is $2\pi L^3 N^{-1} (1 + o(1))$, while the first 2 terms are $O(L^{5/2} N^{-2})$. This gives $\Delta_N(x^*) \leq 2\pi L^3 N^{-1} (1 + o(1))$ and the bound on x^*/u_N . Finally, $rT_N^* = Nx^*/(1 + 2x^*)$ and $N(1 - p_N) = Nu_N/(1 + u_N)$, so

$$rT_N^* - N(1 - p_N) = \frac{N(x^* - u_N)}{1 + 2x^*} + Nu_N \left(\frac{1}{1 + 2x^*} - \frac{1}{1 + u_N} \right) = O\left(\frac{L^4}{N}\right) + O\left(\frac{L^2}{N}\right). \quad \square$$

E Proof of the Fisher information theorem

We keep the notation of Section 8. So $0 < q < 1$, $p = 1 - q$, $\rho = 1 - 2q$, $u = q/p$, $x = Nq$ and $y = Nu = x/p$. Let $\sigma_i \in \{-1, 1\}$ be the direction of step i , with $\sigma_i = 1$ for a step to the right. Then $X = \sigma_1 + \dots + \sigma_N$, and J is the number of $i < N$ with $\sigma_{i+1} \neq \sigma_i$. Put $Z = \mathbb{E}[J \mid K]$. By Proposition 8.1, $e_N = \text{Var } Z / \text{Var } J$.

For an interior bin, Paper A's Bessel estimate makes Z close to $y\sqrt{1 - W^2} + \frac{1}{2}$. When W is small this is close to the quadratic $y(1 - W^2/2) + \frac{1}{2}$, and W is of order $x^{-1/2}$. So Z is close to its projection on the span of 1 and X^2 , and e_N is close to R_N^2 . We first prove the exact moments behind the closed form of R_N^2 .

Exact moments.

Lemma E.1 (exact moments). *For $N \geq 1$ and $0 < q < 1$,*

$$\begin{aligned}\mathbb{E}X^2 &= \frac{Np}{q} - \frac{\rho(1 - \rho^N)}{2q^2}, \\ \text{Cov}(J, X^2) &= -\frac{p}{q^2} [Nq(1 + \rho^N) - p(1 - \rho^N)], \\ \mathbb{E}X^4 &= \frac{3N^2p^2}{q^2} - \frac{Np(1 + 10\rho + \rho^2)}{2q^3} + \frac{\rho(1 + 2\rho)(2 + \rho)(1 - \rho^N)}{2q^4} - \frac{3Np\rho^{N+1}}{q^3},\end{aligned}$$

and $\text{Var } X^2 = 2N^2p^2V/q^2$, with V as in Theorem 8.3.

Proof. Step 1. For real ϕ and χ put $m_N(\phi, \chi) = \mathbb{E}e^{\phi X + \chi J}$. Let A be the 2×2 matrix with entries $A_{ab} = \mathbb{P}(\sigma_2 = b \mid \sigma_1 = a) e^{\phi b + \chi \mathbf{1}_{[a \neq b]}}$ for $a, b \in \{-1, 1\}$, and let $v_a = \frac{1}{2}e^{\phi a}$. Then $m_N = v^\top A^{N-1} \mathbf{1}$ with $\mathbf{1} = (1, 1)^\top$. So, as formal power series in z ,

$$\sum_{N \geq 1} m_N z^N = z v^\top (I - zA)^{-1} \mathbf{1} = \frac{U(z)}{D(z)}, \quad U(z) = z v^\top \text{adj}(I - zA) \mathbf{1}, \quad D(z) = \det(I - zA).$$

Both U and D are polynomials in z of degree at most 2, and their coefficients are smooth functions of (ϕ, χ) . Let ∂ be a partial derivative in (ϕ, χ) of total order m . By induction on m and the quotient rule, $\partial(U/D) = U_\partial/D^{m+1}$, where U_∂ is a polynomial in z of degree at most $2m + 2$. Indeed each derivative multiplies the numerator by D or by a derivative of D , and this raises its degree by at most 2. At $\phi = \chi = 0$ the matrix A has trace $2p = 1 + \rho$ and determinant $p^2 - q^2 = \rho$, so $D(z) = (1 - z)(1 - \rho z)$. Now let $q \neq \frac{1}{2}$. Then $\rho \notin \{0, 1\}$, and the roots 1 and $1/\rho$ of D are distinct. Partial fractions write U_∂/D^{m+1} as a constant plus a combination of $(1 - z)^{-j}$ and $(1 - \rho z)^{-j}$ with $1 \leq j \leq m + 1$. The constant only changes the coefficient of z^0 . The coefficient of z^N in $(1 - z)^{-j}$ is $\binom{N+j-1}{j-1}$, a polynomial in N of degree $j - 1$, and in $(1 - \rho z)^{-j}$ it is the same polynomial times ρ^N . Hence, as a sequence in $N \geq 1$, the derivative ∂m_N at $\phi = \chi = 0$ lies in

$$\mathcal{S}_m = \text{span}\{N^i, N^i \rho^N : 0 \leq i \leq m\},$$

which has dimension at most $2m + 2$.

Step 2. Every sequence in $\mathcal{S}_m$ satisfies the linear recurrence with characteristic polynomial $(t - 1)^{m+1}(t - \rho)^{m+1}$, which has order $2m + 2$. So a sequence in $\mathcal{S}_m$ is determined, for $N \geq 1$, by its values at $N = 1, \dots, 2m + 2$. In particular, if these values are 0, the sequence is 0 for every $N \geq 1$.

Step 3. The moments $\mathbb{E}X^2$, $\mathbb{E}[JX^2]$ and $\mathbb{E}X^4$ are the derivatives $\partial_\phi^2 m_N$, $\partial_\chi \partial_\phi^2 m_N$ and $\partial_\phi^4 m_N$ at 0, so they lie in $\mathcal{S}_4$. The claimed formulas for $\mathbb{E}X^2$ and $\mathbb{E}X^4$ lie in $\mathcal{S}_2 \subset \mathcal{S}_4$. Since $\mathbb{E}J = (N - 1)q$, the claimed formula for $\mathbb{E}[JX^2] = \text{Cov}(J, X^2) + (N - 1)q \mathbb{E}X^2$ also lies in $\mathcal{S}_2$. Hence the difference between each moment and its claimed formula lies in $\mathcal{S}_4$, which has dimension at most 10. By Step 2 it is enough to check that the difference vanishes at $N = 1, \dots, 10$. This is a finite exact computation. The script `verify_sticky_fisher.py` computes the 3 moments exactly for $N = 1, \dots, 12$ by a dynamic program over paths, with q kept as a symbol, and every difference is identically 0. This proves the 3 formulas for $q \neq \frac{1}{2}$. For fixed N both sides are rational functions of q with no pole at $q = \frac{1}{2}$, so the case $q = \frac{1}{2}$ follows by continuity.

Step 4. We have $\text{Var } X^2 = \mathbb{E}X^4 - (\mathbb{E}X^2)^2$. With the formulas of Step 3, the claim $\text{Var } X^2 = 2N^2p^2V/q^2$ becomes an identity between rational functions of 3 independent variables N , q and R , where R stands for ρ^N and $\rho^{N+1} = \rho R$. The same script checks this identity with sympy. $\square$

Proof of the closed form in Theorem 8.3. Let $N \geq 2$. Then X^2 takes the value N^2 on the constant paths and the smaller value $(N-2)^2$ on the paths that switch only at the last step. So $\text{Var } X^2 > 0$, and $V > 0$ by Lemma E.1. The same lemma gives

$$\text{Cov}(J, X^2) = -\frac{Np}{q} \left[1 + \rho^N - \frac{p(1 - \rho^N)}{x} \right],$$

so $\text{Cov}(J, X^2)^2 / \text{Var } X^2$ is the squared bracket divided by $2V$. Dividing by $\text{Var } J = (N-1)pq$ gives the formula for R_N^2 . $\square$

Three lemmas.

Lemma E.2 (convexity). *Let $\delta > 0$, and let f and g be convex and differentiable on $[t - \delta, t + \delta]$ with $0 \leq g - f \leq \eta$ there. Suppose also that g is twice differentiable with $g'' \leq \kappa$ on this interval. Then*

$$|f'(t) - g'(t)| \leq \frac{\eta}{\delta} + \frac{\kappa\delta}{2}.$$

Proof. By the convexity of f and then the bounds on $g - f$, we have $f'(t) \leq (f(t + \delta) - f(t))/\delta \leq (g(t + \delta) - g(t) + \eta)/\delta$. By Taylor's theorem $g(t + \delta) \leq g(t) + \delta g'(t) + \kappa\delta^2/2$, so $f'(t) \leq g'(t) + \eta/\delta + \kappa\delta/2$. In the same way

$$f'(t) \geq \frac{f(t) - f(t - \delta)}{\delta} \geq \frac{g(t) - g(t - \delta) - \eta}{\delta} \geq g'(t) - \frac{\eta}{\delta} - \frac{\kappa\delta}{2}. \quad \square$$

Lemma E.3 (a Bessel ratio). *For $t > 0$ let $R(t) = I_1(t)/I_0(t)$. Then*

$$\frac{\sqrt{1+t^2}-1}{t} \leq R(t) < 1.$$

In particular $0 < 1 - R(t) \leq 1/t$.

Proof. Since $I'_0 = I_1$ and $I'_1(t) = I_0(t) - I_1(t)/t$, the function R satisfies $R' = 1 - R/t - R^2$. From the power series, $R > 0$ and $R(t) = t/2 - t^3/16 + O(t^5)$ as $t \rightarrow 0$. If $R(t_0) = 1$ for a first $t_0 > 0$, then $R'(t_0) \geq 0$, but the equation gives $R'(t_0) = -1/t_0$. So $R < 1$.

Put $s = \sqrt{1+t^2}$ and $\ell(t) = (s-1)/t = t/(s+1)$. A short computation gives $1 - \ell/t - \ell^2 = \ell/t = 1/(s+1)$ and $\ell' = 1/(s(s+1))$. Hence $\ell' - (1 - \ell/t - \ell^2) = (1-s)/(s(s+1)) \leq 0$. For $d = R - \ell$ this gives

$$d' \geq -\frac{d}{t} - (R^2 - \ell^2) = -d\left(\frac{1}{t} + R + \ell\right).$$

Let $\Theta(t) = t \exp \int_0^t (R + \ell)$. Then $(\Theta d)' = \Theta(d' + d(1/t + R + \ell)) \geq 0$. Since $\ell(t) = t/2 - t^3/8 + O(t^5)$, we have $d(t) = O(t^3)$, so $\Theta(t)d(t) \rightarrow 0$ as $t \rightarrow 0$. Hence $\Theta d \geq 0$ and $d \geq 0$. Finally $1 - \ell(t) = (t + 1 - s)/t \leq 1/t$, since $s \geq t$. $\square$

Lemma E.4 (moments of the endpoint). (a) *For each $m \geq 1$ there is a constant C_m such that $\mathbb{E}W^{2m} \leq C_m x^{-m}$ whenever $0 < q < 1$ and $x \geq 1$.*

(b) *If $0 < q \leq \frac{1}{2}$, then $\mathbb{E}W^4 \leq 3p^2/x^2 + 9/(2x^4)$.*

The reason for the order x^{-m} is that X is roughly a sum of about x runs with alternating signs and lengths of order $1/q$. So X is of order $\sqrt{x}/q$ and W is of order $x^{-1/2}$.

Proof. (a) Extend the walk by a further step σ_{N+1} with the same rule, and let $\mathcal{F}_i$ be generated by $\sigma_1, \dots, \sigma_i$. Since $\mathbb{E}[\sigma_{i+1} \mid \mathcal{F}_i] = (p - q)\sigma_i = \rho\sigma_i$, the variables $\xi_{i+1} = \sigma_{i+1} - \rho\sigma_i$ are martingale differences. Summing $\sigma_{i+1} - \sigma_i = -2q\sigma_i + \xi_{i+1}$ over $i = 1, \dots, N$ gives

$$X = \frac{\sigma_1 - \sigma_{N+1} + \Xi}{2q}, \quad \Xi = \sum_{i=2}^{N+1} \xi_i.$$

If step $i + 1$ repeats step i , then $\xi_{i+1}^2 = (1 - \rho)^2 = 4q^2$. Otherwise $\xi_{i+1}^2 = (1 + \rho)^2 = 4p^2 \leq 4$. So

$$\sum_{i=2}^{N+1} \xi_i^2 \leq 4q^2 N + 4J',$$

where $J' \sim \text{Bin}(N, q)$ is the number of switches among the $N + 1$ steps. Burkholder's inequality [26, Ch. 2] bounds the moment of order $2m$ of a martingale by a constant c_m times the m -th moment of the sum of its squared differences. So

$$\mathbb{E} \Xi^{2m} \leq c_m \mathbb{E} (4q^2 N + 4J')^m \leq c_m 8^m ((q^2 N)^m + \mathbb{E} J'^m).$$

Here $q^2 N = qx \leq x$. The factorial moments of J' are $\mathbb{E}[J'(J' - 1) \cdots (J' - j + 1)] = N(N - 1) \cdots (N - j + 1)q^j \leq x^j$, and J'^m is a combination of the falling factorials $J'(J' - 1) \cdots (J' - j + 1)$, $1 \leq j \leq m$, with nonnegative coefficients (the Stirling numbers of the second kind). So $\mathbb{E} J'^m \leq b_m x^m$ for $x \geq 1$, where b_m is the sum of these coefficients. Hence $\mathbb{E} \Xi^{2m} \leq c_m 8^m (1 + b_m) x^m$. Since $|\sigma_1 - \sigma_{N+1}| \leq 2$ and $x \geq 1$,

$$\mathbb{E} X^{2m} \leq (2q)^{-2m} 2^{2m-1} (2^{2m} + \mathbb{E} \Xi^{2m}) \leq C_m q^{-2m} x^m,$$

and $\mathbb{E} W^{2m} = N^{-2m} \mathbb{E} X^{2m} \leq C_m x^{-m}$.

(b) Dividing the formula for $\mathbb{E} X^4$ in Lemma E.1 by N^4 gives

$$\mathbb{E} W^4 = \frac{3p^2}{x^2} - \frac{p(1 + 10\rho + \rho^2)}{2x^3} + \frac{\rho(1 + 2\rho)(2 + \rho)(1 - \rho^N)}{2x^4} - \frac{3p\rho^{N+1}}{x^3}.$$

For $q \leq \frac{1}{2}$ we have $0 \leq \rho < 1$. Then the second and the last terms are at most 0, and the third is at most $1 \cdot 3 \cdot 3/(2x^4)$. $\square$

Proof of the bound in Theorem 8.3. Let $\mathcal{P}$ be the orthogonal projection in L^2 onto the span of 1 and X^2 . Since X^2 is a function of K , we have $\mathbb{E} Z = \mathbb{E} J$ and $\text{Cov}(Z, X^2) = \text{Cov}(J, X^2)$. So $\mathcal{P}Z = \mathbb{E} Z + \text{Cov}(J, X^2)(X^2 - \mathbb{E} X^2)/\text{Var } X^2$, and $Z - \mathcal{P}Z$ is orthogonal to 1 and to X^2 . Hence

$$\text{Var } Z = \frac{\text{Cov}(J, X^2)^2}{\text{Var } X^2} + \mathbb{E}(Z - \mathcal{P}Z)^2,$$

and dividing by $\text{Var } J$ gives $e_N - R_N^2 = \mathbb{E}(Z - \mathcal{P}Z)^2/\text{Var } J \geq 0$. Also $\mathbb{E}(Z - \mathcal{P}Z)^2 \leq \mathbb{E}(Z - G)^2$ for every G in the span of 1 and X^2 . We take

$$G = y \left(1 - \frac{W^2}{2} \right) + \frac{1}{2}.$$

From now on $0 < q \leq \frac{1}{2}$ and $1 \leq x \leq N^{1/4}$, and C denotes an absolute constant that may change from line to line. Then $p \geq \frac{1}{2}$, $x \leq y \leq 2x$, and $\text{Var } J = (N - 1)pq \geq x/4$ for $N \geq 2$. So it is enough to show that $\mathbb{E}(Z - G)^2 \leq C(x^{-2} + x^4/N)$.

For an interior bin $0 < k < N$ put $s_k = \sqrt{k(N-k)}$, $c_k = N/(2s_k)$ and $z_k = s_k u$. On the event $K = k$ we have $k(N-k) = N^2(1-W^2)/4$, so $c_k = (1-W^2)^{-1/2} \geq 1$, $2z_k = y\sqrt{1-W^2}$ and $c_k z_k = y/2$. Let $\mathcal{I}_k(z) = I_0(2z) + c_k I_1(2z)$ and $\tilde{S}_k(v) = 2v \mathcal{I}_k(s_k v)$. This is Paper A's Bessel approximation to $S_{k,N-k}(v)$, which is called $T_{a,b}$ in [2]. By [2, Thm. 6.2],

$$0 \leq 1 - \frac{S_{k,N-k}(v)}{\tilde{S}_k(v)} \leq \frac{Nv^2}{2} + v \quad \text{for all } v > 0. \quad (38)$$

Let $\Lambda_k(t) = \log S_{k,N-k}(e^t)$ and $\beta_k(t) = \log \tilde{S}_k(e^t)$. By Proposition 8.1, $Z = \Lambda'_K(\log u)$ on the event $0 < K < N$.

Step 1. We compare Λ'_k with β'_k at $\log u$. Both functions are convex, since each is the logarithm of a series in e^t with nonnegative coefficients. Let $0 < \delta \leq 1$ and $|t - \log u| \leq \delta$, so that $v = e^t \leq eu$. Then

$$\frac{Nv^2}{2} + v \leq \frac{e^2 y^2 / 2 + ey}{N} \leq \frac{2e^2 + 2e}{\sqrt{N}},$$

using $y \leq 2N^{1/4}$. The right side is at most $\frac{1}{2}$ for $N \geq 1700$. Since $-\log(1-\varepsilon) \leq 2\varepsilon$ for $0 \leq \varepsilon \leq \frac{1}{2}$, the bound (38) gives $0 \leq \beta_k - \Lambda_k \leq \eta$ on this interval, with $\eta = (e^2 y^2 + 2ey)/N$.

Next, $\beta_k(t) = \log 2 + t + \log \mathcal{I}_k(s_k e^t)$, and $\mathcal{I}_k(s_k e^t) = \sum_n a_n e^{nt}$ with $a_n \geq 0$. So $\beta''_k(t)$ is the variance of n under the weights $a_n e^{nt}$, which is at most the mean of n^2 . From the power series, $(z \frac{d}{dz})^2 I_0(2z) = 4z^2 I_0(2z)$ and $(z \frac{d}{dz})^2 I_1(2z) = (4z^2 + 1)I_1(2z)$. So the mean of n^2 is at most $4z^2 + 1$ with $z = s_k e^t \leq ey/2$, since $s_k \leq N/2$. Hence $\beta''_k \leq \kappa = e^2 y^2 + 1$ on the interval.

Now take $\delta = \sqrt{2\eta/\kappa}$. It is at most 1, since for $N \geq 4$ we have $2\eta \leq \frac{1}{2}(e^2 y^2 + 2ey) \leq \kappa$. Lemma E.2 with $f = \Lambda_k$ and $g = \beta_k$ gives, for every $0 < k < N$,

$$|\Lambda'_k(\log u) - \beta'_k(\log u)| \leq \frac{\eta}{\delta} + \frac{\kappa\delta}{2} = \sqrt{2\eta\kappa} \leq \frac{2e^2 y^2 + 2ey + 1}{\sqrt{2N}} \leq \frac{11(1+y)^2}{\sqrt{N}}. \quad (39)$$

The second inequality is $\sqrt{ab} \leq \frac{1}{2}(a+b)$.

Step 2. We compute $\beta'_k(\log u)$. Write $z = s_k e^t$, $w = 2z$ and $R = R(w)$ as in Lemma E.3. Using $I'_0 = I_1$ and $I'_1(w) = I_0(w) - I_1(w)/w$,

$$\beta'_k(t) = 1 + \frac{z \mathcal{I}'_k(z)}{\mathcal{I}_k(z)} = 1 + \frac{wR + c_k w - c_k R}{1 + c_k R} = w + \frac{1 + w(c_k - 1)(1 - R)}{1 + c_k R}.$$

At $t = \log u$ we have $w = 2z_k$. Call the last fraction B_k at this point. So $\beta'_k(\log u) = 2z_k + B_k$. By Lemma E.3, $0 < R < 1$ and $w(1-R) \leq 1$. Since $c_k \geq 1$, the numerator of B_k lies between 1 and c_k . The numerator is also at most $1 + wc_k = 1 + y$, and the denominator is at least 1, so $0 < B_k \leq 1 + y$. Moreover

$$B_k - \frac{1}{2} \leq \frac{c_k}{1 + c_k R} - \frac{1}{2} = \frac{(c_k - 1) + c_k(1 - R)}{2(1 + c_k R)} \leq \frac{c_k - 1}{2} + \frac{c_k(1 - R)}{2},$$

and $B_k - \frac{1}{2} \geq \frac{1 - c_k R}{2(1 + c_k R)} \geq -\frac{c_k - 1}{2}$. By Lemma E.3, $c_k(1 - R) \leq c_k/w = c_k^2/y$. Hence

$$\left| B_k - \frac{1}{2} \right| \leq \frac{c_k - 1}{2} + \frac{c_k^2}{2y}.$$

If $W^2 \leq \frac{1}{2}$ on $K = k$, then $c_k^2 \leq 2$ and $c_k - 1 \leq W^2$, because $(1-v)^{-1/2} - 1$ is convex in v , vanishes at $v = 0$ and is less than $\frac{1}{2}$ at $v = \frac{1}{2}$, so it is at most v for $0 \leq v \leq \frac{1}{2}$. Hence $|B_k - \frac{1}{2}| \leq W^2/2 + 1/y$ there.

The term B_k is the part of the score that does not grow with x . It is close to $\frac{1}{2}$ in the middle of the row, and this is the $\frac{1}{2}$ in G .

Step 3. Let $\psi(w) = 1 - w^2/2 - \sqrt{1 - w^2}$ for $|w| \leq 1$. The power series of $\sqrt{1 - v}$ is $1 - v/2$ minus a series $\sum_{j \geq 2} a_j v^j$ with $a_j \geq 0$, and $\sum_j a_j = \frac{1}{2}$ since $\sqrt{1 - v}$ vanishes at $v = 1$. So $0 \leq \psi(w) \leq w^4/2$. On $K = k$ we have $2z_k = y\sqrt{1 - W^2} = y(1 - W^2/2) - y\psi(W)$. So on the event $0 < K < N$, Steps 1 and 2 give

$$Z - G = (Z - \beta'_K(\log u)) + \left(B_K - \frac{1}{2}\right) - y\psi(W).$$

On the 2 atoms $K \in \{0, N\}$ we have $Z = 0$ and $W^2 = 1$, so $Z - G = -(y + 1)/2$.

Step 4. We collect the moments we need. By Lemma E.4, $\mathbb{E}W^4 \leq 3/x^2 + 9/(2x^4) \leq 8/x^2$ and $\mathbb{E}W^8 \leq Cx^{-4}$. By Markov's inequality, $\mathbb{P}(W^2 > \frac{1}{2}) \leq 16\mathbb{E}W^8 \leq Cx^{-4}$. Also $\mathbb{P}(K \in \{0, N\}) = \mathbb{P}(J = 0) = p^{N-1} \leq e^{-q(N-1)} \leq e^{1/2}e^{-x}$.

Step 5. Write $\|\cdot\|$ for the norm of L^2 and $\mathbf{1}_{\text{int}}$ for the indicator of $0 < K < N$. By Step 3 and Minkowski's inequality,

$$\|Z - G\| \leq \|(Z - \beta'_K(\log u))\mathbf{1}_{\text{int}}\| + \|(B_K - \frac{1}{2})\mathbf{1}_{\text{int}}\| + \|y\psi(W)\| + \frac{y+1}{2} \mathbb{P}(K \in \{0, N\})^{1/2}.$$

By (39) the first term is at most $11(1 + y)^2/\sqrt{N} \leq 99x^2/\sqrt{N}$, since $(1 + y)^2 \leq (1 + 2x)^2 \leq 9x^2$. For the second term we use Step 2 on the event $W^2 \leq \frac{1}{2}$, and $|B_K - \frac{1}{2}| \leq 1 + y$ on the rest of the interior. This gives

$$\mathbb{E}[(B_K - \frac{1}{2})^2 \mathbf{1}_{\text{int}}] \leq \mathbb{E}\left(\frac{W^2}{2} + \frac{1}{y}\right)^2 + (1 + y)^2 \mathbb{P}\left(W^2 > \frac{1}{2}\right) \leq \frac{\mathbb{E}W^4}{2} + \frac{2}{y^2} + \frac{C(1 + y)^2}{x^4} \leq \frac{C}{x^2}.$$

By Step 3 the third term is at most $\frac{y}{2}(\mathbb{E}W^8)^{1/2} \leq C/x$. By Step 4 the last term is at most $Cxe^{-x/2} \leq C/x$. Therefore

$$\mathbb{E}(Z - G)^2 \leq C\left(\frac{x^4}{N} + \frac{1}{x^2}\right),$$

and dividing by $\text{Var } J \geq x/4$ gives $e_N - R_N^2 \leq C(x^{-3} + x^3/N)$.

We used $N \geq 1700$ only in Step 1, so we may take $N_0 = 1700$. Every other constant is absolute, since each bound above holds uniformly in $0 < q \leq \frac{1}{2}$ and $1 \leq x \leq N^{1/4}$. This proves the bound in Theorem 8.3. The expansion of e_N in Theorem 8.3, Corollary 8.4 and the lower bound in Remark 8.5 follow from this bound and the closed form, and their proofs are given in Section 8.

References

- [1] P. Kagey, *Problem 131*, Open Problem Collection, <https://peterkagey.com/problems/131/> (accessed September 2026).
- [2] A. Pemmasani, *Uniform Bessel bounds for endpoint crossings in the persistent random walk*, preprint, 2026.
- [3] A. Pemmasani, *Vertex crossings in a symmetric Markov multinomial model*, preprint, 2026.
- [4] P. Dupuis and Y. Liu, *On the large deviation rate function for the empirical measures of reversible jump Markov processes*, Ann. Probab. **43** (2015), 1121–1156. <https://doi.org/10.1214/13-AOP883>.
- [5] A. Guillin, B. Nectoux and L. Wu, *Large deviations of the empirical measures of a strong-Feller Markov process inside a subset and quasi-ergodic distribution*, preprint (2024), to appear in Ann. Fac. Sci. Toulouse Math., <https://arxiv.org/abs/2411.17216>.

- [6] J. N. Darroch and E. Seneta, *On quasi-stationary distributions in absorbing discrete-time finite Markov chains*, J. Appl. Probab. **2** (1965), 88–100. <https://doi.org/10.2307/3211876>.
- [7] J. N. Darroch and E. Seneta, *On quasi-stationary distributions in absorbing continuous-time finite Markov chains*, J. Appl. Probab. **4** (1967), 192–196. <https://doi.org/10.2307/3212311>.
- [8] F. Colonius and M. Rasmussen, *Quasi-ergodic limits for finite absorbing Markov chains*, Linear Algebra Appl. **609** (2021), 253–288. <https://doi.org/10.1016/j.laa.2020.09.011>.
- [9] Z. Dietz and S. Sethuraman, *Occupation laws for some time-nonhomogeneous Markov chains*, Electron. J. Probab. **12** (2007), 661–683. <https://doi.org/10.1214/EJP.v12-413>.
- [10] Z. Dietz and S. Sethuraman, *Large deviations for a class of nonhomogeneous Markov chains*, Ann. Appl. Probab. **15** (2005), 421–486. <https://doi.org/10.1214/105051604000000990>.
- [11] P. Whittle, *Some distribution and moment formulae for the Markov chain*, J. R. Stat. Soc. Ser. B **17** (1955), 235–242. <https://doi.org/10.1111/j.2517-6161.1955.tb00197.x>.
- [12] R. Dawson and I. J. Good, *Exact Markov probabilities from oriented linear graphs*, Ann. Math. Statist. **28** (1957), 946–956. <https://doi.org/10.1214/aoms/1177706795>.
- [13] L. A. Goodman, *Exact probabilities and asymptotic relationships for some statistics from m -th order Markov chains*, Ann. Math. Statist. **29** (1958), 476–490. <https://doi.org/10.1214/aoms/1177706623>.
- [14] M. Polettini, *BEST statistics of Markovian fluxes: a tale of Eulerian tours and Fermionic ghosts*, J. Phys. A **48** (2015), 365005. <https://doi.org/10.1088/1751-8113/48/36/365005>.
- [15] G. Carugno, P. Vivo and F. Coghi, *Graph-combinatorial approach for large deviations of Markov chains*, J. Phys. A **55** (2022), 295001. <https://doi.org/10.1088/1751-8121/ac79e6>.
- [16] L. Bertini, D. Gabrielli and C. Landim, *Metastable Γ -expansion of finite state Markov chains level two large deviations rate functions*, Ann. Appl. Probab. **34** (2024), 3820–3869. <https://doi.org/10.1214/24-AAP2051>.
- [17] C. Landim, *Metastability from the large deviations point of view: a Γ -expansion of the level two large deviations rate functional of non-reversible finite-state Markov chains*, Stochastic Process. Appl. **165** (2023), 275–315. <https://doi.org/10.1016/j.spa.2023.09.001>.
- [18] E. B. Fox, E. B. Sudderth, M. I. Jordan and A. S. Willsky, *A sticky HDP-HMM with application to speaker diarization*, Ann. Appl. Stat. **5** (2011), 1020–1056. <https://doi.org/10.1214/10-AOAS395>.
- [19] J. Gillis, *Correlated random walk*, Proc. Cambridge Philos. Soc. **51** (1955), 639–651. <https://doi.org/10.1017/S0305004100030711>.
- [20] A. Chen and E. Renshaw, *The Gillis–Domb–Fisher correlated random walk*, J. Appl. Probab. **29** (1992), 792–813. <https://doi.org/10.2307/3214713>.
- [21] A. Chen and E. Renshaw, *The general correlated random walk*, J. Appl. Probab. **31** (1994), 869–884. <https://doi.org/10.2307/3215313>.
- [22] D. Marris and L. Giuggioli, *Persistent and anti-persistent motion in bounded and unbounded space: resolution of the first-passage problem*, New J. Phys. **26** (2024), 073020. <https://doi.org/10.1088/1367-2630/ad5d85>.
- [23] I. Santra, U. Basu and S. Sabhapandit, *Run-and-tumble particles in two dimensions: marginal position distributions*, Phys. Rev. E **101** (2020), 062120. <https://doi.org/10.1103/PhysRevE.101.062120>.
- [24] L. Angelani, *Orthogonal run-and-tumble walks*, J. Stat. Mech. (2022), 123207. <https://doi.org/10.1088/1742-5468/aca588>.
- [25] J. A. Smiga, M. Radaelli, F. C. Binder and G. T. Landi, *Stochastic metrology and the empirical distribution*, Phys. Rev. Research **5** (2023), 033150. <https://doi.org/10.1103/PhysRevResearch.5.033150>.
- [26] P. Hall and C. C. Heyde, *Martingale Limit Theory and Its Application*, Academic Press, New York, 1980.
- [27] A. Berman and R. J. Plemmons, *Nonnegative Matrices in the Mathematical Sciences*, SIAM, Philadelphia, 1994. <https://doi.org/10.1137/1.9781611971262>.

- [28] B. Bollobás, *Modern Graph Theory*, Springer, New York, 1998. <https://doi.org/10.1007/978-1-4612-0619-4>.
- [29] NIST, *Digital Library of Mathematical Functions*, Eqs. 10.25.2 and 10.40.1, <https://dlmf.nist.gov/10.25.E2>, <https://dlmf.nist.gov/10.40.E1>.